UNIVERSIDAD POLITÉCNICA DE MADRID

ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN

VISIONS “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services”

TESIS DOCTORAL

IVÁN LEQUERICA ROCA
Licenciado en Ciencias Físicas
Ingeniero Superior en Electrónica

2013

DEPARTAMENTO DE INGENIERÍA DE SISTEMAS TELEMÁTICOS

ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN

VISIONS “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services”

AUTOR
IVAN LEQUERICA ROCA
Licenciado en Ciencias Físicas
Ingeniero Superior en Electrónica

DIRECTORES
CARLOS MIGUEL NIETO
Doctor Ingeniero de Telecomunicación
PEDRO M. RUIZ
Doctor Ingeniero de Telecomunicación

2013

A Alejandra, Ignacio y Mario.

AGRADECIMIENTOS

A Alejandra y los caimanes, por esos 6 ojos azules que lo iluminan todo

A mis padres, por su apoyo incondicional

A la tower, por mostrarme un mundo diferente

A mis amigos, por compartir una visión de la vida

A Pedro y Carlos por guiarme en el camino.

Al equipo de la UMU por descubrirme la Investigación con mayúsculas. A todos los compañeros de los proyectos: MARTA, m:VIA, SAFESPOT, CARING CARS, TELEFOT, COM2REACT, MYCAREVENT

A Telefónica I+D por impulsar la innovación como motor de cambio empresarial. Al FP7 de la Comisión Europea. A los programas CENIT y PROFIT de los Ministerios de Industria y Ciencia e Innovación, cuando este país apostaba por la Investigación

A la música que me ha acompañado: David Bowie, Miles Davis, Leonard Cohen, Pink Floyd, The Velvet Underground, Vinicius de Moraes, Charlie Parker, Bettye Lavette, Camarón de la Isla, Enrique Morente, Bob Dylan, Chet Baker, Art Tatum, Stan Getz, Pixies, Micah P. Hinson, Kiko Veneno, Daniel Johnston, Ñico Saquito, The Band, Violent Femmes, The Hustlers y tantos otros!

INDEX

ABSTRACT 1
RESUMEN 3
1 INTRODUCTION AND MOTIVATION 5
1.1 Introduction 5
1.2 Vehicular Communications 7
1.2.1 VANET 7
1.2.2 Classification 8
1.2.3 Graphical examples 11
1.2.4 Challenges 14
1.3 Motivation 19
1.4 Objectives of the thesis 20
1.5 Organization of this thesis 21
2 STATE OF THE ART 22
2.1 Introduction 22
2.2 Vehicular platforms in the literature 23
2.3 Challenges in Vehicular Platforms 24
2.3.1 Comprehensive architectures 24
2.3.2 Seamless use of multiple radio technologies 27
2.3.3 VANET routing performance 30
2.3.4 Security 31
2.4 IP Multimedia Subsystem 34
2.4.1 IMS Core 34
2.4.2 Enablers 39
2.5 Radio technologies in vehicular communications 48
2.5.1 WiFi 49
2.5.2 WiMAX 49
2.5.3 UMTS 49
2.5.4 LTE 49
2.5.5 WAVE. IEEE 802.11p 50
2.6 Conclusions 51
3 SOLUTION DESIGN 52
3.1 Introduction 52
3.2 Requirements 53
3.2.1 Requirements for improving Vehicular Communications 53
3.2.2 Requirements for the Service Deployment Platform 54
3.3 Design key decisions 57

3.3.1 An IMS based solution 57
3.3.2 Infrastructure without RSUs 64
3.3.3 On-board architecture based on CALM 65
3.4 VISIONS Architectural Design 65
3.4.1 On-board Architecture 66
3.4.2 Infrastructure Architecture 77
3.5 VISIONS key processes 83
3.5.1 Vehicle data gathering 84
3.5.2 File generation and target area selection for VANET routing improvement 86
3.5.3 Distribution 92
3.5.4 Overall Operation 94
3.6 Conclusions 95
4 VANET ROUTING IMPROVEMENT 99
4.1 Introduction 99
4.2 Related Work 100
4.3 VANET Routing improvement using VISIONS 103
4.3.1 Benefits 103
4.3.2 Architecture 104
4.4 Performance evaluation 106
4.4.1 Qualitative analysis 106
4.4.2 Quantitative analysis 109
4.5 Conclusions 116
5 VANET SECURITY IMPROVEMENT 118
5.1 Introduction 118
5.2 Related Work 119
5.3 Certificate revocation improving using VISIONS 121
5.3.1 Benefits 122
5.3.2 Architecture 123
5.4 Performance evaluation 124
5.4.1 Delta-CRL distribution simulation 124
5.4.2 Base-CRL analytical study 129
5.5 Conclusions 130
6 A SERVICE BASED ON VISIONS 131
6.1 Introduction 131
6.2 Social networks in vehicular scenarios 133
6.3 Challenges and Design issues 134
6.4 Application Platforms 136

6.5 Drive and Share 138
6.5.1 Service features 138
6.5.2 Architectural design 139
6.5.2.3 Connectivity 142
6.5.2.4 Usability analysis 142
6.6 Conclusions 145
7 CONCLUSIONS 147
7.1 Novelties 148
7.2 Fulfillment of the objectives 149
7.3 Future Work lines 150
7.4 List of Publications 151
7.4.1 Journal Papers 151
7.4.2 Conference Proceedings 151
7.4.3 Book Chapters 152
7.5 Patents 153
REFERENCES 155
Glossary of Terms & Acronyms 167

ABSTRACT

Digital services and communications in vehicular scenarios provide the essential assets to improve road transport in several ways like reducing accidents, improving traffic efficiency and optimizing the transport of goods and people.

Vehicular communications typically rely on VANET (Vehicular Ad hoc Networks). In these networks vehicles communicate with each other without the need of infrastructure. VANET are mainly oriented to disseminate information to the vehicles in certain geographic area for time critical services like safety warnings but present very challenging requirements that have not been successfully fulfilled nowadays. Some of these challenges are; channel saturation due to simultaneous radio access of many vehicles, routing protocols in topologies that vary rapidly, minimum quality of service assurance and security mechanisms to efficiently detect and neutralize malicious attacks.

Vehicular services can be classified in four important groups: Safety, Efficiency, Sustainability and Infotainment. The benefits of these services for the transport sector are clear but many technological and business challenges need to be faced before a real mass market deployment. Service delivery platforms are not prepared for fulfilling the needs of this complex environment with restrictive requirements due to the criticism of some services

To overcome this situation, we propose a solution called VISIONS “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services”. VISIONS leverages on IMS subsystem and NGN enablers, and follows the CALM reference Architecture standardized by ISO. It also avoids the use of Road Side

Units (RSUs), reducing complexity and high costs in terms of deployment and maintenance.

We demonstrate the benefits in the following areas:

1. VANET networks efficiency. VISIONS provide a mechanism for the vehicles to access valuable information from IMS and its capabilities through a cellular channel. This efficiency improvement will occur in two relevant areas:

    a. Routing mechanisms. These protocols are responsible of carrying information from a vehicle to another (or a group of vehicles) using multihop mechanisms. We do not propose a new algorithm but the use of VANET topology information provided through our solution to enrich the performance of these protocols.

    b. Security. Many aspects of security (privacy, key, authentication, access control, revocation mechanisms, etc) are not resolved in vehicular communications. Our solution efficiently disseminates revocation information to neutralize malicious nodes in the VANET.

2. Service delivery platform. It is based on extended enablers, reference architectures, standard protocols and open APIs. By following this approach, we reduce costs and resources for service development, deployment and maintenance.

To quantify these benefits in VANET networks, we provide an analytical model of the system and simulate our solution in realistic scenarios. The simulations results demonstrate how VISIONS improves the performance of relevant routing protocols and is more efficient neutralizing security attacks than the widely proposed solutions based on RSUs.

Finally, we design an innovative Social Network service based in our platform, explaining how VISIONS facilitate the deployment and usage of complex capabilities.

RESUMEN

Los servicios digitales y comunicaciones en entornos vehiculares proporcionan herramientas esenciales para mejorar el transporte por carretera; reduciendo el número de accidentes, mejorando la eficiencia del tráfico y optimizando el transporte de mercancías y personas.

Las comunicaciones vehiculares generalmente están basadas en redes VANET (Vehicular Ad hoc Networks). En dichas redes, los vehículos se comunican entre sí sin necesidad de infraestructura. Las redes VANET están principalmente orientadas a difundir información (por ejemplo advertencias de seguridad) a los vehículos en determinadas zonas geográficas, pero presentan unos requisitos muy exigentes que no se han resuelto con éxito hasta la fecha. Algunos de estos retos son; saturación del canal de acceso de radio debido al acceso simultáneo de múltiples vehículos, la eficiencia de protocolos de encaminamiento en topologías que varían rápidamente, la calidad de servicio (QoS) y los mecanismos de seguridad para detectar y neutralizar los ataques maliciosos de manera eficiente.

Los servicios vehiculares pueden clasificarse en cuatro grupos: Seguridad, Eficiencia del tráfico, Sostenibilidad, e Infotainment (información y entretenimiento). Los beneficios de estos servicios para el sector son claros, pero es necesario resolver muchos desafíos tecnológicos y de negocio antes de una implementación real. Las actuales plataformas de despliegue de servicios no están preparadas para satisfacer las necesidades de este complejo entorno con requisitos muy restrictivos debido a la criticidad de algunas aplicaciones.

Con el objetivo de mejorar esta situación, proponemos una solución llamada VISIONS "Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services". VISIONS se basa en el subsistema IMS, las capacidades NGN y es compatible con la arquitectura de referencia CALM estandarizado por ISO para sistemas de transporte. También evita el uso de elementos en las carreteras, conocidos como Road Side Units (RSU), reduciendo la complejidad y los altos costes de despliegue y mantenimiento.

A lo largo de la tesis, demostramos los beneficios en las siguientes áreas:

1. Eficiencia en redes VANET. VISIONS proporciona un mecanismo para que los vehículos accedan a información valiosa proporcionada por IMS y sus capacidades a través de un canal de celular. Dicho mecanismo contribuye a la mejora de dos áreas importantes:

    a. Mecanismos de encaminamiento. Estos protocolos son responsables de llevar información de un vehículo a otro (o a un grupo de vehículos) utilizando múltiples saltos. No proponemos un nuevo algoritmo de encaminamiento, sino el uso de información topológica de la red VANET a través de nuestra solución para enriquecer el funcionamiento de los protocolos más relevantes.

    b. Seguridad. Muchos aspectos de la seguridad (privacidad, gestión de claves, autenticación, control de acceso, mecanismos de revocación, etc) no están resueltos en las comunicaciones vehiculares. Nuestra solución difunde de manera eficiente la información de revocación para neutralizar los nodos maliciosos en la red.

2. Plataforma de despliegue de servicios. Está basada en capacidades NGN, arquitecturas de referencia, protocolos estándar y APIs abiertos. Siguiendo este enfoque, reducimos costes y optimizamos procesos para el desarrollo, despliegue y mantenimiento de servicios vehiculares.

Para cuantificar estos beneficios en las redes VANET, ofrecemos un modelo de analítico del sistema y simulamos nuestra solución en escenarios realistas. Los resultados de las simulaciones muestran cómo VISIONS mejora el rendimiento de los protocolos de encaminamiento relevantes y neutraliza los ataques a la seguridad de forma más eficientes que las soluciones basadas en RSU.

Por último, diseñamos un innovador servicio de red social basado en nuestra plataforma, explicando cómo VISIONS facilita el despliegue y el uso de las capacidades NGN.

1 INTRODUCTION AND MOTIVATION

1.1 Introduction

The boom of wireless communications in recent years has led to its implementation in many diverse scenarios. One of the newest and more relevant is the communication between vehicles, in some cases known as Vehicular Ad-hoc Networks (VANETs). Such communications are in an embryonic stage but it is clear that will bring great benefits to society especially in areas related to safety.

Road network is one of the key factors in global economies. Road transport takes a significant percentage of Gross Domestic Product (GDP) and employs a large number of citizens around the world. Although transportation is one of the most relevant activities in the global economy, the costs due to energy, time and pollution are still high. Moreover, road accidents are one of the higher causes of mortality all over the world. The number of deaths and serious injuries has significantly decreased in the last few years as a result of recent actions taken by authorities in many countries but it is still a challenge to the society as a whole.

The dramatic figures of fatal road accidents are making governments of many countries and the European Commission encourage research in information technology and communications applied to the automotive sector. The resulting technologies are typically known as Intelligent Transport Systems (ITS).

To give an example of the tragedy, here are some numbers from the Road Safety Annual Report for 2011 [1]:

COUNTRY	ROAD FATALITIES	DEATHS/100.000 INHABITANTS
JAPAN	5.745	4,5
GERMANY	3,648	4,5
UK	1.905	3,1
SPAIN	2.478	5,4
US	32.885	10,6

Table 1.1: Road fatalities in 2010

Moreover, the car is the third scenario where a person spends more time in its daily live after home and office. This makes it an ideal environment for the development and deployment of services that improve the quality of people´s life. Services can be offered in several areas of great interest to users, and are typically classified in these four categories: Road safety, traffic efficiency, sustainable transport, information and entertainment (Infotainment).

A typical example of vehicular communications is the exchange of warning messages between vehicles. A vehicle can detect a dangerous situation or be involved in an incident and send information through wireless communication to other vehicles (or elements of infrastructure) to avoid increasing the scope of that accident.

Although there have been many research efforts, the vehicular communication and services are not mature enough to allow a mass market deployment. Due to the lack of effective solutions in this field with enormous potential, we have chosen this topic as the main research area for this thesis.

Our motivation comes up as a response of those challenges, in which we aim to design the technology required to improve road transport, creating an intelligent environment within the roads and the vehicles.

Our main objective is improving vehicular communications using the capabilities of the IMS [2] subsystem and NGN Enablers. We call to our solution “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services” (VISIONS).

We demonstrate the benefits of using VISIONS in two critical areas of vehicular networks: Routing and Security. Additionally, we analyze and propose a platform to deploy vehicular services based on VISIONS.

1.2 Vehicular Communications

It is important to distinguish between vehicular communications and VANET (Vehicular Ad hoc Networks). By vehicular communications we refer to any data exchange in which a vehicle (an on board device) participates.

VANET are a special case of vehicular communications, in which vehicles can only communicate with each other without the need of infrastructure. One of the advantages of VANET is that a node can send information to another node that is outside its radio range through a multi-hop communication.

In this section we introduce the VANET as a relevant group within vehicular communications. We provide some background on vehicular communications and their possible classification methods. Finally, we present their challenges and open issues in order to understand their limitations and how our proposal can contribute to solve them.

1.2.1 VANET

VANET is a special case of ad hoc networks. An ad hoc network is a set of nodes which are able to communicate with each other without the need of infrastructure. When the destination node is outside the scope of the source, intermediate nodes act as a bridge between the two, so that data travels from node to node until the destination is eventually reached. Another important feature of an ad hoc network is that nodes can be mobile, which complicates communication.

In VANET, nodes (vehicles) move at high speed. They have a pattern of restricted mobility, because the vehicles move on roads, following a topology, with certain rules (speed limits, traffic lights, priorities…) etc. An important advantage compared to other ad-hoc networks is that OBUS in vehicles can have great processing power and memory, do not have energy constraints and can have multiple interfaces and technologies for data communications. In addition, vehicles may have navigators on board, so they know their position and the streets around itself.

Vehicular networks opened up a new path in ad-hoc network research because of the mobility of the nodes, the high dynamic topology but mainly because of the restricted requirements they have in terms of performance and security. Let's look in detail at the feature set that characterizes these networks:

- Autonomy. Each terminal is an autonomous node with processing and communication capabilities being able to retrieve information from other nodes in the same network.
- Distributed control. Due to the lack of an online centralized management and coordination entity, an efficient use of the available bandwidth of the wireless channel is hard to achieve. This might result in a less efficient usage of the channel and in a large number of packet collisions. On the other hand, the fact that the basic operation of a VANET does not depend on network infrastructure makes it more tolerant to system failures.
- Routing. It is necessary that all nodes are capable of providing dynamic routing mechanisms. Multi-hop routing is based on the capabilities of each node. Traditional routing protocols do not work in this context as they are not prepared for the fast changes in network topology nor the large amount of nodes in the network.
- Network topology and high-speed variable nodes. In a VANET nodes can move at high speeds, but usually they do follow certain patterns motion, for example by following the trajectories of a highway. Because of the road traffic, the topology of the network can be subdivided into multiple clusters of vehicles and produce significant packet losses. Mechanisms are needed to detect these circumstances and minimize its effects.
- Inherent characteristics of the radio channel. VANET scenarios have unfavorable characteristics for developing wireless communications, i.e., multiple reflecting objects that degrade the strength and quality of the signal. Additionally, the mobility of the surrounding objects, including the own vehicle, may produce fading effects that lead to a variable capacity of the links. Moreover, the multihop transmissions through several vehicles increase the problem.

1.2.2 Classification

For a proper presentation of vehicular communications we firstly classify the different types. There are three different ways in which vehicular communications are commonly classified:

- Attending to the scope in terms of which elements participate in the communication; only vehicles, Road Side Units (RSUs), nodes from other networks (Internet) and/or nomadic devices.
- Attending to the destination; taking into account if the destination is a vehicle, a group of vehicles located inside a certain area, a group of vehicles subscribed to a service, or a combination of these options.
- Attending the type of communication. Here we distinguish the communications in which only vehicles take part through the VANET from those that are supported by RSUs or telecom operator mobile networks. There is a special case here where every option is supported and such network is commonly known as hybrid.

We analyze below each category in more detail and provide some examples:

1.2.2.1 Attending to the scope

- Inter-vehicular. This category contains all the communications in which a vehicle communicates with another one or a group of vehicles regardless the technology and it is commonly represented by Vehicle to Vehicle (V2V). Here we can find a huge variety of services; including safety, efficiency or infotainment. For example V2V may be used to share critical information with other cars that might be in danger or to transmit a picture from one vehicle to another, even if they are not close to each other or in different roads.
- Intra-vehicular. Here we find all the communications inside the vehicle, including communication between users and interaction between users and the vehicle system (commonly known as V2U). Anyway, the communication does not exceed the limits of the vehicle so they are currently provided using Short Range Communications (SRC). Representative examples of this category are messaging between passengers inside public transportation and accessing multimedia content stored in the vehicle equipment from the user´s device.
- With the infrastructure. In this group we can find two different types of communications. Firstly there are those that transmit information between vehicles and RSUs (also known as V2R) but also include services in which vehicles access information from other networks or the Internet (V2I). In those cases where vehicles do not connect directly to the Internet using a cellular technology it is necessary some gateway discovery mechanism to find the most appropriate gateway through the VANET and the RSU.

1.2.2.2 Attending to the destination

- Unicast is the information transmission from a single sender to a single receiver. In vehicular communications is generally associated to non-safety services when the information is private and needs to be sent only to a certain vehicle. If the data is transmitted through the VANET, those vehicles that participate forwarding the packet are not allowed to access the content.
- Broadcast is the act of disseminating information through the entire network. In VANETS this type of communication is also known as flooding. Due to the typical size of the VANET, it does not make sense to completely flood the entire network with a message and normally there are schemes to restrict the propagation. These schemes control the forwarding behavior of the vehicles, either by adapting the time when to forward or by introducing rules on whether a given vehicle should forward the packet or not. (i.e.: maximum number of hops).
- Geocast is a specific case of Broadcast. Is the act of disseminating data only within a given geographical zone. It is very useful to narrow the special scope of a message, for example that informs of adverse weather conditions or an accident. In some situations a group of vehicles will participate in the forwarding chain but as they are not in the target area, they won´t receive the information.
- In Multicast the sender transmits data only to interested destinations by using special address assignments. Nodes commonly subscribe to the service to receive multicast information within a group. Participants of these groups can be far away from each other, even in different roads complicating the mechanism.

1.2.2.3 Attending to the type of communication

- VANET. Only vehicles can participate in the transmission of the information through Ad-hoc links. Depending on the distance between sender and receiver, the communication consists in a single hop or on the contrary by a multi-hop path in which other vehicles act as gateways. Road Site Units can participate in the communication increasing the connectivity and/or the capacity of the VANET. In some cases they also can act as gateways to the Internet.
- Mobile operator networks (3G/4G). Although in certain forums vehicular communications are completely identified with VANETs, to be precise this general term implies every type of communication involving the vehicle. Thus, those services provided through mobile networks in which the source or

destination is a vehicle are included in this category. Mobile networks offer a lot of benefits to vehicular scenarios such as larger coverage, bandwidth and QoS mechanisms. However, the delays caused by the operator infrastructure make them inappropriate for time critical services, commonly safety related ones.

- Hybrid. By hybrid communications we understand those systems in which the technology used (VANET or Mobile) is dynamically chosen taking into account the network situation and the service needs. In fact, there can be some cases in which a service starts transmitting through the VANET but after losing connectivity, it performs a vertical handover to continue sending information via the mobile network.

1.2.3 Graphical examples

For a better understanding of these types, we present some graphical examples of the most important ones:

In Figure 1.1 the red vehicle detects an accident and disseminates a warning message within a geographical area in which there may be vehicles in danger. Notice that the forwarding mechanism uses cars travelling in the opposite direction to progress the message in the broadcasting zone (inside the red rectangle).

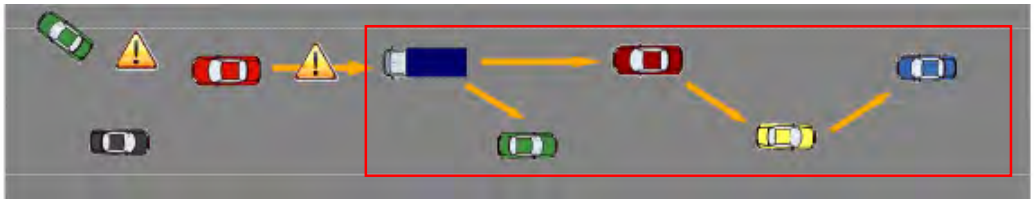


Figure 1.1 Inter-vehicular (V2V) geocast communication through the VANET

In Figure 1.2 the blue lorry sends information to a single vehicle, this can be a private message or even multimedia content. Even if both vehicles have on-board equipment to transmit it through the VANET, the situation of the traffic makes impossible to do it only through Ad-hoc connections. This is a typical scenario of clustering, in which vehicles tend to progress into groups, a clear example of how the nature of traffic patterns influences the communication requirements [3].

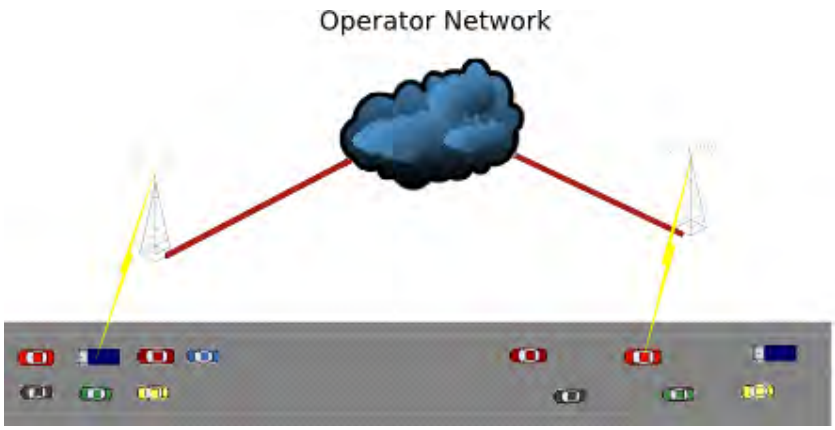
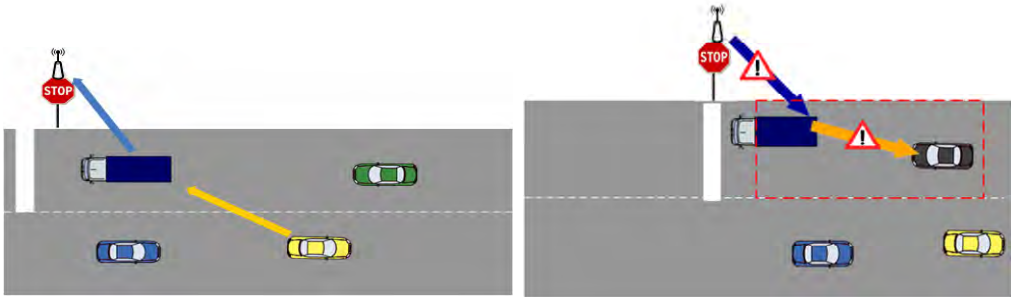


Figure 1.2 Inter-vehicular (V2V) unicast communication through the mobile network

In Figure 1.3a the yellow car sends data to the RSU located in a road junction via the VANET. This can be caused because an incident occurs near the junction and the service decides that the best way to propagate the information to approaching vehicles is through the RSU (Figure 1.3b).



(a) (b)

Figure 1.3 Unicast communication with an RSU (V2R) through the VANET

In Figure 1.4 the grey vehicle accesses an external server, for example for sending an e-mail through an RSU connected via a fixed backbone to the Internet. In order to do this, the VANET must provide a gateway discovery mechanism responsible of selecting the most appropriate gateway in every situation. On the contrary, if the Internet access is provided through the mobile network (Figure 1.5) there is no need to use either an RSU or a gateway mechanism. One of the most important benefits in this option is that users can access external nodes regardless of the situation of other vehicles and the locations or density of the RSUs.

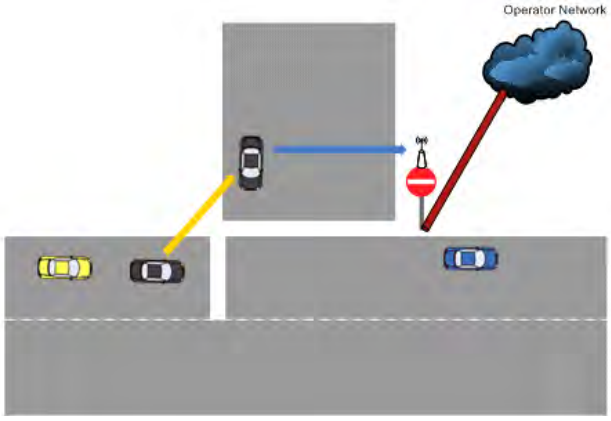


Figure 1.4: Unicast communication with the Internet (V2I) through the VANET

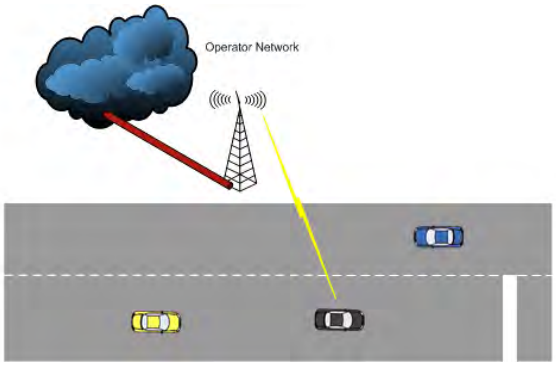


Figure 1.5 Unicast communication with the Internet (V2I) through the mobile network

In Figure 1.6 a hybrid mechanism to connect to the Internet is depicted. The green vehicle uses the VANET to reach the blue car, which sends the information to the Internet using the mobile network. In cases in which not all the vehicles are equipped with mobile capabilities and there is not an RSU nearby (or in order to optimize the number of radio connections), clusters use a vehicle as a gateway to other networks.

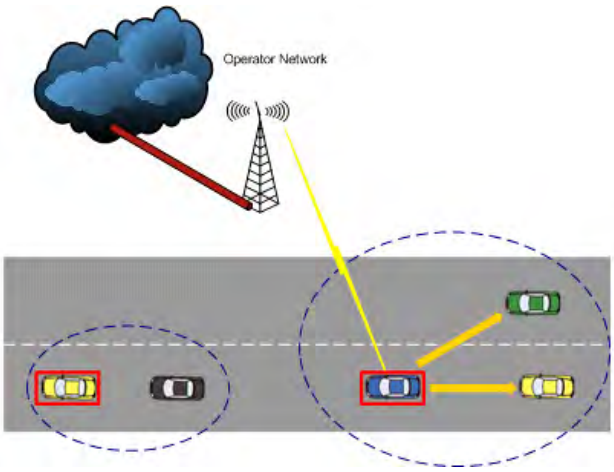


Figure 1.6: Unicast communication with the Internet (V2I) hybrid

We depict in Figure 1.7 an example of intra-vehicular communication in which several nomadic devices of the passengers connect to the On board equipment to share information between them and other nodes outside, for instance for chatting purposes.

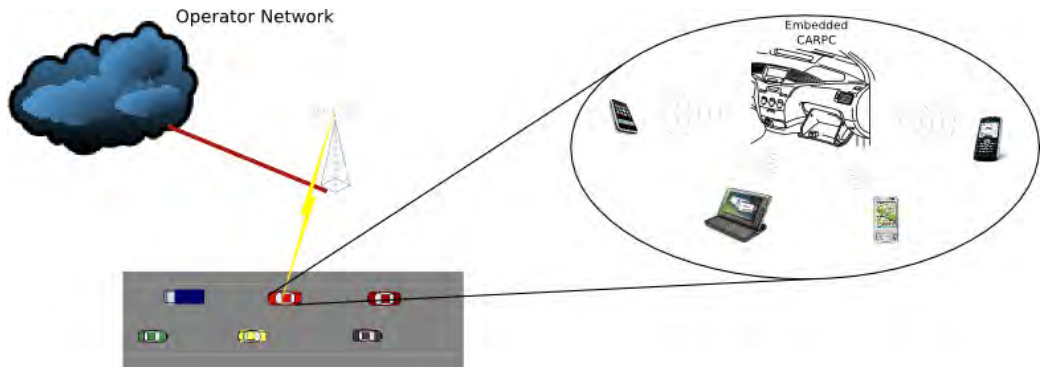


Figure 1.7 Intra-vehicular communications (V2U) with access to the Internet via mobile

1.2.4 Challenges

Vehicular communications in general, and more specifically VANET are one of the more interesting fields in terms of research due to their potential for the transport sector but there are still many open issues and research challenges. We explain the most relevant ones in the following subsections.

1.2.4.1 Connectivity among vehicles

It is difficult to achieve certain connectivity when the network is formed by vehicles on the move. When source and destination are within the coverage range the communication is direct, if they are not, routing protocols will try to use intermediate vehicles as gateways to send the information. In those cases where there are not enough vehicles to provide a routing path, problems arises.

Vehicular communication mechanisms should be able to avoid this fragmentation. Many solutions in the literature that will be analyzed in Chapter 2 propose the use of RSUs to overcome this situation; data packets should be sent to the infrastructure which will be responsible of routing the information to the desired destination. In addition, the use of the infrastructure will also be useful for sending data to other networks and the Internet.

Moreover, this lack of connectivity will be very frequent in the initial phases of the deployment of vehicular communication systems as there will be low penetration rate. The few vehicles that have this technology on board will need some infrastructure to assure the connectivity between them. As these systems popularize; vehicles will be less dependent on infrastructure as the increase of density will mean more scenarios in which vehicles will be able to communicate directly.

There will be some scenarios in which the connectivity has intermittences because of the clustering effect in the roads; for example groups of vehicles travelling at different speeds in a highway. In these cases Delay and Disruption Tolerant Network (DTN) mechanisms can be a good choice [4]. DTN solutions are based on the store and forward principle; a node stores a packet and keeps it until it finds a more appropriate vehicle to which transmit the message. This concept is suitable for many applications. Imagine an emergency notification application after an accident. As soon as the accident is detected the application sends a message to vehicles approaching the emergency location, so they are alerted and stopped. The notification is going through all the coming vehicles until the last one receives the packet; as there are not more neighbors to whom sending the message, it is discarded. If any vehicle arrives after that, it will not be notified and may collide with the last vehicle alerted. Thus, is very helpful to store and forward the packet as soon as it identifies a new neighbor approaching.

Although the use of RSUs and DTN mechanisms definitely help to minimize the problem of fragmentation and lack of connectivity, we are still far away from a solution that effectively covers all the use cases. The huge costs of deploying and maintaining an infrastructure based on RSUs makes non-viable the implementation in real scenarios. On the other hand DTN mechanisms introduce in many cases a delay which is unacceptable for critical services.

1.2.4.2 Connectivity with the Internet

In the vehicular environment, there is also a need to communicate not only with vehicles but also with other nodes and networks. For example drivers are very interested in accessing traffic information in order to decide the optimal route to their destination, and this information is usually stored in centralized servers owned by authorities or service providers. Thus OBUs have to support V2V communication but also connectivity with the Internet, often represented as V2I.

One of the most common solutions for the connection with Internet requires the existence of gateways that act as a bridge between vehicles and Internet in the

infrastructure. This means that some (or all) of the RSU not only act as relying nodes for V2V communication but as gateways for other networks. In this line, gateway discovery protocols responsible for dynamically selecting the most appropriate gateway are a hot research topic nowadays.

However, vehicles can include into their onboard equipment other radio technologies (such as 3G or WiMax) providing direct access to external networks so there is no need to look for a gateway in the infrastructure. In this case there is no need for gateway discovery mechanisms but on the other hand, simultaneous access to 3G networks by many vehicles, especially in some cases like traffic jams can result into very inefficient solutions causing severe network congestion. Taking into account that sometimes the desired information is present in neighboring vehicles and can be accessed through V2V, 3G is not always the best option.

To the best of our knowledge, a solution that dynamically chooses the connection mechanism (including radio technology and route) considering traffic situation, roadside gateways location and 3G coverage/congestion is not yet proposed in the literature.

1.2.4.3 Routing efficiency

In order to provide optimized performance for VANETs in terms of packet delivery ratio and delay it is crucial to have very efficient algorithms in routing protocols. The intrinsic characteristics of VANETs and the mobility patterns of vehicles in different scenarios make the design of routing protocols a very challenging activity. Protocols designed for MANETs are not suitable to the characteristics of a vehicular network, where memory and processing capacity or availability of energy are much higher in a VANET network. There is also relevant scalability issues as a VANET can include millions of vehicles in several countries [5]. On the other hand, vehicles have tracking systems or communication facilities which can provide useful information for the development of hybrid solutions. This information may be the position of the node, the density of the traffic in several routes, or the recommended path calculated by the navigator.

Classical protocols do not behave well when flooding is required, especially in areas of high traffic density, where many nodes try to communicate seamlessly. Each transmission would reach the whole network becoming a problem since it would consume bandwidth on control messages, leaving very few resources for applications or services, which is what really matters to the user. In addition, it may cause the broadcast storm problem [6], leading to packet loss by collision. If we use a Carrier Sense Multiple Access (CSMA) mechanism at the MAC layer such as IEEE 802.11, it drastically increases latency in the message dissemination.

An added problem in VANETs is that given a path between origin and destination, this route will only be valid for a short period of time due to mobility of vehicles. The higher the number of nodes in the route, the easier it is that one of them has shifted its position and, therefore, the route will be broken. A more interesting proposal is to make use of geographic routing, where only local information is used to route a packet hop by hop. In fact, geographic routing has been chosen as the basis of many routing protocols for VANET as we will see in further chapters.

Geographic routing algorithms designed for MANETs are not directly applicable to VANETs because the nodes do not move randomly but follow the paths allowed by the traffic rules. Therefore, there would be cases in which a packet is forwarded to a vehicle that provides good progress but it moves away from the destination.

Thus, routing protocols in vehicular networks should use as much information as possible from the vehicle (location, speed, direction, route…) even information from other vehicles or the road infrastructure. This information is used by protocols for deciding the next hop to the destination. In urban scenarios, the protocols typically tend to determine the path to the destination following a list of streets rather than use a list of nodes.

In this regard, an efficient solution that would provide useful information of the traffic situation (nodes location, density of vehicles in different roads, gateways) in real time, would contribute to optimize the performance of every single VANET protocol designed to operate in these scenarios.

1.2.4.4 Quality of service

Another great challenge of vehicular networks is to provide quality of service. A network which contains a wide range of services must ensure a minimum quality for proper operation. Moreover, there will be high priority applications such as emergency warning messages that we must ensure that the message arrives on time without suffering delays due to the critical nature of this kind of services. A minimum level of QoS is essential for some services, basically safety related, in vehicular scenarios. But this is hard to achieve.

In wireless environment is extremely difficult to provide certain level of QoS due to intrinsic instabilities of radio technologies [7], moreover in VANET it is even harder [8] as the data must travel across several vehicles moving at high speeds. It is almost impossible to assure QoS in terms of classical metrics like bandwidth, delay and jitter but the use of external information would help to choose the best option in terms of route and technology.

1.2.4.5 Security

Security is a central issue in any network, but its importance increases drastically in these environments supporting safety services that could save lives. A malicious user could generate wrong information to redirect other vehicles through other roads, or even worse; delay or remove warning messages from a real accident increasing the severity of it. Recent studies including a categorization of the types of threads and attacks can be found in [9] and [10].

The deployment of vehicular communications in real scenarios demands very challenging security mechanisms. Although we will provide a deeper analysis in further chapters, here is a summary of the main requirements in security within vehicular communications.

- Authentication: For certain services, the system must ensure the authenticity of messages transmitted over the network and the identity of the sender unambiguously. Different users may have different privileges accessing the service, moreover in certain cases only authorized users will be allowed to transmit information; for example an ambulance or the police asking the rest of the vehicles to keep the road clear.
- Data consistency: It is also mandatory to check the consistency of the data to ensure the information has not been modified after the generation by the sender. Without this property a malicious user could alter the information and make other vehicles misbehave.
- Availability: Another aspect to highlight is the communication availability since a malicious user could disable the network in some areas using frequency inhibitors. Is very difficult to avoid completely the effects of this Denial of Service (DoS) attacks but identifying the area in which the network is compromised could minimize the impact.
- Revocation. In those cases where the attacker is a member of the VANET is necessary a mechanism to identify and revoke the credentials of the vehicle to minimize the impact and prevent further attacks.
- Non-repudiation: If an accident or a problem is caused by a malicious behavior from a user, the system should provide mechanisms to identify the offender.
- Privacy: The ability to keep track of the vehicles trajectories or any other data using the system facilities should be only allowed to certain authorities and only in extraordinary cases.

The mechanisms needed to cover all these requirements in a distributed environment like the VANET are far away from a reality nowadays.

1.3 Motivation

The main motivation of this thesis is to improve the operation of vehicular communications and services with an efficient and inexpensive solution being able to be deployed in real environments.

Vehicular services provide the essential communication to improve road transport in several ways like reducing accidents, improving efficiency pathways, optimizing the transport of goods and people, etc.

Many of the services to be deployed in these networks are extremely critical as their proper operation can avoid accidents, saving human lives. This requires that the operation of these systems must be guaranteed anytime under any conditions. It is not acceptable that such systems only work under certain favorable circumstances, like for example with high density of vehicles.

A VANET is generally formed by several vehicles and RSUs equipped with devices providing communication and computation capabilities that allow them to interact without any operator infrastructure. The technologies used for this purpose are basically nomadic networks of the IEEE 802.11 family [11], also known as WiFi. Moreover IEEE has defined and standardized 802.11p to fulfill the requirements of vehicular communications.

To ensure service availability in a VANET under any circumstances, there should be enough points in the road infrastructure to provide connectivity in all geographic areas to be covered. The reduced coverage provided by 802.11 technologies implies an incredibly high density of RSUs, with tremendous costs associated in terms of deployment and maintenance. Therefore, pure VANET are not suitable to fulfill service availability and proper performance, thus a support network is needed to ensure the requirements of these services.

Nowadays, there is a large research activity around communications between vehicles and infrastructure (V2V and V2I), but mostly focuses on pure VANET technologies, routing protocols based on geographic positions, gateway discovery mechanisms, information dissemination through the RSUs, etc. The use of pure VANET approaches lead to unresolved challenges that avoid a mass market deployment of these solutions.

We propose the use of cellular interfaces and backend infrastructure based on the IMS subsystem and NGN enablers to improve fundamental aspects of the vehicular communications like routing and security. As we demonstrate in Chapters 4 and 5, a mechanism that allows the vehicles the exchange of information with the infrastructure, can significantly improve the performance of existing routing algorithms and the

neutralization of attacks. This mechanism does not consume relevant operator's resources in terms of bandwidth, and can be considered inexpensive due to the current costs of consumer electronics and data tariffs. On the other hand, our solution eliminates the need of RSUs as key V2I elements which presented a barrier to entry for the deployment of vehicular communication.

Moreover, we also explore and analyze the potential of using our solution based on IMS and NGN (VISIONS) as a reliable platform for service delivery in vehicular environments.

1.4 Objectives of the thesis

The vehicular environment presents a great opportunity for ICT services. These services can be classified in four important groups: Safety, Efficiency, Sustainability and Infotainment. The whole set of services presents very challenging requirements in terms of communications that haven´t been successfully fulfilled nowadays. Many technological and business challenges need to be faced before launching to the mass market communication systems and services on the road.

After reviewing the characteristics, challenges and opportunities of VANETs, we can state more precisely the objectives of this thesis:

3. Improving efficiency in VANET networks using a cellular channel to access valuable information from IMS and its capabilities. This efficiency improvement will occur in two relevant areas currently unresolved:

    a. Routing mechanism. These protocols are responsible of carrying information from a vehicle to another (or a group of vehicles) using multihop mechanisms. They are one of the more interesting research fields nowadays. We do not propose a new algorithm but the use of VANET topology information provided through our solution to enrich the performance of these protocols.

    b. Security. Many aspects of security (privacy, key, authentication, access control, revocation mechanisms, etc) are not resolved in pure VANET networks and the use of IMS enablers will allow to properly covering the requirements. More specifically we use our solution for efficiently disseminate revocation information to neutralize malicious nodes in the VANET.

4. Providing an ITS service delivery platform based on IMS and NGN capabilities. The aim is to facilitate the development and deployment of new services on the

vehicular environment by using standardized enablers. Our solution will extend the use of ITS maximizing their impact as the majority of telecom operators worldwide are implementing IMS as an enabling technology for the next generation of services.

1.5 Organization of this thesis

The remainder of the thesis is organized as follows: In chapter 2 we analyze the state of the art in those areas worked in this thesis; vehicular platforms, issues in vehicular communications including routing and security, radio technologies, the IMS subsystem and NGN enablers.

In chapter 3 we define the requirements and design our solution: VISIONS. We provide an analysis of the potential and applicability of NGN enablers in vehicular scenarios. We justify the decisions taken and present the architectural elements in both; on board and in the infrastructure. Finally, we define an analytical model that formally represents our solution explaining the operation schemes.

In chapter 4 we analyze the challenges in VANET routing and demonstrate how VISIONS improves efficiency in some relevant routing protocols by distributing topology information of the VANET. We quantify the improvement simulating a urban scenario, also probing that it does not require a lot of bandwidth from the operator network.

In chapter 5 we study security challenges related with certificate revocation mechanisms when one of the members of the VANET becomes an attacker. We simulate an interurban scenario and demonstrate how VISIONS is more efficient neutralizing these attackers than the widely proposed solutions based on RSUs.

In chapter 6 we describe the limitations for bringing vehicular services and applications to the vehicle in a mass-market approach. We justify how VISIONS facilitate the development, deployment and usage of complex services. We design an innovative Social Network application to share relevant information while driving based on our solution.

Chapter 7 presents the conclusions of this thesis and describes future works.

2 STATE OF THE ART

2.1 Introduction

In the previous chapter, we introduced the main characteristics of Vehicular communications and VANET, including the different criteria to classify them and the most relevant challenges. We explained the motivation and objectives of the thesis. As stated, we aim to design a comprehensive platform suitable for improving specific challenges in vehicular communications and for being an efficient service delivery platform. In this chapter we provide an analysis on other vehicular platforms with similar objectives.

We present the most relevant research projects and standardization bodies working on this area. We specifically analyze how these initiatives cover the most challenging issues that our solution aim to improve; the architecture on board vehicles and the infrastructure, the use of different communication channels, the performance of VANET routing and the security issues.

The reminder of the chapter is organized as follows: In section 2 we present the most relevant platforms in the literature, including research projects and national initiatives from different countries like US and Japan. In section 3 we describe how these initiatives cover the challenging issues that we are trying to achieve in this thesis; architectures, use of multiple radio technologies, routing performance and security. In section 4 we present IMS and those enablers, justifying the potential of these technologies for overcoming current challenges in vehicular communications and services. In section 5 we provide a summary of the wireless and mobile technologies

susceptible of being used in vehicular communications. Finally, in section 6 summarize the main findings and conclusions of this chapter.

2.2 Vehicular platforms in the literature

Due to the important benefits that this kind of solutions can bring to the society, there are several initiatives worldwide covering ITS and vehicular platforms.

In the European Union, many research projects like CVIS [12], SAFESPOT [13], COOPERS [14], PREVENT [15], COM2REACT [16], and initiatives like Car 2 Car Communication Consortium [17], iMobility Forum [18] have been funded by the European Commission. The eCall initiative [19] aims to deploy a telematics device installed in all vehicles that, in the event of a serious road accident, will automatically dial 112 and send the location of the car to local emergency agencies. The European Commission is aiming to have a fully functional eCall service to be in place throughout the EU by 2015

The US Department of Transportation (DOT) via the Research and Innovative Technology Administration (RITA), has committed to the use of the Dedicated Short Range Communications technologies (DSRC based on IEEE 802.11p) for active safety for both vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) applications. The Department is also reaffirming its intention to explore all wireless technologies for their applicability to safety, mobility, and environmental applications. The U.S. DOT in collaboration with National Highway Traffic Safety Administration (NHTSA) and some research partners are conducting a Safety Pilot [20] from the fall of 2012 to the fall of 2013 in Ann Arbor, Michigan. The deployment includes approximately 3,000 cars, trucks, and transit vehicles, as well as roadside infrastructure. Vehicles will include fully integrated V2V systems and aftermarket devices.

In Japan, Initiatives in ITS are headed by the Japanese Ministry of Land, Infrastructure and Transportation (MLIT). One of the two main initiates is called Vehicle Information and Communication System (VICS) [21]. VICS brings real-time traffic information into the vehicles and aims to reduce road fatalities, congestion rates and annual CO2 emissions. Currently, 30 million vehicles approximately are equipped with VICS onboard units.

Regarding standardization initiatives, ISO is working in reference architecture for ITS called Continuous Air interface Long and Medium range (CALM) [22]. This architecture provides a set of standard protocols that allow communication over several communications systems (i.e. VANET, 3G, WiMAX, etc) , using the most appropriate

one at any moment based on certain parameters such as quality of service and performance of the media available.

Most of the previously mentioned projects are aligned with the architecture of CALM, who defines the communications in the vehicular environment as a group of wireless technologies, including VANETs, nomadic and short range (SRC). The use of different technologies can be adapted dynamically to attend the requirements of different services and the coverage of technologies. A very important feature is the seamless support of vertical handovers. This means that the system can transmit data using different radio technologies with session maintenance and being transparent for the users.

2.3 Challenges in Vehicular Platforms

As we introduced in previous chapter, the challenges for deploying vehicular communications and services are numerous and difficult to overcome. In this section we focus on four relevant ones, which are central to the work in this thesis; platform architectures, seamless use of multiple access technologies, VANET routing protocol performance and security.

2.3.1 Comprehensive architectures

For a real deployment of a vehicular platform, it is crucial to design a proper architecture on board vehicles, in the roads infrastructure and centralized in the Internet or the operator´s infrastructure. For this purpose, it is important to take into account relevant KPIs like communication performance, deployments costs, modularity, obsolescence, etc.

2.3.1.1 On board equipment

Devices on board are responsible for the communications and service execution within a vehicle. Generally, these devices have the following characteristics:

- Large memory and high processing power nodes capable to handle complex protocols and services, and even storage multimedia content. The vehicles have high power capacity, so energy consumption is not a problem to take into account designing solution for VANETs as it occurs in others wireless networks.
- Several media interfaces to provide different modes of communication; typically WAVE (IEEE 802.11p) for vehicle-to-vehicle, Bluetooth for internal

communications with personal devices or sensors, and 3G for connecting with the Internet.

- Positioning systems and navigators, mainly based on GPS and cartography are very useful not only to guide the driver but as an information enriching capability for routing protocols as will see in further sections.

Basically there are two different approaches within on board configurations:

1. Only a unique device on board responsible for both service execution and communications. When the goal is to optimize the cost of the equipment and provide limited functionality like signaling messages for safety services, there are nodes with limited processing resources and only a WiFi interface to communicate with the surrounding vehicles.
2. The equipment on board consists on several nodes connected for different functions. The On Board Unit (OBU) is responsible for the communications and presents multiple network interfaces (i.e.: 3G/4G, WAVE, WiFi, WiMAX, etc), the Application Unit (AU) manages the services execution and has high processing capabilities and storage. Sometimes, there is also another device for the interaction with the vehicle systems, including sensors and actuators. This is widely proposed by eSafety Forum [18] and the most relevant research European projects (CVIS [12], SAFESPOT [13] and COOPERS [14]). This reference architecture is depicted in the Figure 2.1.

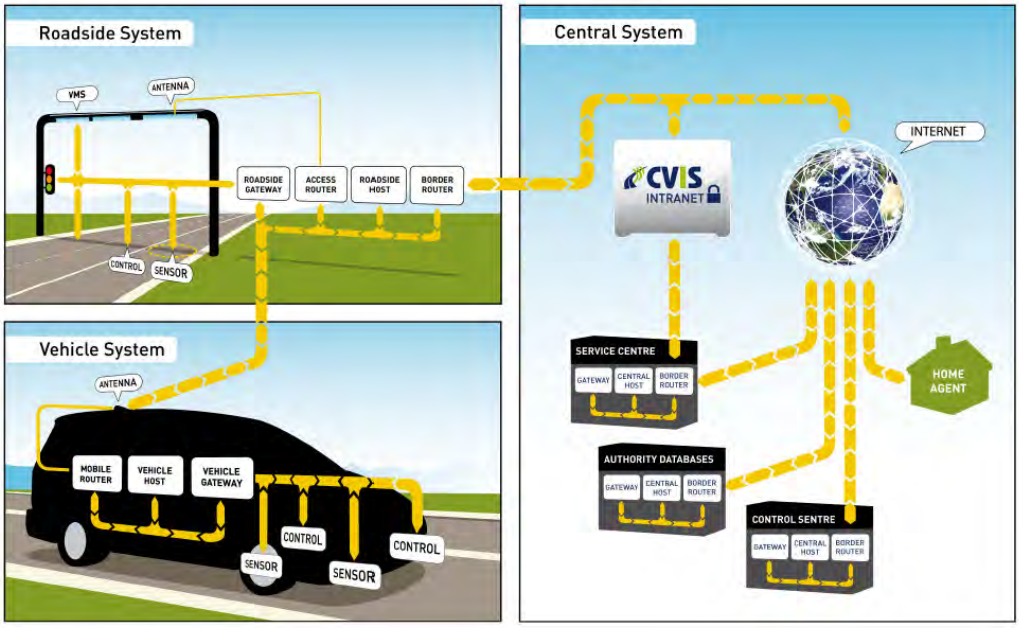


Figure 2.1 CVIS Reference Architecture

2.3.1.2 Road Infrastructure

Most of the initiatives propose the use of Road Side Units (RSUs) in order to connect the VANET to external information sources, or to increase connectivity of the network.

RSUs may be either stand-alone devices that communicate only with vehicles via wireless communication, or they may be also connected to a backbone network, which could use either a wired or wireless/mobile infrastructure (UMTS, LTE, WiMAX…), or even a mesh network of RSUs. The architecture proposed for RSUs by CVIS and eSafety Forum is shown in Figure 2.1. In this scheme, different equipments are dedicated to the communication with vehicles, with the Internet and a host computer that will run the applications; all of them interconnected.

There also some proposals [23] that promote the use of supporting units at road junctions installed at traffic lights, to optimize the efficiency and make road junctions safer. Other studies [24] analyze the performance of different kinds of supporting stations. They consider backbone-interconnected RSUs, a mesh network of RSUs, and stand-alone RSUs comparing the number of units necessary to keep the delay low. The main conclusion is that in order to achieve the same performance as with interconnected stations, the number of stand-alone RSUs must be 5 or 7 times more. In any case, the deployment and maintenance of RSUs with or without connected backbone have significantly high costs.

2.3.1.3 Central Infrastructure

Reference architectures also propose central nodes in the operator´s infrastructure to support communication management and services. Basically the functions from these central nodes are mostly related to connectivity, mobility management, data storage and security issues like Authentication, Authorization and Accounting (AAA).

There are some initiatives like Cooperative Cars project (CoCar) [25] that propose the use of a centralized infrastructure for Geocasting information through cellular channels (UMTS and LTE) using NGN capabilities but for the best of our knowledge there is not a platform that proposes an exhaustive use of IMS and NGN for the improvement of V2V routing and security. CoCar architecture only leverages on central elements and avoids the use of RSUs, its architecture is depicted in Figure 2.2

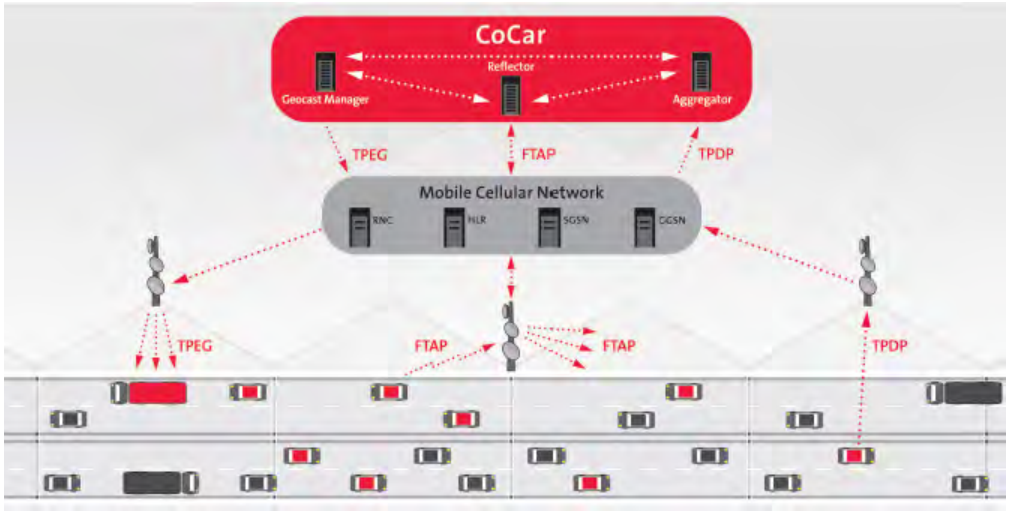


Figure 2.2 CoCar Architecture

2.3.2 Seamless use of multiple radio technologies

There is a need of connectivity with certain guarantees for many different services that demand very low delays, efficient routing algorithms, quality of service, bandwidth, availability and of course high levels of security.

All this requirements make a single radio technology solution unviable, thus all the initiatives worldwide working on this (standardization bodies, authorities, research projects, etc) are proposing reference architectures like CALM [22] that support several radio technologies, and the services will use them attending in real time conditions and communications requirements.

There is not a single technology that can fulfill all the communication requirements of the vehicle services. So, several radio technologies must be provided. Always Best Connected (ABC) solutions are emerging in order to provide the best communication channel available to the mobile nodes in these vehicular networks. We describe the radio technologies susceptible of being used in our system including WAVE (IEEE 802.11p) in section 2.5. ABC refers to being not only always connected, but also being connected in the best possible way, i.e., combining several communications technologies [26]. The definition of 'best' depends on a number of aspects such as personal preferences, limitations of the device (terminal), application requirements, security, policies, or network resources and coverage.

The ABC philosophy involves vertical handovers (VHO), that is, seamless switching between two network technologies maintaining the same connection (e.g., a TCP-connection). Generically, a VHO encompasses the search for a second communication network to switch to, the transmission of information about the terminal to this second network, as well as of temporary channel information of the second network to the terminal, and the control of a temporary channel connection between the terminal and the second network.

There are two main standardization initiatives in charge of specifying vertical handovers: the 802.21 Working Group for Media Independent Handover (MIH) and Interoperability of the Institute of Electrical and Electronics Engineers (IEEE) [27], which applies to generic mobile/wireless networks, and the CALM standards' suite [28], which is specific to the vehicular environment.

2.3.2.1 CALM

Through CALM, the International Organization for Standardization (ISO) is defining a standard framework for V2V and V2I communications. Handing off connections is a key function of CALM. The ISO 21210 standard [29] specifies the network protocols for media management and transfer within CALM.

The CALM Architecture is depicted in Figure 2.3

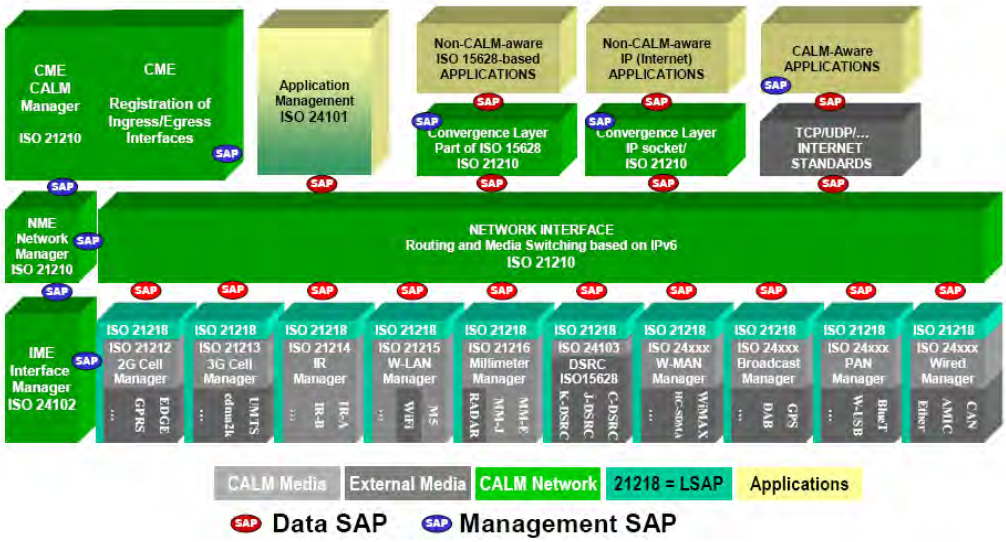


Figure 2.3 CALM Architecture

CALM Management Entity (CME) is responsible for the control of the system. It interacts with the Application Management to gather the application requirements

through management Service Access Points (SAPs). It has three different components; Interface Management Entity (IME), Network Manager Entity (NME) and CALM Manager. The functionalities of these entities are described below.

IME, specified by ISO 24102, interacts with adaptation levels that provide information on the status of the different communication channels, allowing direct control over the media. It is the responsible of coordinating different media avoiding interferences and blocks, and triggering horizontal handovers between two access points of the same network. IME also communicates the status of the different channels to the NME and the CALM Manager.

NME, specified by ISO 21210, gathers the requirements from the application and compares them with the status information of every communication channel available. It determines the election of the media taking into account some media selection policies. The media selection policies have a set of default configurations, but also can be configurable by the user. Typically, this policy will be based on selecting the media that fulfils service requirements at the lower cost. NME manages the process of vertical handovers between different media. If a link goes down, NME should change immediately to operate on other means available without losing information, maintaining the session and updating the routing table.

CALM Manager, specified by ISO 21210, manages the overall function of CALM, but also provides session management with the application. To manage the session it must send messages, information and requests to / from the application to the appropriate element. This entity continuously compares the list of properties of the media available with the list of application requirements, takes the decisions and sends commands to the NME to perform the action.

2.3.2.2 IEEE 802.21

The IEEE 802.21 Working Group is focused on standardizing an interface for interoperability and handover in heterogeneous networks, that is, MIH. MIH relies on an intermediary layer between layers 2 and 3 of the Open Systems Interconnection (OSI) stack, which is commonly referred to as ‘layer 2.5’. This layer is in charge of converting communication-system-specific commands from layer 2 into events used by the network selector to trigger a handover. This selector takes the handover requirements from the applications, considers the status of all the different available interfaces (media) and determines the best medium to support the communication. By itself, this layer neither controls the handover nor manages the mobility, but it provides a complete overview of the available media and their radio conditions.

There are three types of VHOs: those controlled by the terminal, those initiated by the terminal and assisted by the network, and those initiated and controlled by the network.

In the literature we may find several handover solutions based on MIH. For example, the vehicular communication system in [30] integrates the MIH function (MIHF) to provide ABC communications. The media considered in [30], which include IEEE 802.3, IEEE 802.11, HiperLan/2, IEEE 802.16, 2/2.5G and 3G, are set under MIH's 'layer 2.5'. The Mobility Management and Network Selection functions are built on top of the MIH layer in order to switch among the above-listed media according to the events received from the MIH. In [31], the authors implement the IEEE 802.21 framework to test the MIHF in different handover scenarios. Authors in [32] propose a simplified mechanism that operates at the network layer and achieves quasi-continuous communication during the handover process by monitoring the route to the destination node.

2.3.3 VANET routing performance

In VANET, vehicles can communicate wirelessly through multihop paths. This means that vehicles use other intermediate vehicles as relays to reach the final destination of data packets. The routing protocol has the responsibility of finding a path between the source and the destination.

A vehicle has a mobility pattern bounded by roads, traffic signs, the movement of other vehicles, weather conditions, traffic density, traffic jams, etc. Thus, also the distribution of nodes depends largely on the type of road on which they move. We can separate three types of scenarios:

- Rural. The speed of vehicles is slow and traffic density is quite low, therefore the network will appear disconnected with high probability. The most common issue in this type of scenario is network fragmentation described in [33].
- Urban. There is a relative number of vehicles on the road, which makes communication more feasible and, therefore, easier to find a path between origin and destination. In this environment, the speed can vary between low and high, depending on the characteristics of the track. In towns there are buildings or other obstacles to line of sight between vehicles that are not on the road.
- Highway. Vehicles traveling at high speed along the path marking the route, in which there are no intersections or traffic lights, so there is little traffic stops (except traffic jams). Here again, is quite common to find clusters of vehicles travelling at a similar speed so there is again great dependence to the existence of an infrastructure that supports communication.

In any of the three scenarios, vehicles tend to travel forming groups. Thus, routing protocols must solve the issues from having disconnected clusters of vehicles.

Due to the great number of vehicles which may participate on a VANET, routing protocols need to be localized to ensure scalability. Vehicles make routing decisions solely based on information locally available in their close vicinity. Therefore, exchanging information with neighboring vehicles via beacon messages is a fundamental part of routing protocols in the literature.

Usually, vehicles can obtain position information from GPS receivers. Hence, many protocol designers have employed geographic routing as the basis for VANET-specific solutions. By using greedy heuristics, the protocols choose as next hop the neighbor which provides greater advance towards the destination's position. These solutions assume that an end-to-end path between the source and the destination exists at the moment of issuing a data message. However, this is not the common case in VANET, since vehicular networks are formed by disconnected groups of vehicles.

In order to solve this problem, the Store, Carry and Forward (SCF) paradigm can be employed. It consists of storing the message when there is no neighbor which provides advance towards the destination. Only when a suitable neighbor is found, the forwarding takes place.

Some protocols are based on the planned trajectories of vehicles instead of on geographic routing. Vehicles forward the data message to those neighbors whose trajectory is more useful than the one followed by the current holder.

Authors in [34] detail the problems in current routing protocols and outline solutions to solve those problems. In Chapter 5 instead of proposing a new routing protocol, we define a mechanism based on our solution to enrich the information used by current protocols in order to overcome the problems and improve their efficiency.

2.3.4 Security

One of the mayor challenges in vehicular communications is security. Although research community has been intensively working in the last years, many open issues have not been properly covered. It is essential to make sure that life-critical information cannot be inserted or modified by an attacker but at the same time, it should protect the privacy of the drivers and passengers.

These concerns may look similar to those encountered in other kind of networks, but they are much harder to cover. The speed of the vehicles, the huge size of the network,

the relevance of their geographic position, the intermittence connectivity between them and the criticism of the services make the problem very complex and challenging.

In this section we describe the security the most relevant treats in vehicular networks. We provide a general classification of attacks substantiated by a list of attacks, initially identified by [10] and enriched with novel attacks recently discovered. But before describing the attacks we classify the attackers, defining four metrics. This classification was initially presented by [36] and is commonly accepted in the literature:

1. Membership. This dimension denotes if the attacker is an authenticated member of the network that can communicate with other members (insider) or not (outsider). The outsider is considered by the network members as an intruder, thus; the diversity of attacks he can perform is limited.
2. Motivation. A malicious attacker seeks no personal benefits, aims only to harm the functionality of the network. A rational attacker seeks personal profit and usually the attack target and methods are more predictable.
3. Method. An active attacker generates packets or modifies those from legitimate nodes. On the opposite a passive attacker contents himself with eavesdropping on the wireless channel.
4. Scope. An attacker can be limited in scope if the malicious node controls few entities, which makes him local. An extended attacker controls several entities that are scattered across the network, thus extending his scope.

Based on this, we characterize an attacker by Membership.Motivation.Method.Scope where Membership stands for Insider (I) or Outsider (O), Motivation for Malicious (M) or Rational (R), Method for Active (A) or Passive (P), and Scope for Local (L) or Extended (E). For example, I.R.P.E is an insider attacker that behaves rationally, and eavesdrops in extended areas. A star ("*") indicates that the corresponding field can take any value.

2.3.4.1 Common attacks

To illustrate the importance of an appropriate security solution for vehicular communications, we present in this section many of the known attacks including examples, providing an idea of the damage that can cause.

1. Fake information: Attackers are classified as I.R.A.$*$ and diffuse wrong information in the network to affect the behavior of other drivers. For example an attacker can disseminate a false traffic jam event in certain road making other vehicles vary their routes, artificially reducing the traffic on the roads he is using. This type of attack is relevant for road operators and drivers as they can

cause serious congestions in certain roads, while others are practically empty. A particular case of this kind of attacks are the ones that a malicious node disseminate wrong sensor data. Attackers in this case are I.R.A.L, and use this attack to alter their perceived position, speed or direction in order to escape liability in the case of an accident. In some cases, attackers can generate fake vehicles in different locations to generate an illusion, taking advantage of the traffic situation.

2. Privacy Violation. These attacks are I.R.P.E consists in gathering information from the vehicles and passengers. The major vulnerability comes from the periodic and frequent messages transmitted over the vehicular network; context-aware data, transaction-based communications or other control information. In those occasions, messages include by default some data (e.g., time, location, speed, direction, vehicle ID, route description, etc) that facilitates the identification of the originating vehicle as well as the behavior or preferences of the driver or passengers. An attacker compromising one or several nodes in different locations can retrieve critical information including trajectories monitoring of targeted vehicles and use this data for illegitimate purposes (i.e. to control the transactions and movements of relevant people or to track the floats of car rental companies without permission).

3. Denial of Service: The attacker is $*$.M.A.L and usually aims to bring down the network. Typically, these attacks consist of channel jamming if it is produced by an external node or aggressive injection of dummy messages if it is internal. As the network coverage area (e.g., along a highway) is usually well-defined, jamming is a low-effort exploit opportunity. An attacker can easily, without compromising cryptographic mechanisms and with limited transmission power, disable the network in certain area. In those cases where the attack is produced in a critical situation (an accident, congestion, adverse weather conditions, etc) it is extremely dangerous and can even cause (or increment the effects of) an accident.

4. Impersonation: Message generation, modification, and replay can also be used towards impersonation. The attackers are classified as I.*.A.L and actively pretend to be another vehicle by using false identities and can be motivated by malicious or rational objectives. An impersonator can be a serious threat; for example, an attacker masquerading as an emergency vehicle can tell other vehicles to slow down and yield; or an adversary impersonating roadside units can spoof safety messages.

There are many challenges related with security in the vehicular environment [10]. In Chapter 5 we will deeply analyze the problem of effectively neutralize attackers in VANET and how to overcome it using our solution.

2.4 IP Multimedia Subsystem

In this section we describe IMS, describing the potential and benefits of IMS as the base of our service platform. We point out the advantages of the use of IMS for service provisioning in vehicular environments.

IMS enables the communication between vehicles in real time, establishing multimedia sessions with QoS parameters according to the network conditions and the service characteristics. Users also benefit from new integrated QoS-guaranteed multimedia services, accessing independently from their geographical location. IMS is access agnostic, so vehicles can dynamically use the most appropriate technology to transmit data (i.e. WAVE, Wifi, GPRS, UMTS, LTE, WiMAX, etc). Authors in [38] analyze IMS scalability issues and the applicability with 3G networks

IMS provides security through reliable authentication and authorization of users. This feature is a critical and mandatory issue in vehicular services. It is based on well-known and widely used protocols as Session Initiation Protocol (SIP) [39], Session Description Protocol (SDP) [40], Real-time Transport Protocol (RTP) [41] and DIAMETER [42]. The identities are managed through Universal Resource Identifier (URI).

One of the most powerful advantages of the subsystem is the integration with advanced Next Generation Networks (NGN) enablers. Enablers provide centralized common functionalities to services, so they can reduce their logic and delegate critical operations to a trusted reliable entity.

These facts facilitate consistency and reliability in the creation of new services, promoting re-use, reducing development costs and improving usability. Thus, IMS becomes a very attractive framework for a strong growth in the service provisioning market in vehicular environments.

In the following subsections we present the architecture of the IMS Core describing the elements and those enablers with more potential for vehicular scenarios.

2.4.1 IMS Core

A Next Generation Network (NGN) is a network architecture designed to replace the switched telephone networks for voice telephony and multimedia. This architecture brings together in one all technologies to standardize on a single platform.

IMS is responsible of the control and access of common services for all applications in the NGN architectural model. It defines open interfaces for session management, access control, mobility management, service control and charging.

The IMS model architecture has been defined by 3GPP and although it was initially only linked to third generation mobile networks, it can be used in a wide range of networks, fixed, nomadic and cellular. Other bodies have also collaborated on the definition of the IMS model: Internet Engineering Task Force (IETF), International Telecommunication Union (ITU) and European Telecommunications Standards Institute (ETSI).

The NGN/IMS model is based on four fundamental aspects very promising for the success in the vehicular field:

- Access technology agnostic. IMS only requires IP connectivity and therefore inherently involves the convergence of different networks; fixed, nomadic and mobile.
- Full mobility support. IMS provides mobility support between different accesses maintaining the session when the clients perform handovers. These handovers are defined as horizontal when the initial access technology and the final are the same or vertical when technologies are different and IMS supports seamlessly both. It also features customer mobility between different terminals (PC, laptop, Smartphone, mobile, tablet...), also known as User Equipment (UE).
- Use of the Internet protocols (TCP, HTTP, SIP, DIAMETER...), integrating personal communications (mainly voice and messaging) with IT applications. This leads to more power and flexibility for the provision of multimedia services and the development of new applications.
- Use of standardized enablers for many common features required by many services optimizing the efficiency and simplicity of service development and deployment.

The architecture of NGN/IMS follows a layered approach in which we can find:

- Transport Layer: Provides the necessary connectivity converging different access networks. This layer is intended to disregard the mechanism used to transport data acting as the intersection point between the access layers and the IP network above it. It is responsible for doing initial IP provisioning as well as facilitating the registration of devices with the higher layers.

- Control Layer: Responsible for providing the intelligence; it controls the authentication, routing, and distribution of IMS traffic between the transport and the service layers, mostly based on SIP. This layer also provides the capability to interface other services (i.e.: with billing or QoS to ensure the appropriate delivery and handling of the purchased content).
- Service Layer: Responsible for providing services to final users. This includes traditional voice services as well as new applications built on the IMS architecture. In this layer, the access network used is transparent as it has been previously abstracted by network and control layers. This makes the IMS architecture a reality for digital convergence and an ideal environment for the deployment of digital services.

2.4.1.1Architecture

The IMS presents a functional modular architecture which elements are distributed into several functional layers. This type of architecture allows each of the levels to evolve independently, as the market and technology evolve. The architecture is depicted in Figure 2.4.

The heart of the subsystem is the IMS Core. The Core has been designed to enable operators to provide a wide range of real-time, packet-based services and to track their use allowing several types of charging. It also provides a framework for the deployment of both basic calling services and enhanced services. In this section we describe the basic entities needed for managing communications.

2.4.1.1.1 Proxy Call Session Control Function (P-CSCF)

Commonly known as "Proxy", it is the entry point to the IMS domain and serves as the outbound proxy server for clients. The basic philosophy of IMS is to empower the P-CSCF to make the critical decisions. Specifically, it will either route the messages into the system or deny access to the IMS. Its main features are:

- Route messages, including logs, accounting information and signaling to other entities of the Core.
- Provide control information for all the sessions of registered users.
- Provide integrity and security of signaling with UE.
- This element can execute media policing, enabling QoS for different sessions. This can be done examining the content of the SDP payload.

2.4.1.1.2 Interrogating Call Session Control Function (I-CSCF)

It acts as an inbound SIP proxy server. During IMS registrations, It uses the SIP protocol to communicate with the elements: P-CSCF, S-CSCF and MGCF and RADIUS or DIAMETER protocols to communicate with the HSS. Its main features are:

- Interrogate the home subscriber server (HSS) to determine and assign the S-CSCF to every user.
- Forward SIP requests to S-CSCF
- Send account information to CCF

2.4.1.1.3 Serving Call Session Control Function (S-CSCF)

It is the brain of the subsystem being responsible for session control and registration services for the UE. It also maintains the session state and interacts with the service platforms and charging functions as required by the operator. The S-CSCF marries the capabilities of the UE with the services of the application servers. Its main features are:

- Perform session control storing and accessing registration information on the HSS. This information is also required when handling requests from unregistered users.
- Interact with service platforms; the S-CSCF determines if a request or response needs to be sent to an application server for further processing.
- Handle registration requests: This element acts as the registrar for registration requests from the UE. To facilitate this process, the S-CSCF knows the IP of the UE and which P-CSCF the UE is using as the entry point.
- Authenticate users: The IMS Authentication and Key Agreement (AKA) procedures allows for authentication of both the UE and the home network.
- The S-CSCF routes mobile-terminating traffic to the P-CSCF. It also routes mobile-originated traffic to the I-CSCF, the BGCF or the Application Server (AS).

2.4.1.1.4 Multimedia Resource Function (MRF)

It has two elements; the Multimedia Resource Function Controller (MRFC) and the Multimedia Resource Function Processor (MRFP). The MRFC acts as an interpreter in the user plane while the MRFP performs the operations.

- MRFP uses Media Gateway Control Protocol (MEGACO) to communicate with the MRFC. Responsible for providing media resources for services such as

RTP flows Mix (conference), RTP flow injection (phrases), transcoding (conversion of voice codecs), voice recognition.

- MRFC behaves as a SIP "user agent", use the SIP protocol. Using the S-CSCF controls the use of media resources available in the MRFP.

2.4.1.1.5 Breakout Gateway Control Function (BGCF)

BGCF can be defined as a SIP message router, which performs the control of several MGCFs. Their existence in a NGN network is conditional upon the number of MGCFs deployed within the same, since their main task is to facilitate communications between MGCFs and the S-CSCF uses SIP and DIAMETER protocol to communicate.

2.4.1.1.6 Media Gateway Control Function (MGCF)

Analyze and interpret bundled SS7 signaling (SIGTRAN) from the T-SGW and behaves as an "endpoint" for SIP signaling.

Submissions to the Public Switched Telephone Network (PSTN), is responsible for selecting and controlling MGWs suitable for carrying out the mapping between Time-division multiplexing (TDM) voice circuits and RTP streams.

In communications from the PSTN, including selecting the appropriate MGWs to carry out the mapping between TDM voice circuits and RTP flows, is responsible for selecting an appropriate S-CSCF to route the call signaling.

2.4.1.1.7 Media Gateway (MGW)

MGW is responsible for the interconnection between circuit switched networks and packet switched networks. The MGW has the necessary resources to carry out a correct format conversion between data circuit-switched networks (TDM) format and data packet switched networks (IP networks).

2.4.1.1.8 Home Subscriber Server (HSS)

Is the element where the database resides on users of the system stores information related to:

- User Profile: ID, IP address, access permissions, available services, call filter criteria, etc.
- Location of the user.

- User status.

2.4.1.1.9 Application Server (AS)

Based on SIP, it is responsible for providing advanced services within the IMS architecture interacting directly with the S-CSCF.

2.4.1.1.10 Signaling Gateway Transport (T-SGW)

It is responsible for the transport of signaling number 7 on IP packets, using SIGTRAN, M3UA more specifically.

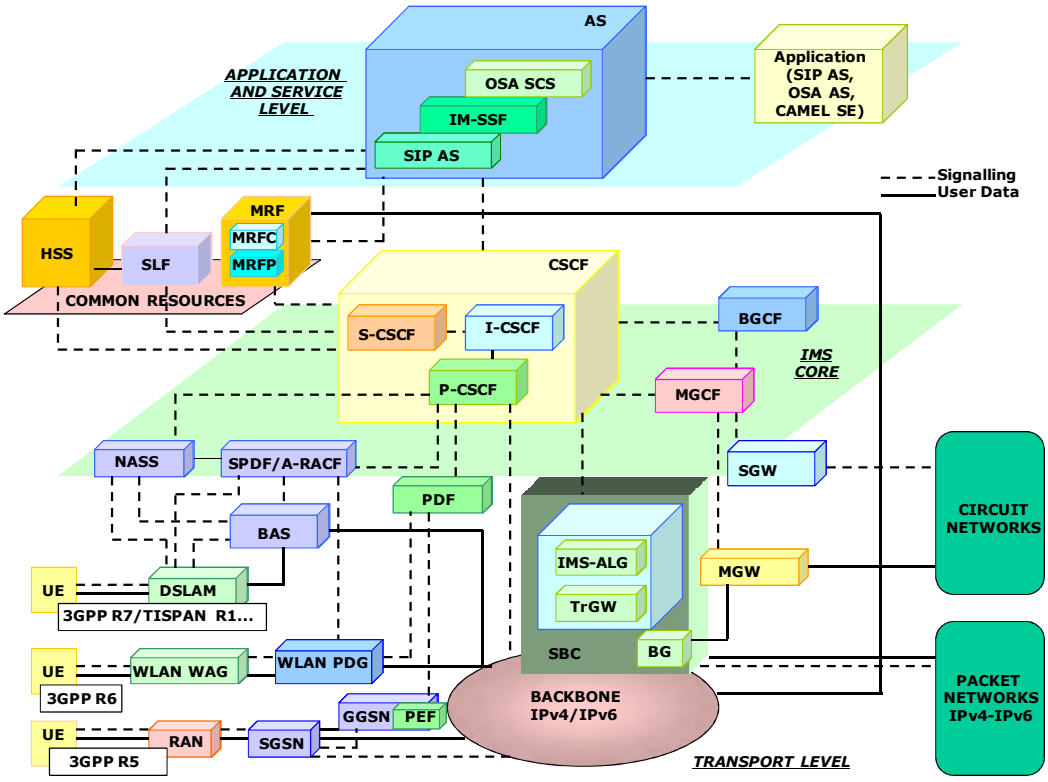


Figure 2.4 IMS Architecture

2.4.2 Enablers

The network infrastructure has many features that help enrich the end-user services without complicated logic. These capabilities are exposed to the services through the interfaces of certain platforms, commonly known as enablers. Capabilities can provide

the service the ability to delegate tasks such as authentication, to verify the user's location or presence status (available, unavailable, etc.) as well as customized information through standardized interfaces and protocols (HTTP, SOAP, SIP, etc). In this way, the service is focusing its efforts on developing only those functions that are intrinsic, making requests to the capabilities when they require any of its functions (i.e.: requesting the multimedia features of the equipment of the vehicle).

Although is the IMS Core and its elements the heart of the system that allows the basic function of our solution, the enablers are the ones that provides the flexibility and power of VISIONS.

There are a huge number of standardized enablers able to operate over the IMS subsystem. Our approach is not only use this enablers as features for enriching the ITS services logic but also to solve many open issues in vehicular communications.

As the total amount of enablers is enormous, in this section we cover those that we identify with more value for our purpose that are available currently. The criterion for choosing them was the advantages in functionality provisioning, in both vehicular communications and services. In some scenarios, the participation of network infrastructure and its capabilities is crucial, while in others, not being essential, it can bring benefits to both business plans for service providers and final users.

The capabilities that have been selected belong all to the standardized NGN reference architectures OSA/Parlay and Parlay X, OSE and Liberty Alliance.

As the community is continuously designing and standardizing more capacities for IMS and NGN systems, we foresee a promising future in this field where they would appear several enablers specifically designed for the vehicular environment.

For every enabler we present an overview, its basic characteristics and, the standardization bodies in charge.

2.4.2.1 User Profile

The main functionality of this enabler is to provide user profile information in terms of static and/or dynamic attributes. This information is available on request or subscribing to a service that monitors profile changes.

The user profile attributes can be stored in containers such as name, address, status, nickname, birth date, contact details, preferred language etc, and may be defined and developed in accordance with the needs of the service or applications that will use it.

On these attributes, the interfaces defined allow queries and modifications. It is not necessary to know the real identity of the user to access them, just using the

pseudonym provided by the Identity Provider enabler (also described in this section). Since these capabilities are integrated within the Identity Web Service Framework (ID-WSF), the protocol used is SOAP.

The functionality of the User Profile (UP) capabilities, specifications and features are defined by Liberty Alliance. There is also a proposal for the definition of the 3GPP Generic User Profile (GUP).

2.4.2.2 Presence

The basic functionality is to provide presence information or availability of the use to network elements, applications or other users. Network elements will be able to publish presence information for users, for example if a user is logged on the network or not. Moreover, the network elements and/or other services can be themselves consumers of presence information.

Users publish their presence information by the utilization of various predefined statements such as: Available, Busy, Out To Lunch, Out of range etc. But these states can be customized for a more targeted application (i.e.: verified, not verified, activated, not activated, emergency or the like). Each state's presence should be associated with a certain period of validity, to ensure that the information remains fresh, not obsolete.

The functionalities of the Presence capacity and the requirements, specifications and features are defined by OMA (Open Mobile Alliance), and by Liberty Alliance, Parlay/3GPP-OSA and 3GPP (IMS).

2.4.2.3 Terminal Capabilities

The basic functionality of Profile Terminal Capabilities (TC) enabler is to allow the collection of the characteristics of terminals or devices used to access the service.

The application must provide the identity of the terminal as input and the result should include the terminal features. There are different attributes for which the application can interrogate like multimedia capabilities, software version, make, model, etc., attributes that may be expandable if necessary

Additionally, a service can subscribe to a feature that will notify when any change on the attributes of a terminal occurs. Of course, this enabler is reliable on privacy aspects according to user preferences. It is therefore necessary to implement certain interfaces to allow:

- Collection of the terminal attributes providing the terminal ID.

- Monitoring, receiving reports due to changes in the terminal capabilities or when the enabler has no information on the capabilities of the device.

TC is defined within the OSA/Parlay, Parlay X and 3GPP bodies [43].

2.4.2.4 Identity Provider Capacity

The identity is defined as a set of attributes relating to an individual or group. Some of these attributes can be identifiers (either pseudonyms or not), others can be static characteristics (ID, name, age) or dynamic (location, presence status). The definition of the attributes that compose the identity of a certain user is based on an appropriate data model, adapted to the needs of the services that make use of them.

In real scenarios where a wide offer of services, not every service should have access to all the attributes of a user's identity either because they are not necessary or because they should not have permission because of privacy policies.

There are some services that only need to identify the user by gathering an identifier without any verification mechanism. These are services where security requirements are minimal, for example the control of the air conditioning is only based on the driver´s identity. However, others require the verification of user identity; in these cases it is necessary to employ authentication mechanisms which guarantee that the user is authentic, this means that the attributes presented identified him/her unequivocally.

The most relevant standardized capacity for Identity Management is the Identity Provider (IdP), a provider of identity defined by Liberty Alliance Project. IdP provides services after authenticating the user; using a pseudonym as their identity it also protects privacy. The pseudonym is created by the IdP in the federation process in order to link user accounts and service provider identity. In addition to the pseudonym, the IdP will provide the service attributes of identity that the user is authorized to access. To deliver the service attributes, the IdP must have access to platforms containing (eg user profile, terminal profile, etc) in case of not having stored therein.

Within the Liberty Alliance architecture, the attributes can be obtained directly by other enablers (i.e.: User Profile, Location, Messaging, Broadcasting) without knowing any real identities, only using the pseudonyms created by the IdP.

2.4.2.5 Location

The Location capabilities provide the position of terminals to the external services. They can locate a fixed or mobile device in different formats and with different qualities.

It generally relies on signals measurements from infrastructure including satellites, access points, cellular antennas, etc.

As this enabler is extremely relevant for the vehicular environment as almost all the services need location information we will analyze many aspects in detail. Our purpose is to provide a complete view of this enabler, including the technologies involved and some arguments to decide which the best option for our solution is.

Location information contains the geographical position of a terminal and responds to the following parameters:

- Accuracy: The accuracy of the position offered may range from exact geographical coordinates of the terminal to just the country or a specific region.
- Reliability: The reliability of the information depends mostly on the technology used and the area in which the terminal moves. Any position is associated with an uncertainty which may be large in some cases or very low in others (GPS).
- Coverage: The area in which terminals can be located.
- Time: It is the time that takes obtaining the target´s position.

Given the requirements for these parameters in a certain service, we will be able to effectively select the location technologies and design the enabler.

The location is generally associated with the access network and therefore depending on tracking mechanisms. In general there are two ways to classify location capabilities:

- Attending the communication channel
- Attending the location technique

When classifying with the communication channel, we can distinguish how the location information is obtained by the network to communicate to a third party. There are two ways:

- Data channel. The measures or data between the server and the terminal travel over an IP-based user channel, making this type of location agnostic to the access network.
- Control channel. The measures or data necessary to carry out the location are exchanged by the signaling channel of the access network, making it more transparent and cost effective to the customer as it has no impact on bandwidth.

According to the location techniques we can find techniques based on:

- The access network. The Cell ID that controls the terminal is the one that provides the location. The accuracy is very dependent on details of the topology of the network (i.e. in urban areas up to 150m). Enriching this mechanism with power measures of the serving and neighboring cells increases the accuracy, allowing triangulation of the position of the terminal.
- Time: the identification of the cell is joined by measurements of time delay between the antenna and the terminal and adjacent antennas times.
- IP or access point: The service gets the position of correlating the IP terminal or access point in the same network with data that have network access.
- Satellite systems. The terminal antenna picks up the signal emitted by satellites (3 required to obtain the position and a fourth for the altitude) and computes its own coordinates. It has a very high accuracy but has the drawback of not having indoor coverage. Depending on what the terminal does with the data, there are two types of positioning.
    a. Calculated in the network: the data are processed by the location server which is the one who gets the final position. Save battery and processing power in the terminal.
    b. Calculated in the terminal: the final position is obtained in the terminal and transmitted to the server with the subsequent needs for battery and processing.

The terminal location system must be able to provide different functionality to services depending on the access network and the capabilities of the terminals:

- Request for immediate localization: the location is only provided on demand by the service.
- Subscription-based location information event: the location system provides the position of the terminal only when the event occurs as specified by the service, this can be triggered by time or geographic conditions.
- Management: the service can register, change preferences or unsubscribe from active subscriptions.
- Location reports. The system can be configured to periodically obtain certain reports of the location of the terminals.

There are several standardization forums involved in the development of location technologies. We just mention the most important.

OMA (Open Mobile Alliance) is working on different fields. Mobile Location Services (MLS): covering two main protocols: Mobile Location Protocol (MLP) and Roaming Location Protocol (RLP); enabling roaming services between location servers. Secure User Plane Location (SUPL) includes features such as triggered services, which provide high functionality at the location. LOCSIP provides location over SIP protocol. Global Location (GL) approach offers a new management layer making the location independent from the complexity of location techniques.

3GPP (3rd Generation Partnership Project) has focused the work on the standardization of location technologies GSM, 3G over signaling channel and is currently standardizing the location for 4G terminals. Stand specifications:

Parlay Group is more focused on defining Web Services to access network capabilities. Open Service Access (OSA) APIs define the interface for accessing location-based services using protocols like CORBA and SOAP.

In further sections, we describe how our Location enabler is designed to optimize the fulfillment of vehicular requirements.

2.4.2.6 Instant Messaging capability

Instant Messaging (IM) capabilities provide the framework and mechanisms to facilitate the exchange of messages in almost real time between users and services. These messages can be sent in as one-on-one or one-to-many. There are two different modes, Pager and session:

- Pager mode is based on the MESSAGE method defined as an extension to SIP for IM. In this mode instant messages are exchanged independently and media broadcast over the SIP signaling channel. This mode is more suitable for scenarios with few participants and small amount of messages. The problem with this kind of IM is the congestion of the nodes in the IMS Core as they share the same resources for signaling and data transfer.
- The Session mode is based on the MSRP protocol [44]. In this mode, messages are exchanged in the context of a SIP dialog and transmitted through different channels. The session mode is suitable for scenarios with large numbers of participants and messages with large contents of different types. Moreover, in session mode IM capacity can be easily integrated with other infrastructure capabilities (i.e. Presence).

Specific functional entities of IM capabilities are explained below:

- IM Client is the entity that accesses the IM capability, it can reside on a mobile device, an M2M device connected to the network, an AS, etc. The IM Client performs the registration with the Core, enabling start/ participation/ completion of the IM sessions. It allows sending and receiving instant messages, notifying the user of incoming messages or invitations and even transmiting commands to other enablers.
- IM Server. Is responsible for sending and receiving instant messages from users in any of the modes. For session mode IM is an MSRP switch and acts as a conferences coordinator, maintaining chat rooms and private messaging. Also performs other functions such as system messaging, storage of conversations, chat history management, etc. It can also act as source or Presence enabler to trigger status changes.
- IM XDMS. XDM is a server that manages XML documents specific for IM functionalities.

The set of specifications known as IETF SIP for Instant Messaging and Presence Leveraging Extensions (SIMPLE) form the basis of the ability to IM. 3GPP is based on the work of IETF to define the framework of IM in IMS 3GPP defining an interoperable solution that takes into account the specific characteristics of the mobile network and the needs of network operators billing, QoS, security, etc. It also defines the access of applications through OSA / Parlay X. OSA provides applications with an interface to access IM features, through a set of APIs defined in Multimedia Messaging (MM).

2.4.2.7 Multicast/ Broadcast Enablers

The ability to Multicast/ Broadcast allows applications to offer services for advanced content distribution, as flows (streaming) or delivery of multimedia files delivery to a large number of users.

There are several standardization fora involved in the development of multi-diffusion technologies advanced content. Here are mentioned the most important:

Parlay X/3GPP-OSA defines two capabilities:

- Multimedia Multicast Session Management that defines a SOAP interface to provide session management.
- Multimedia Streaming Control which defines an interface to provide control of multimedia streams, including the ability to change terminal and adapt network changes (QoS) during a session already in progress.

OMA has standardized the "OMA Mobile Broadcast Service". This capability allows the following functions: Service discovery, provision, protection, content distribution including real-time flows, files and reports, interaction with other enablers and pricing.

The multimedia multicast capability defined by Parlay X/3GPP-OSA defined SOAP interfaces allowing applications to control a multicast session, its members and the flow of multimedia content and presence information to potential participants. These interfaces are:

- Session management and user participation in the sessions. Features include: Creation and deletion of a multicast session, obtaining information associated with a session, inviting users to join a session or to abandon it. It also provides information on participants in a session.
- Notifications/ Events: Provides information to the application on changes in presence, for example, if a user has joined or left a session.
- Management of Notifications/ Events: Allows the application to subscribe to receive notifications about the presence of individuals or multicast groups.

The Flow Control Capability is also defined by Parlay and provides a SOAP interface for content providers to control the flow available to users. X/3GPP-OSA Parlay defines the following interfaces for that capacity:

- Flow Control, which allows the application to control the transmission parameters and methods of a session (pause, log off, etc) and information about the status.
- Flow notification management, which allows the application to receive notifications about new content applications.
- Flow Notifications, which allows through a subscription to send notifications to the application including: errors and changes occurring in the establishing of a session such as; terminal changes, changes in the capabilities or in the network characteristics. These notifications allow the application to adapt to new conditions allowing transmission to continue with the session has already begun or to ensure the delivery of content with a certain quality.

This capability defines a series of logical entities and connections between them, providing the following features:

- Guide and Discovery: Provides mechanisms that allow users to discover and access services available in its broadcasting region.
- Notification: Provides mechanisms for the user to specify the preferences when receiving broadcast content, and notes informing the media content of interest.

- Interaction: Provides point to point communication between the application and the broadcaster or terminal.
- Protection of services and content, allowing several ways to support digital rights and accounting schemes.
- Provision to manage user subscriptions.
- Terminal Provision, which manages the settings of the terminals.
- Distribution of flows (streaming) and files to the terminals using encryption mechanisms, which is usually done on the broadcast channel, but can also be performed on the interaction channel.

As, we have seen there are many standardization fora working on this area [45] but we remark the Multimedia Broadcast Multicast Service (MBMS) [46] standardized by 3GPP.

MBMS is a broadcasting service that can be offered via existing GSM and 3G cellular networks. MBMS is flexible and cost effective. By flexible we mean that it can be configured to use only a portion of a cell carrier, leaving the rest for other services. It is cost effective because radio transmission is independent of the number of subscribers in the cell. It only introduces minor changes to the existing network and terminals, reducing implementation costs and making mobile broadcast a relatively inexpensive technology.

MBMS enables operators to define broadcast and multicast services for geographical areas at fine granularity, down to the size of radio cells. The file downloading service in MBMS consists of three phases: service advertisement, file transmission, and post-delivery repair in order to ensure correct delivery to all subscribers. MBMS adapts the IP multicast model for cellular networks. Data delivery is done based on IP multicast addresses, which are associated with the different flows of a service. Terminals subscribe to the IP multicast addresses of those services they are interested in using the Internet Group Management Protocol (IGMP).

2.5 Radio technologies in vehicular communications

In this section we present a brief summary of the wireless and mobile technologies susceptible of being used in vehicular communications, describing their main features and suitability for this scenario. We do not include Short Range Communications (SRC) that would allow Intra-vehicular connectivity as it is out of the scope of our work.

2.5.1 WiFi

Based on the IEEE 802.11 standard operates in the non-licensed bands. The b and g versions have been greatly extended to the point that most nomadic devices (laptops, tablets, smartphones…) incorporate the technology. WiFi provides coverage of hundreds of meters and a bandwidth of up to 54 Mbps. 802.11n significantly increase coverage and reach rates of up to 500 Mbps. Although initially had serious security problems and did not have QoS support, identified the 802.11i standard is 802.11e responsible for these aspects. Some studies [47], [48] analyze the performance of WiFi technologies with nodes in movement at high speeds.

2.5.2 WiMAX

Based on the IEEE 802.16 standard can operate in bands with or without license. Provides transfer rates of up to 75Mbps with ranges coverage up to 50Km. The 802.16e variant supports mobility. The appearance security is well settled by controlling access based on 802.1X, EAP and AES encryption (Advanced Encryption Standard). It supports QoS natively. It was designed to support high speed of nodes but does not work properly in Ad-hoc mode.

2.5.3 UMTS

Universal Mobile Telecommunications System (UMTS) is one of the third-generation (3G) mobile telecommunications technologies offered by network operators. The first deployment of the UMTS is the release99 (R99) architecture specified by 3GPP and is part of the global ITU IMT-2000 standard.

UMTS, using 3GPP, currently provides in deployed networks a transfer rate of up to 384 Kbit/s for R99 handsets, and 7.2 Mbit/s for HSDPA handsets in the downlink connection. It operates in licensed bands, and the coverage is quite important, especially in urban environments and roads. It was designed to work at the speeds reached by vehicles but the performance reduces as the speed increases [49].

2.5.4 LTE

3GPP Long Term Evolution (LTE), is the latest standard in the world of mobile network technologies. This radio technology was designed to increase the capacity and speed of mobile data networks. LTE is a set of enhancements to the UMTS System (UMTS) which was introduced in 3GPP Release 8 focusing on the adoption of 4G technology, including an all-IP flat networking architecture.

The LTE specification provides downlink peak rates of at least 100 Mbps, an uplink of at least 50 Mbps and RAN round-trip times of less than 10 ms. LTE supports both frequency division duplexing (FDD) and time division duplexing (TDD). It is also endorsed by public safety agencies in the US as the preferred technology for the new 700 MHz public-safety radio band.

2.5.5 WAVE. IEEE 802.11p

Wireless Access for the Vehicular Environment (WAVE) is standardized by IEEE 802.11p group. This technology has been specially designed for Ad-hoc communication between vehicles and for supporting ITS applications.

802.11p is the groundwork used for Dedicated Short Range Communications (DSRC), within a U.S. Department of Transportation project [50] based on CALM. The objective is to deploy a nationwide network that enables communications between vehicles and RSUs. In Europe, 802.11p is used in the ITS G5 standard by ETSI ITS, supporting the geo-networking protocol for V2V and V2R communication.

WAVE is an evolution of the IEEE802.11a standard physical layer with modifications and MAC to improve their performance in the vehicular environment. As IEEE802.11a, WAVE uses OFDM, but with transmission rates of 3, 4.5, 6, 9, 12, 18, 24, and 27 Mbps in 10 MHz channels using 52 sub-carriers modulated using BPSK, QPSK, 16-QAM or 64-QAM and coding of ratios 1 / 2, 2 / 3 or 3 / 4.

As for the pipeline, the standard defines 7 non-overlapping channels in the 10MHz the 5.9GHz band: 6 service channels (SCH) and a control (CCH). The CCH is used as a reference channel for an initial detection nearby vehicles as a prelude to the establishment of communications. At same time, this channel is used to advertise the services available in the SCH channels (Internet access, content downloading, etc.). Given that CCH channel is the reference channel, WAVE uses it to transmit mode broadcast road safety messages. Given the importance of the messages transmitted and the need to ensure proper reception, the channel transmits CCH with a data rate of 6Mbps, corresponding to a QPSK modulation with a coding ratio 1 / 2.

In the MAC layer, WAVE also uses the IEEE802.11 access method based in CSMA / CA (Carrier Sense Multiple Access with Collision Avoidance). To mitigate the hidden terminal problem, WAVE keeps the exchange of messages RTS / CTS (Request-To-Send/Clear-To-Send), although it is disabled on the CCH channel to transmit always in broadcast mode. As a result, all nodes using the control channel are subject to this problem, increasing the risk of packet loss and channel congestion.

2.6 Conclusions

In this chapter we have presented the most relevant initiatives worldwide that propose vehicular platforms. We have considered reference research projects in the EU and also national initiatives from US and Japan to deploy ITS solutions in real environments. Regarding standardization activities, we remark the architecture of CALM, proposed by ISO and followed by the majority of the proposals.

We analyze four of the most important topics of vehicular platforms; architectures, the use of multiple radio technologies, routing performance and security issues. Within the architectural design there is a discussion regarding the on board equipment; if it has to be distributed in several units or only one. The use of RSUs in the road infrastructure is also a critical issue as the deployment and maintenance costs are very high. For the support of multiple radio technologies and vertical handovers, we analyze the most proposed alternatives CALM and IEEE 802.21. After analyzing the current trends in routing protocols for VANET we justify our decision of bringing a new mechanism that enriches the information that protocols use for taking local decisions. We analyze the possible attacks attending membership, motivation, method and scope before presenting the most common attacks in VANET.

We present IMS, describing the potential and benefits of the subsystem and NGN enablers as the base of our vehicular solution. We point out the advantages of the use of IMS for service provisioning in vehicular environments. Those enablers identified with high potential for our purposes are: User Profile, Terminal Capabilities, Identity Provider, Location, Instant Messaging and Multicast/Broadcast Enablers. There is also a brief summary of the wireless and mobile technologies susceptible of being used in vehicular communications.

The main conclusion of this chapter is that a new vehicular solution based on IMS, NGN enablers, standard technologies like CALM and aligned with reference architectures can overcome many current problems in vehicular communications and services. In the next chapter, we justify our decisions and provide a detailed design of the solution.

3 SOLUTION DESIGN

3.1 Introduction

In the previous chapter we provided a complete analysis of the Status of the Art in vehicular communications and services. We have sated the complexity of the environment and the difficulty of proving good and reliable solutions to some of the existing challenges.

For bringing specific and tangible contributions to this area, we present in this chapter the design of our solution “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services” (VISIONS).

VISIONS is a vehicular system with two main objectives. The first one is improving the intrinsic challenges in vehicular networks such as routing and security issues. The second one is being an efficient and reliable platform for deploying services in the vehicular environment. These services will be typically related with Safety, Security, traffic efficiency and Infotainment.

VISIONS includes elements on board the vehicles, an IMS CORE, some standard capabilities and some enablers specifically designed for our purposes. It is also based in the standardized CALM architecture, so other services developed by 3rd parties can

be integrated and deployed over VISIONS system. The solution is radio technology agnostic, but we provide some examples of the current technologies in order to demonstrate the feasibility of the solution.

Taking into account the objectives of this thesis, in section 2 we describe the requirements of the system. In section 3 we justify the rationales behind the main design decisions taken in order to meet the requirements. In section 4 we present the architecture of the solution, detailing the features of the modules present on board vehicles and in the infrastructure. In section 5 we formally define the key process in VISIONS which supports the improvement of VANET challenges regarding routing and security. In section 6 we summary the conclusions of the Chapter.

3.2 Requirements

Due to the specific characteristics of the vehicular environment, a feasible solution that improves communication challenges and provides a service deployment platform is not easy to achieve. The requirements for these two categories are analyzed separately.

3.2.1 Requirements for improving Vehicular Communications

As established in the objectives of the thesis, the aim is to improve the state of the art in the performance two specific communication challenges; VANET routing and Certificate Revocation Lists (CRL) distribution for security purposes.

3.2.1.1 Routing

Typically, a VANET can include a high number of vehicles communicating simultaneously. Thus, routing mechanisms need to take decisions locally to ensure scalability. The more accurate these decisions are the better will perform the routing protocol. Therefore, exchanging information with neighboring vehicles via beacon messages is a fundamental part of routing protocols in the literature. There are some protocols that use not only position of the vehicles but also digital maps of the area and trajectories [51], [52], [53].

Our purpose is not to define a new routing protocol but to provide an efficient mechanism that enriches the information for taking routing decisions used by existing protocols. Thus, the requirements for this mechanism can be quantified by the improvement of the following indicators:

- Increase Packet Delivery Ratio (PDR). This means that more information will arrive to the destination, reducing packet losses.
- Reduce the delivery delay.
- Reduce resource consumption. The mechanism must be efficient in terms of resources required to distribute the information.

3.2.1.2 Security

Security is a critical aspect of vehicular communication. More specifically, the distribution of CRLs is one of the areas in which there is still margin to improve. A window of vulnerability (WoV) is the time frame within the vehicle defensive measures can be reduced or compromised. For our case, we define WoV as the time between the system identifies an attacker and the legitimate nodes get notified of this situation after receiving and processing a new CRL.

Our objective is to provide a centralized process that, without the need of RSUs, can distribute efficiently revocation information. For this purpose we define the following requirements:

- The distribution of the information must be fast to reduce the WoV of nodes.
- The distribution must be effective, which means that a high percentage of the nodes must be notified.
- The mechanism must be efficient in terms of resources required to distribute the CRL.
- The processing time of checking if a certificate is in the CRL must be fast and computationally light.

3.2.2 Requirements for the Service Deployment Platform

Digital services in the vehicular environment have completely different requirements from regular ones like home or office. We describe the main challenges in those categories crucial for the design of our vehicular SDP.

3.2.2.1 Safety

Priority number one on the road is safety. Our solution must have all the mechanisms to provide services avoiding any possible distraction for the drivers. Human Machine Interface (HMI) technologies play a fundamental role. Solutions based on speech

technologies, Head-Up displays and integration with the controls of the car are the most proposed in the literature. HMIs must have control mechanisms to prioritize some services (i.e.: Safety related) or deactivating others (i.e.: Infotainment) under certain circumstances.

Related with this topic, the regulation in many countries is very restrictive, forbidding sometimes any manipulation of an external device in the car. Moreover, the regulation is very dependent on the geography (US, Europe, Japan…) and maybe even different in states form the same country making difficult the deployment of global services.

On the other hand, the on-board equipment and the services, especially those connected to the CAN bus, must ensure the correct performance to avoid any potential interference with the systems of the vehicle.

3.2.2.2 Security

Due to the critical importance of some services related to safety, the system must be secure and reliable. Malicious nodes sending wrong information can cause relevant problems in the roads; like traffic jam or even accidents. A security system for vehicular communications should satisfy the following requirements:

1. Authentication and Authorization: These mechanisms are mandatory to ensure that only trusted entities access the system and can send or receive information. Vehicle reactions to critical events should be based on legitimate messages (generated by legitimate senders). Therefore we need to authenticate the senders of these messages. Not every user of the system will be granted the same privileges. For example emergency services may have permissions to activate green corridors changing the behaviour of traffic lights. Thus, policy mechanisms regarding authorization are needed.
2. Non-repudiation: Those nodes sending false information should be reliably identified; a sender should not be able to deny the transmission of a message. This may be crucial for investigation to determine the author and content of messages exchanged before an accident.
3. Verification of data consistency: The legitimacy of messages also encompasses their consistency. Even in those cases that the sender has been legitimated, the message could contain false or wrong data due to a malicious node that manipulates the content of the message.

4. Availability: Even assuming a robust communication channel, some attacks (i.e.: DoS by jamming) can bring down the network. Therefore, availability should be also supported by alternative means.
5. Privacy: Location and routes of the vehicles are sensitive information that users won´t accept to disclose. The concerns about privacy is increasing these days, and a system that doesn´t guarantee it will be rejected by end users. Moreover, authorities are regulating this at regional and local levels. For example, the Data Protection Directive 95/46/EC [54] regulates the processing of personal data within the European Union and pays special attention to the location based information. The privacy of digital data from vehicular services against unauthorized observers should be guaranteed.
6. Real-time constraints: The complexity of all the described requirements increases taking into account that some services require real-time exchange of information. Not only because of the safety related services, but also because nodes move at very high speeds and only have very short intervals to communicate with each other through the VANET. Thus, strict time constraints mechanisms must be applied.

3.2.2.3 Mobility support

Vehicles are by definition moving elements, thus the only solution is to communicate them through wireless technologies. However, there is not a technology able to fulfill the challenging requirements of vehicular scenarios and services. The solution must include several communication technologies including short, medium and large range. Moreover the system must be able to maintain communications switching between different technologies, so the service is not interrupted.

CALM [28] and IEEE 802.21 [27] initiatives, described in Chapter 2, have been designed and standardized to cover this specific goal.

3.2.2.4 Reliability and Feasibility

Automotive industry can only rely in mature technologies. Original Equipment Manufacturers (OEMs) and aftermarket providers won´t accept marketing digital services that may cause problems in the vehicles. Any problem will lead to high costs to solve it and, even more important, an impact on the credibility of the brand. Thus, the solution must be based on well-proven technologies that ensure a great performance.

Feasibility is from the beginning a socio-economic requirement for this work. The objective is to design a solution that can be implemented and commercialized in the midterm. Thus, technologies, on board elements and network infrastructure must be realistic, reliable, and cost effective. This is hard to achieve due to the fact that there are many stakeholders involved: Authorities, infrastructure managers, car manufacturers, network operators, service providers and end users with different priorities and objectives.

Authorities are focused on regulation aspects, pushing towards systems that improve socioeconomic factors like accident reductions, traffic efficiency, transport sector growth, etc.

To foster the real deployment of services from the technical side; the solution must use standard technologies facilitating the reuse of similar existing services in other areas and attracts the community of developers for building more applications. This has also an impact on the commercial side, an appropriate business model must be in place so companies are profitable and end users can benefit from the services. For this purpose flexible accounting and billing mechanisms are needed.

3.3 Design key decisions

After understanding the challenges and requirements in vehicular scenarios, we describe the most important design aspects explaining the decisions we take and the process followed.

These key design decisions are:

- Solution based on IMS and NGN enablers.
- Eliminate the need of RSUs as key V2I elements.
- On board architecture based on CALM.

3.3.1 An IMS based solution

Besides providing backup communications for vehicles, it has been demonstrated [57] that the third-generation (3G) mobile network infrastructure is capable of ensuring timely message dissemination throughout large areas. Moreover, mobile broadcast enables efficient message dissemination for many-user scenarios.

We justify the election of IMS as the basis of our solution, detailing the advantages of using IMS for overcoming communication challenges and service provisioning in vehicular environments.

IMS has been widely proposed as a Service Delivery Platform (SDP) [56], but in this work we demonstrate the benefits of using it in vehicular systems. IMS can enable the communication between vehicles in real time, establishing multimedia sessions with QoS parameters according to the network conditions and the service characteristics. Users also benefit from new integrated QoS-guaranteed multimedia services, accessing independently from their geographical location. IMS is access agnostic, so vehicles can dynamically use the most appropriate technology to transmit data (i.e. WAVE, Wifi, GPRS, UMTS, LTE, WiMAX, etc).

IMS provides maximum security through reliable Authentication, Authorization and Accounting (AAA) mechanisms. IMS is based on well-known and widely used protocols as SIP [39], SDP [40], RTP [41] and Diameter [42]. The identities are managed through URIs (Universal Resource Identifier).

One of the most powerful advantages of the use of IMS is the integration with advanced Next Generation Networks (NGN) enablers. As described in Chapter 2, these enablers provide centralized common functionalities to services, so they can reduce their logic and delegate critical operations to a trusted reliable entity.

These facts facilitate consistency and reliability in the creation of new services, promoting re-use, reducing development costs and improving usability [57]. Thus, IMS becomes a very attractive framework for a strong growth in the service provisioning market in vehicular environments.

The benefits from using an IMS/NGN based SDP against other legacy and proprietary approaches are clear. Main standardization bodies (3GPP, ISO, IETF, ITU, ETSI, Parlay, OMA among others) and network operators (Telefónica, Vodafone, France Telecom, Deutsche Telekom, etc) are widely deploying NGN/IMS. This directly implies more effort in improving and building new NGN enablers and attracts the development community for launching services based on IMS. Joyn initiative [58] is a good example of how IMS based platforms can facilitate the deployment of services, even making the service interoperable for many operators.

On the other hand, using NGN/IMS approach for solving networking issues is an innovative approach and the performance of our system compared with other proposed solutions is demonstrated in Chapters 4 and 5.

3.3.1.1 Qualitative analysis of the NGN Enablers

In this section we provide some use cases in which the proposed NGN enablers can be applied in the vehicular environment. As we summarize in Table 3.1 most of the enablers have high potentials in both of the areas in which this thesis aim to contribute:

- Networking issues. The enablers help vehicles to have more information of the network and distribute it more efficiently to improve the performance of routing protocols and make the communications more secure.
- Service deployment facilities. Services can leverage on existing and standardized enablers, reusing the logic and making the development and deployment of services easier and cost effective.

Although we found this an interesting analysis, the most valuable use cases are the ones in which many enablers work together, described in Chapters 4, 5 and 6.

3.3.1.1.1 Terminal Capabilities

The following use cases are some examples of the applicability of this enabler in the vehicular environment. Both use cases assume that Terminal Capabilities has been enriched with information about the vehicle and data collected from different sensors.

Use Case TC1: Fleet management. With this enabler, control centers of fleet companies can automatically monitor the profile of vehicles including data in real time. This will allow obtaining data as the model and vehicle characteristics, average or maximum speed during a travel, the fuel level, the number of hours or kilometers since the last stop, tire pressure, engine temperature, etc.

Use Case TC2: Remote diagnosis. Monitoring the vehicle status insurance companies can prevent possible failures. Moreover, if the car breaks they can evaluate the severity of the breakdown to decide if they send a technician to fix it or on the contrary needs to be towed to a garage. Information like the year of purchase, last technical review and previous breakdowns are also valuable in this scenario.

Use Case TC3: Networking. Knowing networking interfaces, IP addresses and estimated routes in the vehicle navigation system is valuable information for routing protocols to decide how to target the destination.

3.3.1.1.2 User Profile

Use Case UP1: Data Management in accidents. After a car accident, a safety service alerts emergency bodies and requests passenger's profiles. Using this enabler the service can retrieve important information like age, sex and medical history of the occupants and automatically send to the emergency team to improve the medical treatment of the people injured. A ubiquitous health monitoring system based on NGN/IMS is proposed in [59].

Use Case UP2: Road controls. Fuelling appropriately the User Profile enabler can lead to automate road controls. Authorities may request and obtain automatically the mandatory documentation, such driver's license, circulation taxes receipts, insurances, etc. This would lead in a much more effective way of controlling the traffic regulations without disturbing drivers who have not made any infraction.

Use Case UP3: Road toll. A professional driver is on the highway and decides to leave at the next exit. When approaching a toll gate, the infrastructure gets driver profile information such as the pay mechanism, the customer type, possible discounts etc.

Use Case UP4: Management of fleet companies employees. This enabler combined with a location server can empower a service to control and optimize the performance of logistic routes, assigning specific deliveries to the most appropriate driver in real time attending driver’s situation and merits (years of experience, background…).

3.3.1.1.3 Presence

Use Case P1: Fleet Management. Companies related to the transport sector that manage taxis, trucks, ambulances etc, have a control center, to locate and optimally manage their resources. These companies may use the capacity of presence to automatically collect or monitor the status of the vehicles; available, busy, emergency, heading a hospital, etc. In this sense it is interesting to define mechanisms to detect the current situation of the vehicles and automatically publish it (i.e: driving, traffic jam)

Use Case P2: Driving situation. When drivers are under adverse weather conditions (or any other situation in which they do not want to be disturbed), they can establish their presence state to “Not available” for any application, service or user attempting to establish any communication with them. Safety services, by being so critical, are an so they will grant the communication. Similarly, the capacity may provide applications with information whether a certain vehicle is connected to the network (even a VANET) or not.

Use Case P3: Networking. A vehicle can publish available interfaces and capabilities (i.e.: if a vehicle is acting as a gateway through one of its interfaces) improving connectivity mechanisms.

3.3.1.1.4 Identity Provider

The following use cases are some examples of the applicability of this enabler in the vehicular environment.

Use case IdP1: Secure billing. A service for road tolls on highways may use this capacity offered by a trusted operator in accordance with the bank, to charge automatically the fee for the vehicles without knowing the real identity, so a third party could not track users' movements. This example can be extended to those applications that serve multimedia content (downloads, advertisement subscriptions, information, etc). Identity Provider can authenticate the user who is requesting these services, and provide access to a billing mechanism through certain attributes. The user may choose what attributes are accessible for each service.

Use Case IdP2: Medical Information. Some of the attributes susceptible to be stored by this enabler are related to the medical history of the users. In case of accident only emergency services would be authorized to access them even before arriving to the place. And, of course this information remains secure and invisible for other service providers or third parties.

Use Case IdP3: Networking. Generating network identities and pseudonyms for the vehicle enable efficient access control and authentication while covering privacy issues.

3.3.1.1.5 Location

Use Case Loc1: Emergency services. Studies have shown that the arrival of emergency services at the scene of an accident in the first few minutes can save millions of lives. The problem is that the driver, when making an emergency call, do not always know their exact position or able to communicate. In this case the knowledge of the exact position of the vehicle by emergency services is a mandatory added value to the call.

Use Case Loc2: Navigation. A server that controls the location and speed of the vehicles enables services that provide optimum route to the destination at any time.

Use Case Loc3: Fleet Management. By knowing the location of several vehicles belonging to a group or a company, services can optimize the transport in terms of costs and time including an efficient resource allocation.

Use Case Loc4: Information Services. Associated with the position of a vehicle, services can provide useful information for the driver like adverse weather conditions, road works, accidents, traffic information, POIs (restaurants, shops or hotels near), etc.

Use Case Loc5: Geofencing. A service can be programmed to generate alarms if any vehicle trespass certain region or exceed speed limits in a certain road.

Use Case Loc6: Networking. Obtaining accurate positions of neighboring vehicles improves the performance of VANET routing protocols.

3.3.1.1.6 Instant Messaging

The following use cases to illustrate possible examples of the use of IM capabilities in the vehicular environment.

Use Case IM1: Incident notification. A user drives through a certain road and detects some incidents (accident, traffic jam, roadwork, etc). Using the IM capability a warning message can be sent to a previously defined group of users that also drive through the same road at the same time.

Use case IM2: Taxi management. Once a Taxi company receives a request to provide the service to certain address, they could use a service in which they get the status of every taxi driver from a Presence capability, their location and after selecting the more appropriate, send a message to the driver.

Use Case IM3: Networking. Credential and key exchange through a cellular channel using IM before starting a communication through the VANET makes the system more secure and reliable.

3.3.1.1.7 Multicast/ Broadcast

The following use cases to illustrate possible examples of the use of Multimedia Multicast capacity in vehicular environment.

Use Case MB1: Road Safety info dissemination. Service providers can facilitate the creation of groups for those users that drive through certain roads or routes. These users will receive updated info via the Multicast enabler through their mobile device or through the OBU, depending if they are on the road or not.

Use Case MB2: TV Broadcasting. Passengers can subscribe to specific multimedia content and enjoy in the car.

Use Case MB3: Traffic Cameras. In order to decide the best route in real time, drivers may be interested in accessing images from traffic cameras of the possible routes.

Use Case MB4: Networking. Geocasting useful real-time information (position, certificate revocation lists, existing gateways, etc.) to a group of vehicles allows the vehicle to optimize routing decisions and security management.

In Table 3.1, we summarize the use cases in which the proposed NGN enablers improve networking aspects (routing efficiency and security) and service performance in vehicular scenarios.

Enabler	**Vehicular networking**	**Vehicular Services**
Terminal Capabilities	Knowing networking interfaces and capabilities of vehicles (e.g. a vehicle acting as a gateway), IP addresses, provided services, estimated routes in the vehicle navigation system, etc.	Facilitate control by authorities (e.g. cars driving without insurance, cars not having passed technical inspections, remote diagnosis, etc.).
User Profile	N/A.	Accident management, road controls by authorities, user data collection on a toll road infrastructure.
Presence	Publishing available networking interfaces and on-board system status.	Manage Fleet companies (e.g. Taxi, transportation) resources attending real time information of the driver status.
Identity Provider	Generation of network identities and pseudonyms for the vehicle, enabling efficient access control, authentication, privacy, etc.	Secure electronic toll, secure gathering of private information by medical services after a crash, etc.
Location Server	Obtain relevant information for communication protocols (vehicle density, positions, mobility predictions, etc.).	Fleet management, enhanced traffic management by authorities (e.g. avoiding traffic jams), etc.
Instant Messaging	Credential and key exchange through a secure channel before starting a communication through the VANET.	Incident notification to those users that also drive through the same road. Taxi request management.
Broadcast/Multicast	Geocasting useful real-time information to a group of vehicles for optimized routing and security management (position, certificate revocation lists, existing gateways, network service discovery, etc.).	Dissemination of critical information (safety warnings, traffic congestion, etc.).

Table 3.1 Applicability of operator enablers in the vehicular context

3.3.2 Infrastructure without RSUs

The use of RSUs is widely proposed in many academic proposals and reference projects for VANET routing improvement ([60], [61]), security management ([62], [63], [64]) and as a basic element in the infrastructure for service deployment and V2I gateways ([12], [13], [14]).

The benefits from using RSUs are clear; they are fixed, reliable and trustable nodes of the VANET that can act as gateways and inter-communicate vehicles with other networks and the Internet. As they are placed in well-known positions of the road, vehicles can import preconfigured lists of gateways and be prepared to use them as they move within a specific route. RSUs have been deployed by road administrators and authorities, making them reliable for security management purposes.

But RSU deployment has a huge disadvantage that is the high costs associated to deployment and maintenance of the road side infrastructure. Some works propose methods for establishing RSU deployment policies based on cost and coverage parameters [65].

In our solution we propose a different approach, deciding not to use fixed RSUs as core elements of the infrastructure in order to reduce costs and foster real deployment of vehicular services. Because some of the vehicles include cellular interfaces, they may offer it as a backbone for other vehicles acting as dynamic and mobile gateways. This can be incentivized by network operators or service provider to have a more efficient use of their resources. In [66] authors analyze the feasibility of mobile gateways in VANETs and their results suggest that using an appropriate gateway election mechanism, each vehicle should be able to connect to at least one gateway for a majority of time.

There is also another usage of the RSUs that is information dissemination in the geographical area influenced by the RSU. For this use case, we propose a combination of NGN enablers (basically Location Server and MBMS multicasting features) for efficiently covering it. A specific application of this mechanism is detailed and simulated in chapter 5 for certificate revocation list distribution.

We conclude that functionality that typically is delegated to RSUs can be substituted by elements in the infrastructure and on-board vehicles, eliminating the need of expensive RSU deployments which is a clear stopping element in current proposals.

3.3.3 On-board architecture based on CALM

Dealing with the architectural design of the on-board modules we decided to follow CALM. There are several reasons behind this decision; here we present the most important ones:

1. CALM is standardized by ISO [22] and widely adopted by worldwide initiatives ([12], [13], [14]). Compared to non-standard and proprietary architectures, CALM will attract more efforts form developer communities. The resultant services by 3rd party providers can be deployed in our system, increasing the offer and making the overall system more attractive to end customers.
2. CALM facilitates the interface selection process with the Network management Entity (NME) which is a basic feature for the communication platform. Although MIH mechanisms [27] can also cover the need for vertical handover management, using CALM we provide architectural coherence in the different OSI layers.
3. CALM allows adding or replacing communications technologies in a transparent way for the services. This reduces obsolescence which is a relevant issue as Automotive and Information Communications Technology (ICT) sectors have completely different production cycles.
4. The CALM API allows having basic communication functionalities and service related features in different on board HW elements; the OBU and the AU respectively. This is widely proposed by standardization bodies (ISO, ETSI) and relevant projects. Although all the functionality may be implemented in a single physical unit, splitting between two help fulfilling HW requirements more accurately as service management and communication management have quite different requirements. This also reduces obsolescence in those cases that one of the two elements runs out of capacity sooner than the other.

3.4 VISIONS Architectural Design

After describing the objectives of our solution, the requirements that it must fulfill and the key design decisions for the solution the vehicular, we present our design. We describe our design by detailing the modules of the architecture and their main characteristics. These modules can be based on standardized elements (NGN capabilities, CALM, radio technologies, etc) or new ones specifically designed for our purposes.

3.4.1 On-board Architecture

Services in vehicular environments provide rich functionalities to users improving safety, traffic efficiency, logistic operations, and making the travels more enjoyable.

Due to those design reasons described previously in this chapter, the onboard elements are located in different entities:

- The On Board Unit (OBU). It is responsible for every communication aspect, including connectivity intra-vehicle (V2U), with other vehicles (V2V), with the road infrastructure (V2R) and with other networks (V2I). It has secure access to the vehicle system through the On Board Diagnostics (OBD-II) interface. A firewall is key to ensure that only the right services can access the proprietary vehicle system.
- The Application Unit (AU): the platform for applications and services within the vehicles. This element includes all the features and capabilities in the vehicle needed for the deployment and usage of digital services. Due to the nature of the services, this element must provide data storage and high processing capabilities.
- Human Machine Interface (HMI). The function of the HMI is the management of the interaction between vehicle users (driver or passengers) with the system. It has a range of interfaces that will solve the interaction of the most appropriate for each type of relationship-Vehicle User-Application. Due to the difficulty to interact with digital services while driving a good HMI is key for a successful deployment of a vehicular platform. A deep analysis of HMIs is out of the scope of this thesis but the most common proposals include Voice recognition systems, touch screens, dashboard and audio integration, and more recently Head-up Displays (HUD) that present data on the windshield without requiring users to look away from their usual viewpoints. [67].
- Sensors and actuators. Services will be able to gather information from different sensors (i.e.: Temperature, humidity, tire pressure, accelerometers, gyroscopes, etc) to enrich the information and also, always with the appropriate permissions, to interact with the vehicle actuators (i.e.: brake, switch on the emergency lights, start the car, with on the air conditioner, etc).
- Interconnection. These elements can be connected through the fixed buses used in Automation or through short range wireless technologies. The buses are typically Controller Area Network (CAN) [68], Local Interconnect Network (LIN) [69], and Media Oriented Systems Transport (MOST) [70]. CAN and LIN are basic buses designed for communication between vehicle components

while MOST is designed for high bandwidths requirements like multimedia streaming. While OBU, AU and HMI must be connected through the buses, information from the sensors is typically collected wirelessly [71].

This decentralized architecture is optimal because each element has different requirements in terms of HW capabilities (processor, storage, interfaces) ensuring scalability and reducing the obsolescence of the whole system, as it can be independently upgradable. This comprehensive on board architecture is depicted in Figure 3.1. From all the components we will focus on those related with the communications and the services (i.e. OBU and AU), which are central to the scope of this thesis.

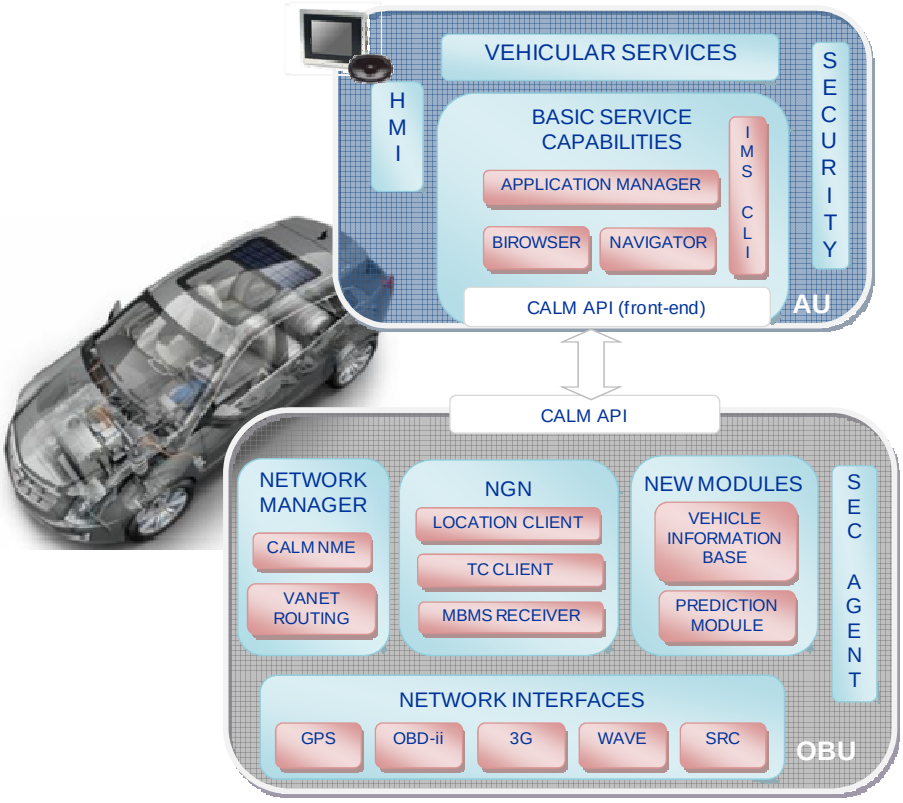


Figure 3.1 VISIONS On-board Architecture

3.4.1.1 On Board Unit

The OBU is the element responsible for all the communications of the vehicle. Thus, it must include all the interfaces and access technologies to fulfill requirements from

services, routing protocols, APIs to communicate with the services running in the AUs, and all the infrastructure modules.

3.4.1.1.1 Network Interfaces

Our solution is access technology agnostic, but in order to provide a realistic approach nowadays, we provide specific examples of the access technologies to be included in order to be powerful and efficient.

1. Wave Interface. IEEE 802.11p is, as described in Chapter 2, the wireless technology specifically designed for Ad-hoc communication between vehicles and for supporting ITS applications. As the link between the vehicles might exist for only a short amount of time, the IEEE 802.11p amendment defines a way to exchange data without the need to wait for the association and authentication procedures to complete.
2. 3G Interface. Although LTE is commercial nowadays, the cost, the coverage and the bandwidth requirements make the 3G technology the most appropriate for our purposes. UMTS and CDMA2000 are typically branded as 3G. The decision to choose between these two depends on the deployment region; UMTS is used primarily in Europe, Japan and China while CDMA2000 is used in North America and South Korea
3. SRC Interface. As some services may require short range wireless communication with other devices in the vehicle (i.e. Tablets of the passengers) it is recommendable to include SRC technologies. Depending on the configuration, it might be required;
    a. Zigbee, mainly oriented for sensors.
    b. Bluetooth, if the range is only few meters.
    c. Wifi, for further ranges, typically used in public or private buses.
    d. Near Field Communications (NFC), very proposed for payments.
4. OBD-II interface. If services need information from the vehicle system or they require interaction with it, a connection to the CAN bus through an OBD-II interface is recommendable. Most of the protocols in the bus are Propietary and every OEM has freedom to use them, but basic diagnosis information is standard accessible.
5. GPS module. It is the most widely used technology for outdoor location nowadays and has enough accuracy to fulfill our requirements.

3.4.1.1.2 CALM API

The API of the communication system is compliant with CALM reference architecture and has two interfaces; management and communication.

A service using the management interface can specify and register the following parameters that the communications must comply:

- Destination: Depending on the communication type (Unicast, multicast, broadcast or geocast) the destination will be defined by IP address and port or by coordinates (latitude, longitude) if the destinations are all the vehicles in a certain geographic area.
- QoS levels; including bandwidth, delay, jitter or the use of specific physical communication interfaces when services demand real time requirements or certain QoS guaranteed.

The API communication interface allows the applications to perform typical operations like:

- Create a connection for both IP and geographic communication.
- Send and receive data through that connection.
- Access the communication parameters associated to a specific connection.
- Request the subscription to certain events that will be received and notified asynchronously.
- Terminate a connection and release resources.

3.4.1.1.3 Network Manager

This element plays a very important role in the system. It is responsible for selecting the most appropriate channel attending the services´ requirements and also includes the routing module for the VANET. It supports IPv4 and IPv6 protocol stacks and geo-routing addresses.

CALM NME gathers the application requirements, previously registered by the CALM API and compares them with the status information of every communication media. It determines the election of the interface taking into account media selection policies. In our case we define the policies for selecting the media that fulfils service requirements at the lower cost. This process is described in Figure 3.2. NME manages the process of vertical handovers between different media. If a link goes down, NME should change

immediately to operate on other interface available without losing information, maintaining the session and updating the routing table.

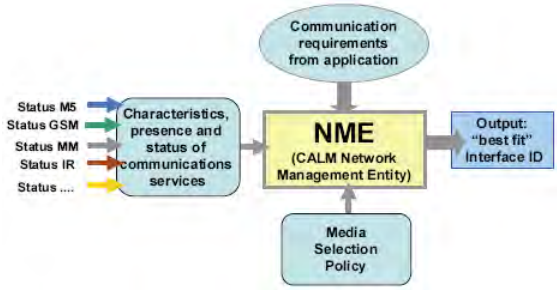


Figure 3.2 Interface selection process

The module responsible of VANET routing can use one or several routing protocols, and it retrieves information from the NGN modules to improve the knowledge of the network in order to improve the performance of the protocols. The main idea is that the interface to use in every hop is decided dynamically using data from the database that includes information from the network (status of the vehicles) and available interfaces.

When the vehicle receives a message, the network manager uses the information to determine what to do. Possible actions are; delete the message, store the message in the buffer, transmit the message using a specific interface and specific parameters (i.e.: specific protocol and route through the VANET).

Thanks to the information available, the algorithm will adapt its operation to different network conditions and traffic. However, we cannot assume that all vehicles in the network have the same amount of information. Therefore, the algorithm must also adapt to the environment information available in order to optimize performance when present. If there is not additional information, the protocol must still provide basic communication services, even with lower performance. For example, if the route of the vehicle itself and its neighbors are known, the algorithm can make smarter decisions when transmitting a message. If this information is not known, the protocol must work properly but can take a route with more delay or number of hops.

This element also supports address auto-configuration protocols required to configure unique addresses on those interfaces which cannot access a centralized configuration service. Standard solutions are IPv6 Stateless Address Autoconfiguration [72] and DHCPv6 [73].

This element also supports Delay/Disruption Tolerant Networking (DTN) where the characteristics of the communications allow it (i.e.: when the service is not very

sensitive to the delay). To do this, the messages can be stored in the buffer until a certain event occurs or before the expiry of its lifetime, when must be reevaluated.

Depending on destination and traffic characteristics of an application, the network manager will use traditional stack of IPv4/IPv6 protocols or it will add VANET routing protocols headers. In this case, the same component in other vehicle is responsible for removing these headers if at any time the routing algorithm decides to route the message through an IP network like 3G.

3.4.1.1.4 Security Agent

This component manages the network layer functions related to privacy, confidentiality, non-repudiation, authentication, authorization and communication accounting.

To authenticate the vehicle in a given network, the Security Agent manages certificates, typically provided by the equipment manufacturer. These certificates are issued by a Certification Authority (CA). One of the main features of the Security Agent is to support authentication against other vehicles or against a gateway of the VANET.

For anonymity and privacy issues but also providing non repudiation, the use of pseudonyms is widely proposed [74], [75]. In case a 3G network is accessible, the Security Agent can use it to verify that the certificate of a given node in the network has not been revoked and therefore is still valid. This process is detailed and demonstrated in Chapter 5.

One of the proposed scenarios is the use of the OBU as a hotspot for other users who want to connect to the Internet. The Security Agent supports this scenario preventing users who have not been previously authenticated from accessing the network. To perform this authentication process, the OBU implements the functionality of a Network Access Server (NAS) to delegate the authentication, authorization and accounting AAA Server deployed in the operator's network. The agents that perform this function are commonly known as enforcement points.

3.4.1.1.5 NGN Enablers Clients

VISIONS provide a source of useful information to the decision-making process of the routing algorithm and the overall network level. Because it is intended to support communication processes some clients of the NGN Enablers are located in the OBU. Moreover, VISIONS can extract information from the vehicle and export it to the operator's network using NGN capabilities. This allows operators and 3rd parties to

deploy services and applications based on information collected by the vehicular network.

1. IMS protocols support. Although the IMS Client resides in the AU, we need in the OBU the support of all the protocols needed by the Enablers we use for communication improvements. These protocols are SIP [39], SDP [40], COPS [76] and DIAMETER [42].
2. The MBMS Receiver gathers valuable information from the infrastructure (CRLs, location of other vehicles, etc) and stores them in the Vehicle Information Base (VIB)
3. The Location client periodically obtains information from the GPS receiver (position, speed, heading and timestamp), stores it in the VIB and sends it to the location server using the cellular interface according to the SUPL standard.
4. The Terminal Capabilities (TC) client obtains certain attributes from the OBU, stores them and sends them to the terminal capabilities enabler. These attributes include Vehicle ID, VANET IP Address, 3G IP Address, the Mobile Station International Subscriber Directory Number (MSISDN) and the estimated route of the vehicle.

3.4.1.1.6 New Modules

Some of the modules have been specifically designed from scratch in order to accurately meet the specific requirements of this work:

1. The Vehicle Information Base (VIB) consists of an information repository of the elements in the area where the vehicle is located. This information is stored in the data base and consists in the following:
    a. Network topology: Position, speed and trajectory of every vehicle in the VANET.
    b. Information of the Gateways. The gateways can be RSUs or vehicles that use different interfaces (typically 3G) as the backbone connection.
    c. Mapping. Information related with the digital map of the area of the vehicle; including roads, optimal route to a destiny, parkings, points of interest, etc
    d. Communication requirements from the services. Registered by the application, we can find data rate, maximum delay, jitter, cost, etc.

e. Interface status. Information related to the capacity and current status of the supported interfaces in the vehicle; maximum data rate, delay, current data rate, current average delay, medium access time, etc.

2. Prediction module. Given that the location of a vehicle changes continuously but the database cannot be updated continuously, this module is responsible of accurately maintaining the current location of a large number of vehicles while minimizing the number of location updates. It is in charge of estimating the position of the nodes when required, based on the position, speed, heading, trajectory from the VIB and time elapsed since the last update. It is based on the work of [77] that proposes a method of generating synthetic spatio-temporal trajectory information of moving objects. Authors in [78] propose to use the road network within which the vehicles are assumed to move for predicting their future positions. Prediction module has an API responsible for providing the information to VANET routing protocols.

3.4.1.2 Application Unit

The AU is the service environment container, including capabilities and services. The capabilities are a group of functionalities needed for the deployment and execution of the services; like the application manager, or the IMS Client. The IMS Client supports IMS session establishment and allows the services to access IMS capacities. The AU requests the OBU for communication provisioning through the CALM compliant API, specifying a set of communication parameters (i.e. QoS, delay-time, packet loss tolerance, jitter...). Once the communication channel has been created, the experimental services can send and receive information regardless the communication technology used by the OBU.

Services show the relevant information through the HMI, which involves a touch screen and speech technologies for a secure and efficient use by the driver and passengers. We now go deeper in every element of the AU.

3.4.1.2.1 Basic Service Capabilities

Here we group some modules which are basic for running services in the platform:

1. Application manager. Accessing the Service Repository, this element is responsible for downloading, installing and configuring new services in the AU. Moreover it offers a layer for 3rd party service providers to offer services

remotely. From the user perspective it allows them to perform the following actions:

a. Display all the available services.

b. Update services already installed.

c. Download new services.

d. Start/stop a service.

e. Uninstall a service.

2. IMS Client. It provides IMS capabilities to all the services running on the platform. It also allows the addition of new functionalities creating a hierarchical structure of modules interacting with each other, but maintaining a low coupling level. Thus, it will lead to a stable, reusable and adaptable asset for future services. Moreover, the communications API based on CALM will give the IMS client the ability to choose the communication interface.

3. CALM API front-end. It enables the use of the communication features in the OBU by the services in the AU.

4. Browser. Not only intended for accessing Webs but also as a framework for many services provided through HTTP.

5. Navigator. One of the most demanded features for in-vehicle use is navigation for having the best route to a destination. The navigator can update the mappings and access the real-time traffic information to select the best route according to road conditions. It is also a good framework for other services like Parking reservation or POI searching. As we will see in future chapters, the routes calculated by navigators can be used also by VISIONS to optimize the communications.

3.4.1.2.2 HMI Interface

This element provides the assets for the driver to interact with the on-board system and the services. An on board HMI has special requirements due to regulations as drivers must avoid any distraction that could represent a risk of having an accident. For an optimal use, the HMI has three different components.

1. Graphical User Interface (GUI) Manager. It allows presenting data through a touch screen and collecting user interactions (pulsations) in the screen. The information can be shown upgrading the screen or via emerging windows ("PopUp") mainly for warnings or alerts. Although it is not the recommended

entry, the GUI manager also includes a virtual keyboard that allows introducing commands or data into the system without a physical keyboard. Both the menu and the services must be designed reducing complexity and facilitating the usability while driving. An example of this is provided in Chapter 6.

2. Speech Technologies. This module allows the user to interact with the system using voice commands, including the navigation through the menu and entering data. The Text to Speech (TTS) feature also offers a way to receive information from services through the audio channel without the need of presenting it in the display. This is the preferred option as minimizes driver distraction.

3. Priority manager. It is responsible for selecting which applications have preference to display/reproduce their messages. The priority of messages can be established upfront depending on the service priority. But applications can prioritize their messages attending on the relevance of them in real time. This element is responsible, for example, of resuming an Infotainment service to inform about a potential danger in the road.

3.4.1.2.3 Security

The module responsible of the security of the service is one of the most relevant as a failure or attack in the system may have tragic consequences. There are many functions that must be considered for service execution to ensure security. Below are the most important:

1. Policy Manager. This element provides uniform management of security policies for all the services offered by the platform. This ensures that all the services comply with the policies attending their category.

2. Signatures. The platform provides tools to sign components and prevent them from installing or updating components without the corresponding certificate by authorized CAs.

3. Access Control. The platform centralizes the authentication and authorization of the users against AAA servers. Only those users with the right permissions are allowed to run the services in the AU.

4. Resource Management. Only those services that have the appropriate permissions to access certain components (especially those related with critical tasks and lifecycle management). For example some services will not be allowed to access the CAN bus or the sensor of the vehicle.

5. Firewall. An application layer firewall is mandatory to prevent attacks from malicious nodes in the Internet. Based on a predetermined set of rules, it controls the incoming and outgoing traffic by analyzing the data and determining whether it should be allowed through or not.
6. Antivirus. Although all the applications and services downloaded into the AU must be properly signed and offered by a trusted service provider, we may end with malicious code running into our system. The antivirus is intended for preventing, detecting and removing malware.

3.4.1.2.4 Vehicular Services

On top of all the described modules run the services. There are many ways to classify vehicular services; attending the benefit for the user (Safety, Infotainment, Mobility, Traffic Efficiency, Comfort, e-Commerce, Fleet Management, etc), attending the communicated entities (V2V, V2R, V2I, V2U), attending the communication requirements (real-time, non-real-time, minimum QoS, etc) and many others.

In this section we propose a classification based on the common characteristics of digital services attending business models and in accordance with legal requirements and standardization procedures. This classification consists in the following groups:

1. Safety-of-life. All services considered "safety-critical" or with any safety implication, although indirect, are considered within this group. There are services that clearly include within this group (i.e.: Advanced Driver Assistance Systems (ADAS) or emergency management) while others, although they could not be considered "safety-critical" may produce a reduction in the number of people injured or dead in road accidents.
2. Liability Critical. Within this group are encompassed all services that have any commercial or legal relationship between the service provider and end users, and also those that may affect 3rd parties. These services must be offered ensuring certain conditions and comply the Service Level Agreements (SLAs) under any circumstance.
3. Non Safety-of-life and non Liability critical. All services that present no commercial involvement, legal or safety are considered within this group.

3.4.2 Infrastructure Architecture

In this section we describe the modules present in the infrastructure. We classify the modules attending to their origin and purpose. Thus we find Security, Service Repository, NGN Enablers and new modules. We pay special attention to the last category that includes those modules specifically designed for our solution from scratch and the modifications applied to the standard ones for our purposes. The architecture is depicted in Figure 3.3.

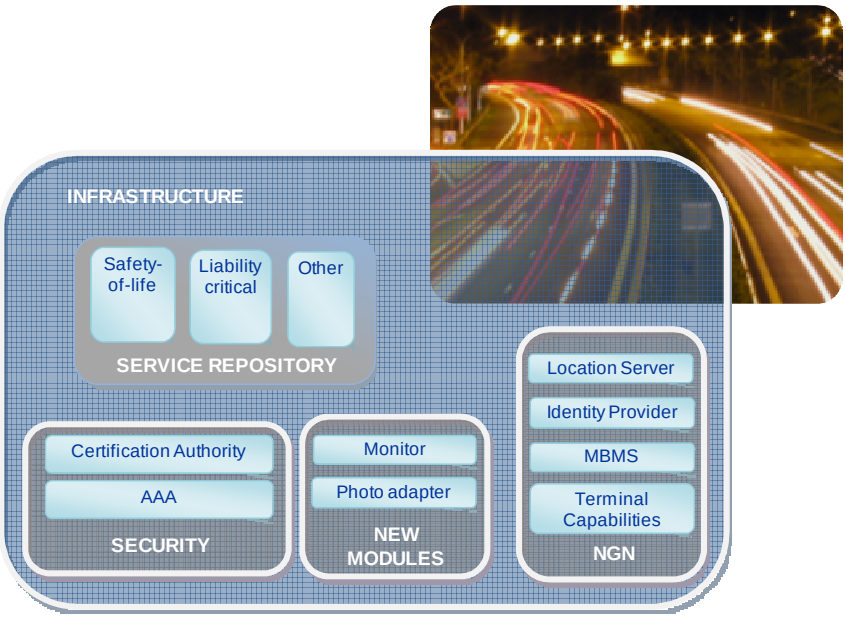


Figure 3.3 VISIONS Infrastructure Architecture

3.4.2.1 Service Repository

The vehicular platform to be useful must include an appropriate an exhaustive catalogue of services. These services are offered through the service repository which includes a set of features, supporting:

- Indexing and discoverability. Users must be able to search and find new services; this includes categorization and a good indexing process. Moreover, there are some initiatives [79] that automatically offer applications to the users attending the ones that already have been downloaded.
- Service Lifecycle management. This feature allows service providers to upload new services, update different versions and remove the old ones.
- Portfolio management. Ensuring that users who have downloaded services can continue enjoying them, even in different devices, this feature controls the service portfolio making it available anytime.

- Payment support. Some services will be offered free of charge but the platform must support several monetization models including; one shot fee, monthly fees, pay per use, etc. This feature communicates with the most common digital payment gateways (i.e.: Pay pal, VISA and MNO mechanisms)
- Legal issues. Contracts, Service Level Agreements (SLAs), Terms and Conditions (T&C) must be centrally managed.

All this features must be standardized and offered through APIs to 3rd parties (developers, device manufacturers, OEMs…) in order to maximize the impact of the platform. All the previously described features, the APIs and a Software Development Kit (SDK) help attracting developer communities to bring new services.

3.4.2.2 Security modules

3.4.2.2.1 Certificate Authority

The CA is responsible for the provision and maintenance of digital certificates and the Certificate Revocation Lists (CRLs). The certificate certifies the ownership of a public key that allows other to rely upon signatures made by the correspondent private key. In this Public Key Infrastructure (PKI) scheme, the CA is the element trusted by both the subject owner of the certificate and the party relying upon the certificate.

We use a standard commercial CA with the basic functions:

- Identity Verification: It validates the identity of the entity who requested a digital certificate before issuing it.
- Certificate Issuing: Digital certificates can be used for encryption, code signing, authentication etc. These certificates are only valid until certain date configured by the CA attending the entity who requested it and the purpose.
- CRL Maintenance: The CA can also revocate the certificate before the expiring date if it has proven facts that the node is infringing the terms of use, for example becoming an attacker in the VANET. A CRL is a list of digital certificates which are no longer valid and have been revoked and therefore should not be relied anymore.

3.4.2.2.2 AAA Server

This element based on DIAMETER protocol is responsible for Authentication, Authorization and Accounting of the platform.

- Authentication is the process where an entity's identity is validated, typically by providing evidence that it holds a specific digital identity such as an identifier and the corresponding credentials. Examples of these credentials are user/passwords, one-time tokens, certificates and digital signatures. Our proposal leverages on SIP Digest authentication procedures as defined in Diameter SIP Application [80].
- The authorization function determines whether a particular entity has permission to perform a given activity. Authorization may be defined based on a range of restrictions, for example restrictions against multiple accesses by the same user. This functionality is key in vehicular services as there are some critical assets, like traffic lights and Dynamic Route Information Panels (DRIPs) that can only be accessed by authorities. Our proposal leverages on the procedures and requirements defined in [81].
- Accounting functionality is responsible for tracking resource consumption by users for analysis, cost allocation or billing purposes. It can be done in real time or batch. This functionality is based on the Accounting state machine defined in Diameter Based Protocol [42].

3.4.2.3 NGN Enablers

3.4.2.3.1 Location Server

This module collects position, speed, heading and timestamp from the vehicles and stores them in the database. The functionalities are offered through a standardized API as this module is based on the Mobile Location Service defined by OMA [83].

Our location service is based on a mechanism that periodically retrieves the position of every node using the Telco operator network in the most accurate way optimizing the use of bandwidth. We propose this mechanism because location services for mobile technologies still have many challenges [84] and lack the required accuracy. The accuracy in those systems depends basically on the number of cells of the scenario, and in interurban scenarios with low density of cells, the accuracy (>100meters) is not enough to provide a clear view of the situation of the traffic.

Because of the large number of nodes, sending unicast information simultaneously through the uplink, the solution must be standard and very efficient, thus we have based our LCS solution on Secure User Plane Location (SUPL) [85]. SUPL is positioning architecture based on the IP protocol stack instead of the traditional SS7 signaling, standardized by OMA.

SUPL offers real-time location access over an all-IP connection, regardless of the wireless signaling layer. User plane location architectures are designed to enable rapid-fix location updates. In user plane location architectures, application intelligence usually resides within the mobile client rather than the server-side. This approach based on secure IP connections independent of the circuit-switched network presents advantages including the ability to quickly update location locally on the device with minimal network costs, inexpensively push location to servers for processing. It also allows the positioning in a mixed 2G/3G or in networks where we do not have full support for control plane standard positioning.

According to SUPL operation, a mobile node can send its position (and extra information) to a centralized server periodically, so the location of all the nodes can be stored and accessed only by authorized nodes and services. The messages defined in SUPL contain a common part and a specific part, with some mandatory fields and some optional ones. This fact allows sending not only position but other vehicle information.

SUPL START (sending info from the node to the server) and SUPL END (ACK from the server to the node) are the minimum set of messages for the proper operation of a SUPL based system. To reduce the resource consumption, basically bandwidth, these will be the only messages used in the Location Service by VISIONS.

3.4.2.3.2 Terminal Capabilities

TC enabler based on the 3GPP standard [43] is responsible for the collection and storage of the following attributes from the vehicles.

- Traditional attributes like make, model, software/firmware version, multimedia and processing capabilities.
- New fields specifically included for our purposes like:
    - ID: Unique identification number of the node.
    - IMSI: Unique identification associated to the SIM that provides access to the cellular network. Very valuable for remote connectivity management through M2M processes.
    - IP Address of the VANET interface.
    - IP Address of the 3G interface.
    - GW. This parameter represents If the node, can act as a gateway or not.

The first group of attributes is key for the deployment of services, it can be used by users and service providers to understand if some applications can be executed in the vehicle or not. The second group of attributes and are intended for improving the efficiency in the communications as we will demonstrate in future chapters.

Using The TC Client, a service can subscribe to the feature that notifies when any change on selected attributes occurs. This can be used, for example, for updating the list of vehicles that can act as a gateway in the road.

3.4.2.3.3 Identity Provider

This element based on [35], is in charge of authentication and authorization functions, covering privacy issues by progressing to services only required attributes, not the real identity. Once the vehicle has been authenticated, IdP generates some credentials, avoiding third parties to obtain the real identity and track the user movements.

This module is the master of the Single-Sign-On session, this means that a user, after been authenticated, can switch between services seamlessly. For the authentication process, IdP relies on the AAA server.

IdP provides a pseudonym and only those attributes of user´s identity that the service is authorized to access.

The benefits for users and service providers using this enabler are tangible: Services do not need to implement authentication and identity management mechanisms anymore, they can rely on IdP. It also improves the user experience because providers can federate services using Single-Sign-On protecting real identities. Users must only provide their relevant information once and then select the services that will be able to access some attributes.

3.4.2.3.4 Multimedia Broadcast and Multicast Service (MBMS)

In this section we describe the enabler responsible for distributing information from the infrastructure to all the nodes within a specific region using a broadcast/multicast mechanism.

Broadcast/multicast is a synonym for point-to-multipoint communication where data packets are simultaneously transmitted from a single source to multiple destinations; all the nodes in broadcast, only subscribed nodes in multicast. To simplify we will now refer to Broadcasting Mechanism.

We have designed the broadcast solution based on MBMS, standardized by the 3GPP [86], because it only introduces minor changes to existing radio and core network protocols. It does not require new receiver hardware in the terminal and additional investments in the network infrastructure, reducing implementation costs in terminals and in the network, making mobile broadcast inexpensive compared to non-mobile broadcast technologies. Moreover, MBMS provides value added features to mobile networks like efficient radio bearers for point-to-multipoint radio transmission within a cell, protocols and media codecs for the delivery of multimedia data and enables operators to define broadcast and multicast services for specific geographical areas at very fine granularity, down to the size of radio cells.

MBMS was designed for streaming services over the mobile network, not for the delivery of files, and this case requires special attention. Broadcast and multicast are one-way transmissions in the downlink. Therefore, the Transmission Control Protocol (TCP) cannot be employed because it requires a bidirectional unicast connection. However, IETF provides a framework for File Delivery over Unidirectional Transport (FLUTE) [37]. FLUTE employs the User Datagram Protocol (UDP) as its underlying transport protocol and employs Forward Error Correction (FEC) [87] as a framework for protecting encapsulated data against packet loss.

MBMS in UMTS presents two important characteristics; radio transmission cost is independent of the number of subscribers in the cell and flexibility. MBMS can use only a portion of a cell carrier, leaving the rest for other services such as regular voice and data. This technology has a capacity advantage over point-to-point connections when several subscribers reside in the same cell. When there are few MBMS users in a cell, it might be more efficient to use a point-to-point connection to each. Considerable effort has been made to determine when a single MBMS point-to-multipoint bearer becomes more efficient than multiple, dedicated, point-to-point connections. To provide more flexibility, a procedure has been introduced in the standards to enable the network to keep track of the number of MBMS users in a cell. The analysis performed in [88] demonstrates the efficiency of multicasting files with MBMS over High-Speed Downlink Packet Access (HSDPA).

3.4.2.4 New Modules

These are the modules in the infrastructure designed from scratch to support VISIONS operative.

3.4.2.4.1 PHOTO Adapter

This module, designed explicitly for our solution, accesses location and terminal capabilities enablers through their standardized APIs for retrieving information in order

to generate a set of digital files called PHOTOS. Each PHOTO contains all the attributes of the vehicles within a specific region.

PHOTOS are distributed in those regions where the information contained in them is useful. There is a compromise between the size of the region and the size of the PHOTO to be distributed in that region. The larger the region, the bigger the PHOTO as more vehicles will be included in it. This module is responsible for the following functionalities:

- Selecting the size and shape of the regions. This will vary from urban to interurban scenarios taking into account real time traffic situation and road morphology.
- Retrieving information from the vehicles and distributing the PHOTOs at certain frequencies in the regions.

These processes are detailed in the next section and Chapter 4.

3.4.2.4.2 Monitor

This element is responsible of the control of the certificate revocation mechanism, monitors the location of the compromised nodes and manages the efficient dissemination of revocation information using the MBMS service.

Once a compromised node has been identified, the Monitor triggers the mechanism informing the CA about the malicious node to prepare new Certificate Revocation List (CRL) to distribute. In parallel, the Monitor queries the location server in order to gather information about the location, the road and the direction of the attacker. This data is obtained using the legitimate neighboring nodes. Finally, the monitor module uses MBMS to disseminate the information to the appropriate zones where it is required. We elaborate on the determination of those zones in the next sections. The whole process is also simulated in Chapter 5.

3.5 VISIONS key processes

After describing the architecture of the solution, we now define in this section the key process of the system for improving VANET routing performance and enhancing the distribution of Certificate Revocation Lists for security.

Both improvements are based on a mechanism that automatically:

1. Collects data from the vehicles.
2. Analyzes the information and prepares an aggregated file.

3. Selects dynamically geographic regions in which the file is valuable.
4. Distributes the file in the geographic regions.

We now describe each phase of the process defining a formal model for both cases; routing and security improvement. In the last section we provide a detailed graphical, step by step operation of this mechanism.

3.5.1 Vehicle data gathering

The mechanism to gather data from the vehicles is a key element of our solution. The information collected includes:

- ID: Unique identification number of that node (e.g. license plate).
- IMSI: Unique identification associated to the SIM.
- IP Address of the VANET interface.
- IP Address of the 3G network.
- GW; If that node, can act as a gateway or not.
- Position.
- Speed.
- Heading.
- Serial number of the latest CRL (or delta-RCL) received.
- Timestamp, indicating the moment in which this information is gathered from the vehicle.

Some of these fields do not vary often, so unless they change during the trip they are only sent and stored in the Terminal Capabilities once. These fields and their size are represented in the Table 3.2.

ID	IMSI	IP VANET	IP 3G	GW
4	4	4	4	1

Table 3.2 Sizes of the fields (in bytes) sent to the TC

The other parameters vary as the vehicles moves so it is necessary to send them periodically to the Location Server according to the SUPL operation. The Certificate Revocation List Serial Number (CRL SN) is only sent when it changes, so unless the vehicle receives a new CRL, that field is not included in the location update. We represent the frequency with which the location update is sent as *Freq_Loc*. The fields and their size are represented in Table 3.3:

ID	POSITION	SPEED	HEADING	TIMESTAMP	CRL SN
4	8	1	1	4	4

Table 3.3 Sizes of the fields (in bytes) in a location update

Assuming that the parameters gathered by the TC are only sent once and that new CRLs only will be received sporadically, the system must only send the size of the location update which we represent by *Size_Loc.* Taking into account that IP header is 20 bytes and UDP header is 8 byte, *Size_Loc* is approximately 46bytes.

With the current data tariffs, sending 46 bytes periodically (even at high frequencies) is inexpensive but as 3G has limited resources it is important to determine the use of resources. For this purpose we define the variable CU_*UL* as the percentage of the Uplink Channel of the 3G network that is used for sending the location updates. CU_*UL* depends on *Size_Loc and Freq_Loc* but also on the number of vehicles using a Node-B which we represent by *Veh_NB.*

$$CU_UL = function_CU_UL\ (Size_Loc,\ Freq_Loc,\ Veh_NB)$$

According to the SUPL standard there is also data transmission in the downlink channel. We define the size of these packets that basically acknowledge the Location update as *Size_Loc_ACK*. CU_*DL* is the variable representing the percentage of the Downlink Channel of the 3G network that is used for acknowledging the location updates. For this parameter *Size_Loc* is not relevant.

$$CU_DL = function_CU_DL\ (Freq_Loc,\ Veh_NB)$$

Note that *CU_UL* and *CU_DL* must also take into account the overhead of the SUPL protocol headers [89].

To optimize *CU_UL* and *CU_DL,* we put in place an additional mechanism that adapts *Freq_Loc* depending on the speed of the vehicle and the trajectory. *Freq_Loc* will decrease if the speed is too low, becoming 0 if the vehicle does not move. If the vehicles increases speed and/or the trajectory varies (typically in urban scenarios) Freq_Loc must be higher.

3.5.2 File generation and target area selection for VANET routing improvement

The file and the area to be distributed are deeply related so we analyze both steps in the same section. The key aspects in this phase are identifying and defining to which vehicles the information (PHOTO or CRL) must be sent and which data should include. As the challenges in our two objectives are quite different in this phase of the process, we analyze them separately.

3.5.2.1 PHOTO generation and Region definition

Following the description of the information gathered from the vehicles, each PHOTO must include all the described parameters except the CRL SN. In Table 3.4 we can see that the size of the information from each vehicle in a PHOTO is 35 bytes.

ID	IP VANET	IP 3G	GW	POSITION	SPEED	HEADING	TIMESTAMP
4	4	4	1	8	1	1	4

Table 3.4 Sizes of the fields (in bytes) included in a PHOTO

We define a Region as the geographical area in which a single PHOTO containing information from all the vehicles in the area is distributed. Regions must fit the morphology of the roads that vehicles are using, we define a region using a pair of coordinates and a heading, representing the opposite vertices of the rectangle and the orientation.

There is a compromise, between the size of the region represented by *Reg* and the size of the PHOTO, *Size_Pho,* to be distributed. If *Reg* is large, many vehicles will receive the same information avoiding the partitioning of the information into many

PHOTOS, but it will increase the *Size_Pho* and some of the vehicles may receive unnecessary information of other vehicles which are far away.

For a given scenario, to determine the optimal size of the PHOTOS we can take into account the number of vehicles in the Region (which of course depend on the size of the Region) and the MBMS channel resource consumption represented by *CU_MBMS*. This is formally stated by the following formula:

*Size_Pho = function_Size_Pho (280bytes * Veh_Reg (Reg), CU_MBMS)*

For obtaining accurately the number of vehicles within the region (*Veh_Reg*), the Photo Adapter queries the Location Server.

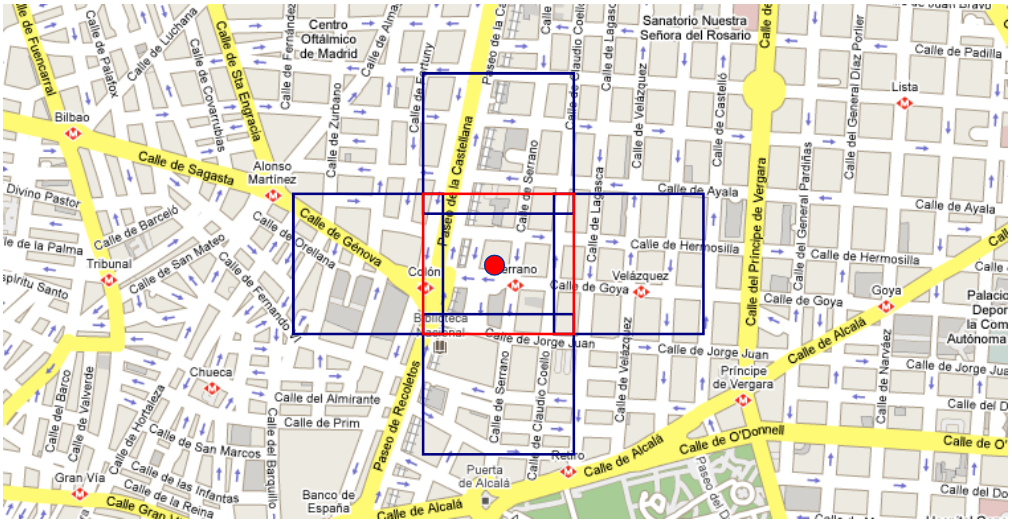
To provide a quantitative example we consider for the MBMS a single channel (out of the 16 available ones per cell and per sector) operating at its basic rate of 64 Kbps with Spreading Factor (SF) = 32, Transport Block Size (TBS) = 5120 bits, Transport Transmission Interval (TTI) = 80ms as in [90]. With these hypotheses, our MBMS channel has 64kbps capacity and each vehicle adds 280 bits to a PHOTO. Thus, a MBMS channel can send a PHOTO with information of 220 vehicles in less than 1 second.

Veh_Reg = 220 vehicles *Size_Pho* = 61.600 bytes

Now we describe the algorithm used by the Photo Adapter to select the size and morphology of the Regions; *Reg.*

1. The area is partitioned in several rectangles. The coordinates of the geographic centre of the rectangle are calculated taking into account the location of the different Nodes B. In urban scenarios the heading of the rectangle is not so relevant but in interurban, the heading must take into account the morphology, adjusting the rectangle to the road cartography.
2. Starting from this centre, the database is queried to obtain the coordinates of a *Reg*, where there are currently *Veh_Reg* vehicles.
3. Considering the typical number of users served by a Node B in [109] the more likely situation is that a Region will be covered by several Nodes B. These Nodes B will be in charge of distributing the same PHOTO.
4. When *Reg* is determined, the neighbouring areas are subdivided in squares with the same size. Regions are overlapped to avoid spots without coverage.
5. The frequency with which *Reg* is calculated is represented by *Freq_Reg* and depends basically on the traffic density variation.

This algorithm is shown graphically in Figure 3.4 for urban and in Figure 3.5 for interurban scenarios.

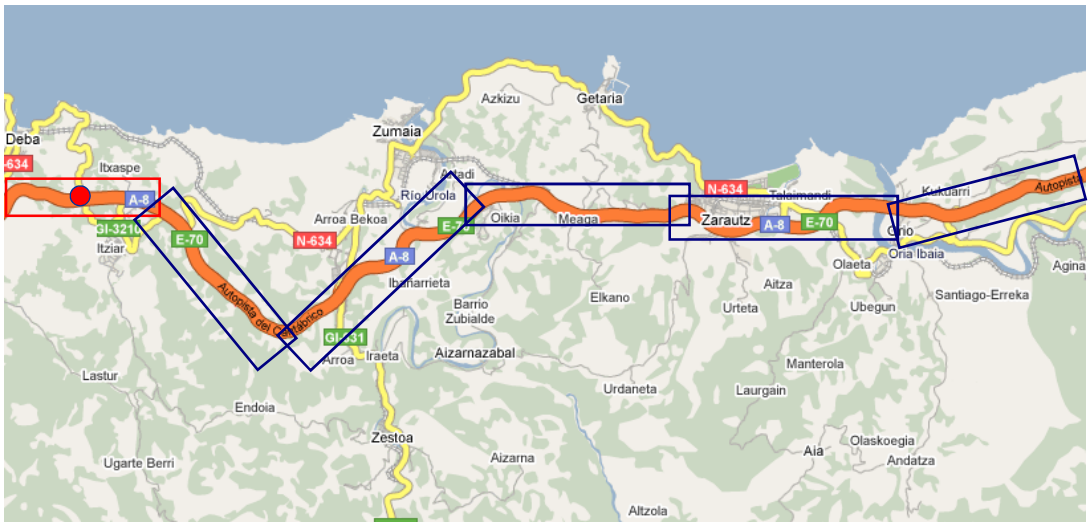


Geographical Centre calculated with the location of the Nodes B

1st square, that defines *Reg* attending the number of vehicles

Subsequent squares with the same size than *Reg*, and overlapping

Figure 3.4 *Reg* selection algorithm in urban scenarios.



Node B that covers the first part of the highway, considering the urban border

First rectangle, attending the cartography and number of vehicles

Subsequent rectangles, attending the cartography and number of vehicles, with overlapping areas

Figure 3.5 *Reg* selection algorithm in interurban scenarios

The algorithm can be enriched with historical data in order to optimize the calculation of *Reg*. Regions are overlapped to avoid those situations where a vehicle is arriving at a region border and has no information from the region it is going to enter.

It is important to remark that the prediction module in the OBU helps to determine the new position of the vehicles. This can be a difficult task in urban scenarios but a relatively easy in interurban scenarios, in which the trajectory and speed of the vehicles can be predicted.

3.5.2.2 CRL generation and Region definition for Security improvement

For the effective CRL generation we distinguish between two different types of Region:

- An Administrative Area (AA) is a region whose security administration can be done with minimal communication with other neighbouring areas.
- A “D-Zone” is a distribution zone around a compromised vehicle where we distribute revocation information as soon as possible when the attack is detected.

And two different type of CRL to distribute:

- Base CRLs keep a complete list of revoked certificates
- Delta CRLs include only those certificates that have been revoked since the last publication of a base CRL.

We assume that any of the attack detection mechanisms proposed in the literature [91], [92], [93] is being used. Once a compromised node has been identified, the Monitor enabler triggers the revocation mechanism. For the revocation to take place, the vehicle detecting the attack reports to the CA the identifier of the malicious node. In fact, when the detection is distributed it can be one node among those who detected the attack, or they may all report the CA and based on that the CA decides whether the certificate needs to be revoked or not. When the CA decides that revocation is needed, it generates a delta-CRL including the certificate of the compromised vehicle and prepares the next base-CRL. The delta-CRL only includes those certificates whose revocation status has changed since the last base-CRL. In parallel, the monitor module in the infrastructure queries the location server in order to gather information about the location, the road and the direction of the attacker. This data is obtained using the legitimate neighbouring nodes.

AA are defined at the beginning of the life of the system taking into account the mobility of vehicles is exploited. That is, a very high proportion of vehicles move within that area compared to those that move to other areas. This can be done based on traffic statistics. There are two options to manage the CRLs in administrative areas:

1. Every AA is controlled by a single CA as proposed in [94]. The generation of CRLs is immediate but the migration of vehicles to/from other regions implies numerous and simultaneous generation of certificates. Besides, the process to revoke a certificate has higher complexity because requires exchanges of information between different zones.
2. The same CA controls many AA, simplifying the management of certificates between zones although the CA must know the vehicles within each region to generate specific base-CRLs. For this purpose, real time information of the VANET is obtained from the location server.

For the definition of D-Zones we present an innovative approach. As we cannot trust the information sent by the compromised node, we have designed an alternative method in which legitimate nodes periodically report the position of the attacker when they see it as a neighbour. This functionality helps the system in locating the compromised node more accurately and it only adds 4 bytes to the Location Update packet including the ID of the attacker.

Therefore, being aware of the position of the attacker, as well as the rest of the nodes of the VANET, a D-Zone is defined by the coverage of the Node B in which the attacker is moving. If at some point a zone lies between the coverage of multiple Nodes B, the delta-CRL is sent out by all of them to cover all the MBMS-enabled vehicles in that zone. This process is depicted in Figure 3.6.

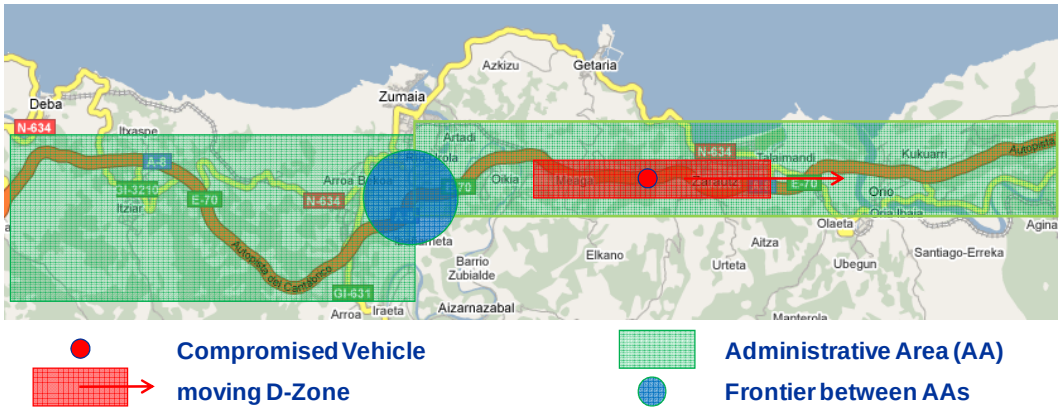


Figure 3.6 CRL Dissemination zones

The sizes of both CRL files; Base and Delta to be distributed in AA and D-zones are represented by *Size_Base_CRL* and *Size_Delta_CRL*. To determine these sizes, we consider the components in X.509 v2 CRL [95], analyzed in [96]:

1. The CRL Header includes:

- CA´s name: 32 bytes (if X.500 name used)
- Date/time of CRL issuance: 6 bytes
- Date/time of next CRL issuance: 6 bytes

2. List of revoked certificates (9 bytes per entry)

- Serial number: 3 bytes
- Revocation date: 6 bytes
- CRL entry extensions (i.e. revocation reason)

3. CRL general extensions (i.e. CRL number)
4. Signature of CA (RSA 1024 bit signature, approx 130 bytes)

We assume that this structure is valid for both; base and delta CRLs. According to this model:

$$Size_Delta_CRL = Size_CRL_Header + Attackers_D_Zone * (Size_Revoked_Certificates_D_Zone) + Size_CA_Sign$$

Size_Delta_CRL depends on the number of malicious nodes in the D-Zone identified since the last Base CRL generated represented by *Attackers_D_Zone*. This parameter directly depends on the size of the D-Zone and the density of the vehicles in that area represented by *Veh_D_Zone*. Higher densities imply more probability of having new attackers since the last Base CRL distributed. Assuming only one (*Attackers_D_Zone* =1) new certificate is revoked in each D-zone:

$$Size_Delta_CRL = 44 + 1*(9) + 130 = 183 \text{ bytes}$$

Accordingly, Size_Base_CRL depends on the number of new malicious nodes in the AA identified since the last Base CRL generated represented by *Attackers_AA*. This parameter directly depends on the size of the AA and the density of the vehicles in that area represented by *Veh_AA*.

$$Size_Base_CRL = Size_CRL_Header + Attackers_AA * (Size_Revoked_Certificates_AA) + Size_CA_Sign$$

$$Size_Base_CRL = 44 + (Attackers_AA) * 9 + 130$$

3.5.3 Distribution

3.5.3.1PHOTO distribution

We define *Freq_Pho* as the frequency with which the MBMS enabler distributes a PHOTO in a Region *Reg*. There is a balance between the use of the MBMS channel and the obsolescence of the information vehicles use for improving their routing decisions.

The maximum delay of information (*Max_Delay)* is calculated taking into account the frequency with which the vehicles send Location updates (*Freq_Loc*), the frequency with which the MBMS enabler distributes a PHOTO (*Freq_Pho*) and the server processing time. With the current process capacities, we depreciate the server processing time.

$$Max_Delay = 1/Freq_Loc + 1/Freq_Pho$$

$$Freq_Pho = 1/(Max_Delay - 1/Freq_Loc)$$

Max_Delay is limited by the tolerance routing algorithms have to inaccurate location of the vehicles and the performance of the Prediction module in the OBU, that can infer the updated position of the vehicles from information previously gathered (position, heading, speed, etc) as we have seen in the Architecture section.

It is also important to remark that *Freq_Pho* must be equal or higher than *Freq_Loc*. If not, the system would be distributing PHOTOS without new information for the vehicles.

3.5.3.2 CRL distribution

The frequency with which the system distributes Delta CRLs in the D-Zones is represented by *Freq_Delta_CRL* and it is crucial because vehicles in a D-Zone are close to a compromised vehicle that they are not aware of. Other possible solution to overcome this challenge would be sending the last delta-CRL to the D-Zone only when

a new vehicle which is not aware of the attack enters the D-zone. This situation would be detected by the infrastructure because vehicles send using SUPL the serial number of the latest CRL (or delta-CRL) they received and their current position, heading and speed. We discard this possibility because in both urban and interurban scenarios vehicles will be continuously entering in the risk zone and the delay of the whole process puts in risk. As we mentioned, these D-Zones are mobile so the monitor enabler is responsible for selecting dynamically the appropriate station to broadcast the information.

Freq_Delta_CRL must be established considering the time that takes to a vehicle entering D-Zone to meet the attacker coverage area. As D-Zone is selected taking into Nodes B, there is a dependency on the coverage of the Node B (one or several) that defines de D-Zone represented by *Coverage_Node_B (D-Zone)* and the maximum speed at that vehicles can move in the D-Zone represented by *Speed (D-Zone)*

Freq_Delta_CRL = function_Freq_Delta_CRL (Coverage_Node_B (D-Zone), Speed (D-Zone))

Within an AA, the system broadcasts updated Base CRLs with a dynamic frequency *Freq_Base_CRL*. When vehicles move to a new region, they must update the last Base CRL in the new region, thus MBMS is broadcasting the Base CRLs in the frontiers of the regions as we can see in Figure 3.6.

Freq_Base_CRL is reasonably low, as vehicles should have a delta-CRL if they are close to an active attack. It is also dynamic; the Monitor can adjust it taking into account new identifications of attacks or variations in the flow of compromised nodes with other regions. As every new attacker identified will be covered by the distribution of Delta CRLs in the D-Zones, the distribution of a new Base CRL eliminates all the Delta CRLs distribution within the AA. *Freq_Base_CRL* will basically depend on the number of malicious nodes identified in the AA since the last Base CRL distributed.

Freq_Base_CRL = function_Freq_Base_CRL (Attackers_AA (Veh_AA (AA))

As part of the future work, we would like to investigate alternative mechanisms in areas without cellular coverage or for those vehicles without a cellular interface. The objective will be to define a mechanism to exchange revocation information (Base/Delta CRLs) between the vehicles in AAs, D-Zones or frontiers through the VANET. The mechanism

can include the identifiers of the last base/delta CRLs in the beacons. When a node receives a beacon with higher ID, it will ask the sender of the vehicle to broadcast the base/delta CRL. Those nodes with an obsolete base/delta CRL will be able to update it. Although the sizes of base-CRLs are larger than delta-CRLs, this mechanism works properly in AAs, because partitioning the whole administrative area leads to reasonable sizes of the CRL files.

3.5.4 Overall Operation

The overall operation of VISIONS in the case of routing improvement is depicted in Figure 3.7. This illustrates the provision of enhanced information to the VANET routing module of an OBU employing the previously described enablers. The operative for security improvement is very similar. The operation consists in the following sequential steps:

First of all, the OBU connects (if it is not already connected) to the cellular network using its 3G interface. The second step is divided in three phases; the TC client (Terminal Capabilities in Fig. 3.1) collects attributes from the OBU and stores them (step 2a) in the Vehicle Information Data Base. Then, attributes are sent to the corresponding enabler (step 2b), located in the VISIONS infrastructure modules. Finally, the Terminal Capabilities enabler in the infrastructure acknowledges the operation (step 2c).

The location process is also divided in three phases: the location client in the OBU retrieves information from the GPS receiver, stores it in the VIB (step 3a) and sends it to the Location Server (step 3b). As in the previous step, finally the Location Server acknowledges the operation (step 3c).

The PHOTO generation mechanism consists of two steps; first, the PHOTO adapter retrieves information from the Location Server (step 4a) and from the Terminal Capabilities (step 4b) and generates the PHOTOs. Each PHOTO is distributed into the geographical region *Reg* through MBMS (steps 5 and 6). When a vehicle within a region *Reg* receives the corresponding PHOTO, it stores the information in the local VIB (step 7). The information contained in a PHOTO may be pre-processed in the infrastructure to get a more compact representation suited specifically for the particular applications. For instance, for routing it could be better just to send the IDs of the streets with connected chains of cars. However, that information would only make sense to improve VANET routing. Instead, in our evaluation we include in PHOTOS the positions of other vehicles in the region. Position information is more general and can useful for other applications beyond VANET routing.

In each vehicle, the VANET routing protocol in the OBU queries the Prediction module (step 8) through the API. This module calculates the current position of the surrounding vehicles and answers the query (step 9). Finally, the VANET routing protocol is provided with relevant information (step 10) and takes an informed routing decision towards the targeted destination.

Note that step 3 is repeated with the frequency *Freq_Loc*. Steps 4-7 are also repeated with the frequency *Freq_Pho* for every region *Reg*.

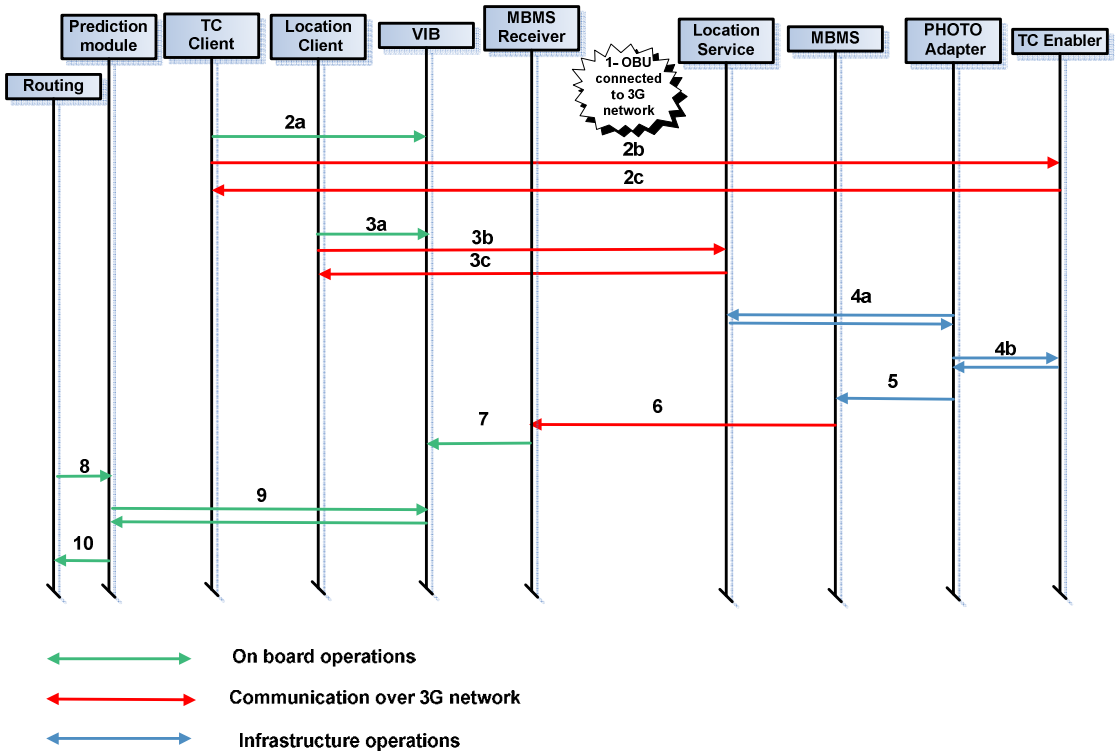


Figure 3.7 VISIONS key process for routing improvement operation scheme

3.6 Conclusions

In this chapter, we have presented the design of our solution called VISIONS. It is a vehicular platform with two main objectives:

The first one is being an efficient and reliable platform for deploying services in the vehicular environment. These services are classified in Life-critical, liability critical and others. Some examples are ADAS, Navigation with real time traffic information and Infotainment applications like video streaming and advertisement. The second

objective is improving the intrinsic challenges in vehicular networks such as VANET routing and security issues.

With these objectives we have defined the requirements that the solution must comply with. Most of the requirements are technical; basically related with Safety, security, mobility support and reliability, but there is also a socio-economic parameter which is feasibility: The solution must be deployable in terms of costs and profitable supporting the use of 3rd party service providers with several business models.

After establishing the objectives and defining the requirements we have designed a modular solution which includes elements on board vehicles in the infrastructure based in standard elements of network operators. We justify the three most important design decisions of VISIONS which are:

1. Leverage on IMS subsystem and NGN standard enablers
2. Follow the CALM reference Architecture in the On board elements
3. Avoid the use of RSUs as basic infrastructure elements

For showing the potential of IMS and NGN enablers in the vehicular scenario; we provide explicit examples of the applicability of the enablers to both vehicular communications and services.

We now summarize the most relevant decisions in the different elements of the architecture:

There are two different units on board vehicles; the OBU, responsible for all the communications of the vehicle and the AU in which is the service environment container. The rest of the elements are centralized in the operator infrastructure, avoiding the need of RSUs which are complex and not effective in terms of deployment and maintenance costs.

OBU is based on standardized CALM architecture, which supports mobility through different communication interfaces seamlessly attending service requirements and status of the networks. Different radio technologies can be easily integrated reducing the risk of obsolescence. A clear example of this is that we have used 3G as the cellular interface due to wider coverage and smaller costs (module and communications) but this can be substituted by LTE (4G) if needed without impacting the rest of the platform. The OBU also includes VISIONS modules that enable the use of information from infrastructure capacities for improving the communications (i.e.: improving the knowledge of the network for taking better routing decisions) and a Security Agent that covers privacy, confidentiality, non-repudiation, authentication, and authorization at the network layer.

CALM also helps to reduce obsolescence issues which are typical as automotive and digital sectors have completely different production cycles.

The AU contains capabilities, services and controls the HMI. The capabilities are a group of functionalities needed for the deployment and execution of the services; like the application manager responsible for downloading, installing and configuring new services, or the IMS Client which allows services to access NGN enablers in the infrastructure. Services communicate with external elements through the OBU using the CALM API. As many 3rd parties develop their services following the CALM reference architecture, it facilitates the execution of those services from external providers in our platform. The HMI is a central part of the solution because it ensures a safe usage of the services on board. There is also a security module in the AU that performs access control, firewall and antivirus functions. Safety, security and reliability are the most important premises in this environment.

Modules in the infrastructure leverage on standard elements of the network operator like IMS subsystem, NGN enablers and basic security servers like CA and AAA. Only two modules have been designed from scratch, while the others are used as standardized or only require light adaptations. The location server and a distribution element based on MBMS are core features within VISIONS. A Service Repository has been included in order to offer different types of services from different providers in order to make the platform attractive for end users and profitable for 3rd parties.

In the last section we have formally defined the mechanism used for both purposes; improving VANET routing performance and enhancing the distribution of Certificate Revocation Lists. We define the Key Performance Indicators (KPI) for the different phases of the process and justify the modus operandi. In summary, the process collects data from the vehicles, analyzes the information in the infrastructure, prepares an aggregated file, selects dynamically geographic regions in which the file is valuable and distributes the file in those regions.

In the following chapters we evaluate our solution and demonstrate how the platform meets the objectives of the thesis. In chapter 4 we analyze the challenges in VANET routing and demonstrate through simulations how our platform improves efficiency in some relevant routing protocols. In chapter 5 we study security problems and particularly those related with certificate revocation mechanisms when one of the members of the VANET becomes an attacker. We simulate an interurban scenario and demonstrate how our platform is more efficient neutralizing these attackers than the widely proposed solutions based on RSUs. In chapter 6 we describe the limitations for bringing vehicular services and applications to the vehicle in a mass-market approach

and propose an innovative Social Network service based in our platform, explaining how VISIONS facilitate the deployment and usage of complex services.

4 VANET ROUTING IMPROVEMENT

4.1 Introduction

As we showed in previous chapters, routing protocols in vehicular scenarios still have challenges in terms of performance. In this chapter, we analyze those challenges, studying which are the proposed solutions in the literature and demonstrate how our solution VISIONS can improve the performance of several routing protocols.

Many VANET-specific routing solutions have been proposed in the last years [97], [98]. The key to all these protocols is to take advantage of additional information available in the vehicles, such as position, digital maps or even the planned routes. By using this additional information, these new VANET protocols improve the performance of routing mechanisms. However, their performance depends on the amount of information in the vehicles – the higher, the better – and is hence limited when the information present in the vehicles is scarce, as well as when this information cannot be disseminated, e.g. across partitioned clusters of vehicles. One option to improve the performance of these routing protocols is the use of a backup network to ensure a data channel for vehicular communications, as proposed in the CALM reference architecture we follow [99]. This network, typically cellular, acts as a backup link when the destination node is not reachable through the VANET.

Besides providing backup communications for vehicles, it has been demonstrated [57] that the 3G mobile network infrastructure is capable of ensuring timely message dissemination throughout large areas. Moreover, the mobile broadcast enables efficient

message dissemination for many-user scenarios. Finally, end-to-end delays of below 500 ms can be achieved in today's 3G networks, although they cannot be guaranteed. The main innovation in this chapter is the use of cellular networks to gather and distribute relevant control information. To accomplish that, we follow the solution designed in Chapter 3 and have tested its particular application to the case of VANET routing improvement in urban scenarios.

The remainder of the chapter is organized as follows: In section 2 we describe the related work in the literature. In section 3 we review those modules in the architecture that participate in the process and describe the benefits of using VISIONS to enrich the information used for taking routing decisions. In section 4 we evaluate qualitative and quantitative the performance of our solution using the KPIs defined in the requirements section; Packet delivery ratio, delay and resource consumption. In section 5 we summary the conclusions of the Chapter.

4.2 Related Work

To understand the designs of the existing routing protocols on vehicular ad hoc networks, the characteristic and mobility patterns of the vehicles in different situations is very relevant. As we have seen, the main characteristic is that the mobility of the nodes is restricted by the road layout, the movement of other vehicles and traffic rules. In addition, the difference between urban (cities) and interurban (highways) scenarios lead to different distribution of vehicles. Another peculiarity of the vehicle mobility scheme is that vehicles tend to move into groups, forming clusters, and portioning the network. All these factors make traditional ad hoc routing not to be a very appropriate solution for vehicular settings. Thus, specific protocols have been proposed accordingly [101], but there is not a single protocol that has proven optimal performance for every traffic scenarios. In [102] identify the pros and cons of routing protocols which can be used for further improvement or development of any new routing protocol.

Because of the great number of vehicles that can be participating on a VANET simultaneously, routing mechanisms need to take decisions locally to ensure scalability. Therefore, exchanging information with neighboring vehicles via beacon messages is a fundamental part of routing protocols in the literature.

We can classify existing VANET routing protocols in four groups based on the information used for the routing decision. These groups of protocols use:

- Position of the vehicles.
- Digital maps of the area.

- Store-Carry-Forward (SCF) paradigm.
- Trajectories of the vehicles.

In the first group, protocols use the position of the nodes to decide which neighbour is the most appropriate for carrying the information closer to the destination. By using greedy heuristics, the protocols choose as next hop the neighbor which provides greater advance towards the destination (i.e., the one which is closer to the destination). GPCR [103] and CAR [104] are examples of these approaches.

Greedy Perimeter Coordinator Routing (GPCR) is based on the idea of coordinators, which are basically nodes placed on the junctions of the streets. The algorithm prioritizes these coordinators as next hop because they usually have more possibilities to progress data using different (at least two) streets. Then, coordinators choose the best street attending the road where is situated the neighbour closer to the destination. Each node must decide and inform if it is a coordinator or not based on the position of its neighbours, if the position of its neighbours have a lineal relation or not.

Connectivity-Aware Routing (CAR) determinates a route to the destination in which there is enough vehicles to carry the message before sending the information. It uses a flooding mechanism to find the destination, then, the destination responses to the source informing its position, speed and the list of junctions the message has gone through. The answer goes back using the same path in the opposite direction. The source finally sends the information through the selected path, selecting in each hop the neighbour closer to the destination.

There are other protocols that use digital maps to forecast which will be optimal path for the packet to reach the destination node. In this way, the map provides information about topology of streets. This is employed by the source node to compute a list of junctions which the data message must traverse to get to the destination. In order to reach each junction, the protocols apply geographic routing along each street. GSR [105] and A-STAR [106] are examples of geographic-based VANET routing protocols that employ map information.

Geographic Source Routing (GSR) presents a solution to the standard geographical algorithms problems, more specifically for the the recovery of local minima. Authors propose as the method for locating the destination node Reactive Location Service (RLS), an algorithm based on flooding. The GSR algorithm aims to determine the intersections that the message must traverse towards its destination. This applies Dijkstra's algorithm on a digital graph that models the area map. The vertices represent intersections and the links between vertices, the streets. The path used by the

message between two junctions of the road is calculated using a voracious algorithm assuming no obstacles. Authors do not take into account that between two junctions may not exist enough vehicles to forward the message.

In Anchor-based Street and Traffic Aware Routing (A-STAR), authors propose a solution to determinate the best path to the destination using digital maps and information from the bus lines using BUSNet [107]. This algorithm assumes that those streets with more density of bus lines will have more density of vehicles, thus more probability to route the packet towards the destination. It is very similar to GSR as it calculates beforehand the path including in the header some points that the packet must traverse, called anchor points.

A-STAR uses an special protocol based on Dijkstra and a digital map to calculate the routes. The information from the bus lines is used to assign weights to each street, and these weights can be updated with the real time status of the traffic. If the route to the destination breaks it uses a route recovery scheme based on RLS calculating an alternative route to the next hop of the original route. Moreover, that street is marked temporary as not valid for routing including this info in the packet as it travels to the destination. Because of this all the vehicles that receive the packet can mark that street as not valid and they do not use it for calculating new routes.

The aforementioned solutions assume that an end-to-end path between the source and the destination exists at the moment of issuing a data message. However, this is not the common case in VANET scenarios, since vehicular networks are formed by disconnected groups of vehicles. In order to solve this problem, the SCF paradigm can be employed. It consists of storing the message when there is no neighbour which provides advance towards the destination. When a suitable neighbour is found, the forwarding takes place. SAR [108] and VADD [51] rely on this kind of routing.

Spatial Aware Routing (SAR) works like GSR, following a previously calculated list of junctions that a packet must traverse. The contribution of this protocol is the recovery mechanisms offered in situations when there are no more vehicles to route the message towards the destination. In those situations, the algorithm can choose between three options:

- Store the packet in the vehicle, trying periodically to reroute as it discovers new neighbours. If the packet remains for long time or the buffer is full, it will be discarded.
- Use greedy forwarding until source routing can be used again.
- Use an alternative route to the next hop calculated using source routing.

Vehicle-Assisted Data Delivery in Vehicular Ad Hoc Networks (VADD) also employs store and forward paradigm but uses more information from the digital map (roadmap topologies and speed limits). Thus, the functionality that selects the next forwarder takes into account the estimated time of travel to the destination of all the neighbours. The route is recalculated at every hop.

On the other hand, some other protocols are based on the planned trajectories of vehicles instead of on geographic routing. Vehicles forward the data message to those neighbors whose trajectory is more useful than the one followed by the current holder. Hence, protocols like Geographical Opportunistic Routing for Vehicular Networks (GeOpps) [52] try to get to the destination as fast as possible, while others like Motion Vector (MoVe) [53] look for the greatest advance towards the destination according to current velocity vectors.

4.3 VANET Routing improvement using VISIONS

In this section, we describe how the use of VISIONS over 3G networks can be used to improve the performance of VANET routing protocols as one of the possible applications. In particular, we focus in this problem to show how our proposed idea is even able to offer advantages to multihop routing which is one of the most researched areas in VANETs. We analyze how the routing protocols can benefit from the functionality provided by the enablers, describe in detail the architecture and provide an example of the operational mode of the proposed solution.

4.3.1 Benefits

Vehicles use VANET routing protocols to deliver messages between a source and a destination using other intermediate vehicles as relays. The constrained mobility of vehicles along streets, intersections, traffic lights and the likes results in a partitioned network with uneven network density in which the routing task becomes very challenging. As a matter of fact, even traditional routing protocols for highly mobile scenarios such as mobile ad hoc networks fail at providing an acceptable performance.

For this reason, VANET-specific routing solutions have been proposed [101] and analyzed [102]. Their main innovation is that they are particularly designed to exploit available information regarding the scenario and other vehicles. For instance, they can be based on the use of street maps, location and speed of other vehicles, temporal storage of data packets to deal with partitions, etc. The general idea behind VANET routing protocols in city scenarios consists of finding the most appropriate list of streets (or junctions) connecting the source and the destination. Data packets are sent from

the source vehicle to the position of the destination by traversing those streets/junctions. To do that, each vehicle participating in the forwarding selects neighbouring vehicles being closer to the next junction than itself. This process, which is called greedy routing, can fail if a message gets stuck at a vehicle travelling along a street with insufficient density to guarantee message forwarding. So, it is highly valuable for VANET routing protocols to count on real-time information about which streets have enough connectivity to facilitate packet forwarding.

To overcome routing challenges, we propose an efficient mechanism based on existing enablers to provide vehicles with information about which streets have enough connectivity among cars so as to provide a multihop path. Note that even highly populated streets may get partitioned. This mechanism is based on the process for gathering information from every vehicle and distributing aggregated information in certain geographies described in section 3.5. We now describe which modules participate in this use case, how they interconnect and the operation.

4.3.2 Architecture

Based on the general architecture of VISIONS, we describe which modules are used and how they are connected for the objectives in this chapter. Infrastructure and Onboard modules that take place in this specific mechanism are depicted in Figure 4.1.

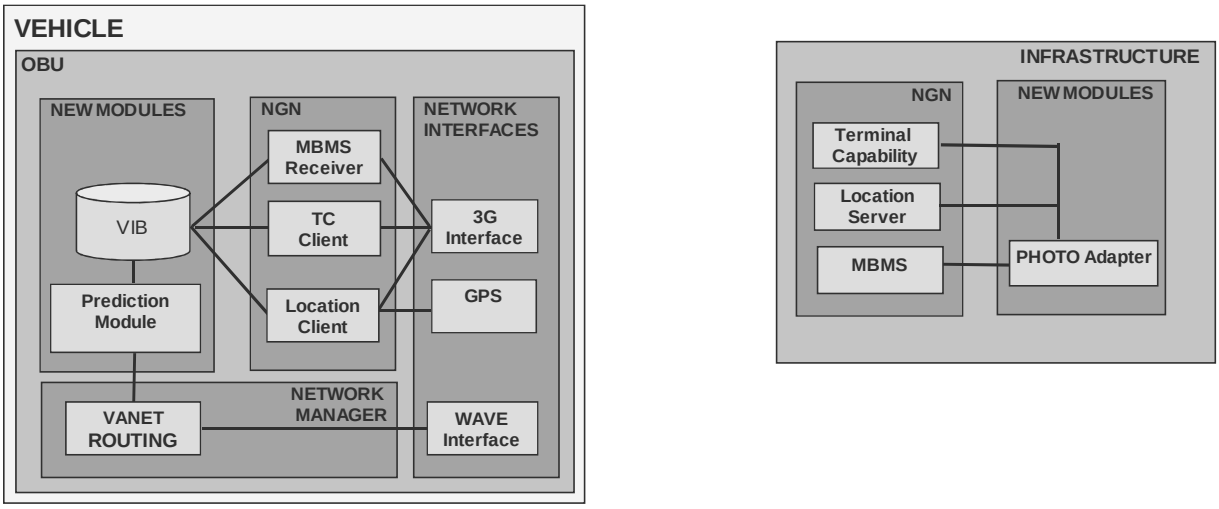


Figure 4.1 VISIONS modules for improving VANET Routing (nueva arch)

4.3.2.1 On board modules

The elements on board are responsible for collecting information of the vehicle, sending it to the server, receiving road situation information and enriching routing mechanisms with the received information.

The hardware modules are a GPS receiver to get the position of the vehicle and two communication interfaces; WAVE (IEEE 802.11p) for the VANET and 3G cellular.

The Vehicle Information Base (VIB) consists of an information repository storing updated data of the nodes in the area where the vehicle is located. The Terminal Capabilities (TC) client obtains a set of attributes from the OBU, stores them and sends them to the TC enabler. These attributes are defined in Table 3.2 and are: Vehicle ID, IMSI, VANET IP Address, 3G IP Address and the GW mode (if the node is acting as a gateway for other vehicles).

The Location client periodically obtains information from the GPS receiver (position, speed, heading and timestamp). This information is stored and sent out to the location server enabler over the user channel of the cellular interface according to the SUPL standard [85].

The MBMS Receiver gets the information about other cars broadcasted by the operator infrastructure and stores it in the VIB. Prediction module is in charge of estimating the position of the nodes when required, based on the position, speed, heading, trajectory from the VIB and time elapsed since the last update. This module has an API responsible for providing the information to VANET routing protocols.

4.3.2.2 Infrastructure modules

These modules, based on the operator enablers, are responsible for collecting data from the vehicles, processing it and sending back efficiently aggregated information to the vehicles inside geographical regions. These regions are defined by the coverage area of one or more radio cells

The Location server collects position, speed, heading and timestamp from the vehicles and stores them in the database. The functionalities are offered through a standardized API as this module is based on the Mobile Location Service defined by OMA [83]. The Terminal Capabilities (TC) enabler is responsible for the collection and storage of the rest of the attributes. This element is based on the standard [43] by the OSA/Parlay and 3GPP groups and data model has been extended for our particular needs.

PHOTO Adapter is a new module, developed explicitly for our solution that accesses Location and TC enablers through their standardized APIs for retrieving information in

order to generate a set of digital files called PHOTOS. Each PHOTO contains all the attributes of the vehicles within a specific region.

MBMS enabler, standardized by the 3GPP group, provides broadcasting capabilities and is used to efficiently distribute PHOTOs to all the vehicles inside regions defined by a set of one or more adjacent cells.

4.4 Performance evaluation

To validate the routing improvements provided by our proposed solution, we have done a simulation-based performance analysis of major VANET routing protocols with and without the information provided by VISIONS. Our goal is not only improving the performance of each routing protocol, but also demonstrating that the proposed idea of using operator's enablers for signalling can certainly be a promising research direction. In fact, we foresee that the performance gains can be even higher when applied to other networking problems such as security (deeply analyzed in Chapter 5), traffic management, etc.

4.4.1 Qualitative analysis

In this section we describe what VISIONS mechanism can provide to the routing protocols in order to enhance the performance.

As we have seen previously, routing protocols use information like position, digital maps and trajectories. Each vehicle can obtain the data of the own vehicle directly consulting the GPS, the cartography and the navigator respectively but is much more difficult to obtain data from other vehicles. Routing protocols have two possibilities to obtain this information:

- Through beacons. This solution increases the overhead in the VANET and each vehicle only knows the data of the vehicles within direct radio range.
- Through a specific request to the destination. This also increases the overhead of the VANET, implies an initial delay and doesn't guarantee the retrieval of all the possible paths.

As we have described in a previous section of this chapter, VISIONS can provide up-to-date relevant information about the nodes within a specific geographic region using an alternative link, not the VANET. This information includes basically IP addresses, GW functionality, position, speed, and heading of all the vehicles in the area.

As an additional improvement, VISIONS could access the navigator of the vehicles, and if it is available, gather and send the selected route to be also distributed in the PHOTOS. This would increase the size of the PHOTO but may be relevant for the improvement of some protocols. This has been considered in the Table 4.1 with an (*).

Now, we describe the added value provided by VISIONS to routing protocols:

- Regarding the group of protocols that uses the location information, VISIONS provides the position of all the nodes in the area. Thus, in GPCR [103] the source can calculate the coordinators by selecting the closer nodes to the junctions without sending any information through the beacons. In CAR [104], the routing algorithm can estimate connectivity to the destination without sending request packets to the destination.
- Protocols using digital maps can access VISIONS information to locate the nodes. Then, GSR [105] will retrieve the density of nodes among different streets obtaining the best list of junctions without the flooding mechanism, reducing considerably the overhead. In A-STAR [106], the protocol will not only estimate the density of vehicles with the bus lines, it will get an accurate density on every street on real time, so it will be more efficient.
- All the protocols based on the Store-Carry-Forward paradigm can optimize the performance by accessing the position, velocity and heading of the rest of the nodes, so the decision of the path to traverse can be taken more accurately without sending any message through the VANET.
- Protocols that use estimated trajectories can select the route not only knowing the rest of the vehicles trajectories (assuming we gather also this info), but their current position, speed and heading, once again without any extra beacon or request packet, reducing overhead and delay.

The table 4.1 shows for every protocol analyzed in the state of the art:

- The information used in the routing decision
- If the protocol uses Store-Carry-Forward (SCF) approaches
- The relevant data that can be provided through VISIONS mechanism attending to the features of the protocols, without sending more information through the beacons or request packages. Note that those fields (Position, speed, heading, trajectories) refer to the info from all the vehicles in the Region *Reg*.
- The qualitative benefits expected.

Protocol	Based on	SCF	Info provided by VISIONS	Benefits
GPCR	Position	No	Position	Delay, overhead, delivery ratio
CAR	Position	No	Position	Delay, overhead
GSR	Digital Map	No	Position	Delay, overhead, delivery ratio
A-STAR	Digital Map	No	Position	Delay, overhead, delivery ratio
SAR	Digital Map	Yes	Position	Delay, overhead, delivery ratio
VADD	Digital Map	Yes	Position, speed, heading, trajectories*	Delay, overhead, delivery ratio
GeOpps	Trajectories	Yes	Position, speed, heading, trajectories*	Delay, overhead, delivery ratio
MoVe	Trajectories	Yes	Position, speed, heading, trajectories*	Delay, overhead, delivery ratio

Table 4.1 Routing protocols and profits using VISIONS

The conclusions of the qualitative analysis are the following:

- All the protocols analyzed benefit from VISIONS, reducing the average delay due to the decrease of the route calculation time.
- All the protocols analyzed benefit from VISIONS, reducing the overhead in the VANET because request packets are not necessary and information broadcasted through the beacons is reduced.
- Most of the protocols analyzed benefit from VISIONS improving their packet delivery ratio due to a better stability of the selected routes.
- Most of the protocols analyzed only require the location (position) of the rest of the nodes to obtain benefit from VISIONS, thus the information collected and broadcasted can be sensibly reduced.

4.4.2 Quantitative analysis

4.4.2.1 Simulation scenario

Our performance analysis is based on the Network Simulator ns-2 (version 2.33) [110], together with Simulation of Urban Mobility (SUMO version 0.98) [111] and Traffic and Network Simulation Environment (TraNS version 1.0) [112].

The scenario we used is depicted in Figure 4.2. Region *Reg* is a 1x1km^2 area with a set of main streets, which have enough traffic density to ensure connectivity among them, and a set of secondary streets with lower traffic density. Vehicles move along different predefined routes at a maximum speed of 50km/h, providing a scenario with different traffic densities. Moreover, the simulations are able to fulfil common traffic situations such as overtakes or stops at intersections, which cause disconnected networks. The map includes 26 streets and 15 junctions. Secondary streets have a variable density of 10% to 50% of the main streets density. We called this parameter Differential Density Ratio (DDR). The main streets density has been fixed to 1/10 vehicles per route per second. That means one vehicle is injected into a route every 10 seconds. This way we check the evolution of the performance difference of densities between main and secondary streets.

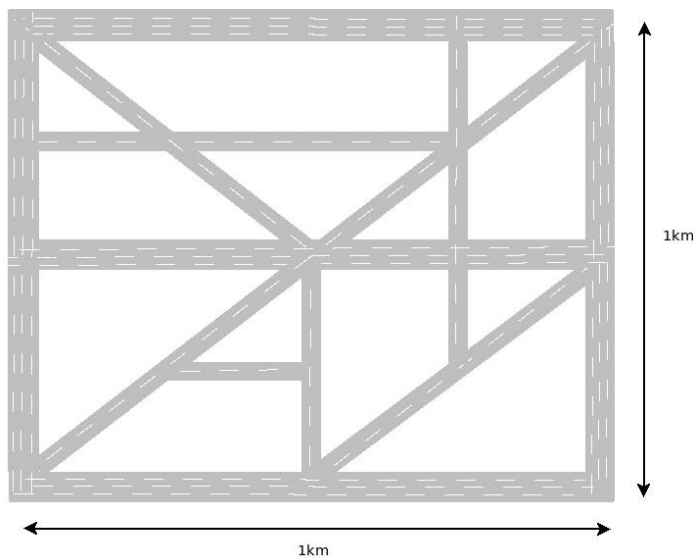


Figure 4.2 Urban scenario simulated

We use the two-ray-ground propagation model. Vehicles employ an 802.11p interface card, implementing the enhanced ns-2 802.11 physical and MAC models. The maximum transmission range is fixed to 250m.

We have simulated 10 independent experiments for each configuration (figures in simulation results section show the average of such experiments). For each experiment there are 100 different random data sources. Each vehicle sends a 512Bytes data message toward a static vehicle located in the centre of the map.

We consider the following configuration for VISIONS:

- Vehicles register their position and heading every 2 seconds, which is also the period that we use of sending PHOTOS from the infrastructure. Freq_Loc = Freq_Pho = 0,5Hz
- Each vehicle in a PHOTO contributes with 39 bytes: *Size_Pho= 39 * Veh_Reg*
- The size of the location update transmitted through the Uplink is 105 bytes. *Size_Loc* = 105
- The size of the location update ACK transmitted through the Downlink is 50 bytes *Size_Loc_ACK* = 50
- The whole Region is covered by 1 Node B: *Reg* = 1x1km2

All evaluated routing protocols (GSR, SAR and A-STAR) employ a beacon interval of 2 seconds. Each message can be kept in the same vehicle for 30 seconds. In existing evaluated protocols path computation is based on assigning a weight in each street and applying Dijkstra's algorithm. To take advantage of VISIONS, the weight of each street is set to $P = d/X$, being X a Boolean variable obtained from the current distribution of vehicles provided by VISIONS and d the distance to the destination. A node sets $X = 0$ if vehicles along a certain street are not connected and $X = 1$, otherwise. Note that traffic is just routed through the VANET and the cellular network is used to provide signalling information to take better routing decisions.

Like in similar MBMS performance evaluation papers [90] we have considered a hexagonal UMTS R6 macrocell with 1000m between centres of adjacent cells. That is, in our case, our scenario fits perfectly into such a macrocell. Given that we want to show that VISIONS is viable, we use a single sector in the cell as well as low rates configurations for both unicast and multicast channels. In particular, for uplink unicast we consider an HSUPA common channel at its lowest rate of 780Kbps (1 single code, spreading factor (SF) of 4, transport block size (TBS) of 7296 bits and transport transmission interval (TTI) of 10 ms). For unicast downlink we consider an HSDPA channel at its lowest rate of 1800 Kbps. For the MBMS channel we consider a single

channel (out of the 16 available ones per cell and per sector) operating at its basic rate of 64 Kbps as in [90] (SF=32, TBS=5120 bits, TTI = 80ms).

We have chosen GSR, SAR and A-STAR protocols for our simulation analysis because they all are designed for VANETS and share a similar philosophy based on the use of digital maps to decide which will be the optimal path. This allows us to clearly quantify and assess the performance benefit of VISIONS over directly comparable protocols. However, VISIONS is also perfectly suited for other protocols in the literature which require real-time traffic information like Greedy Traffic Aware Routing (GYTAR). For further information about these protocols, please refer to [98].

4.4.2.2 Simulation results

In our simulations, we analyze the GSR, SAR and A-STAR protocols with and without VISIONS. The metrics used for the performance comparison are:

- Packet delivery ratio (PDR)
- Packet drops
- End-to-end delay.
- Operator resource consumption

Note that the “V” next to the protocols name in the next figures indicates that this protocol operates in the “enhanced” mode (with VISIONS). For completion, we also include a study of the cellular network load caused by VISIONS. In our evaluation, we compare the performance of the protocols for varying the difference in density between primary and secondary streets with the previously defined DDR parameter. DDR =10 means that secondary streets have only 10% of the rate of vehicles in primary streets. Note that the higher the DDR, the larger the overall density of the scenarios.

Figure 4.3 (data in Table 4.2) illustrates the PDR for the three VANET routing protocols with and without VISIONS. It clearly shows that PDR results are better using VISIONS in all cases. In the best case, VISIONS obtains an increase of 18% in the PDR, which corresponds to almost doubling the number of packets being delivered. This is because the information provided by our solution helps to avoid those streets not having enough density of vehicles. As expected, the difference between a protocol in “default” and “enhanced” modes is reduced as the Differential Density Ratio (DDR) increases because, both primary and secondary streets tend to have connectivity, which means that protocols with and without VISIONS tend to choose the same paths. Note that the Confidence Interval (CI) for a 95% confidence level is approximately 10%.

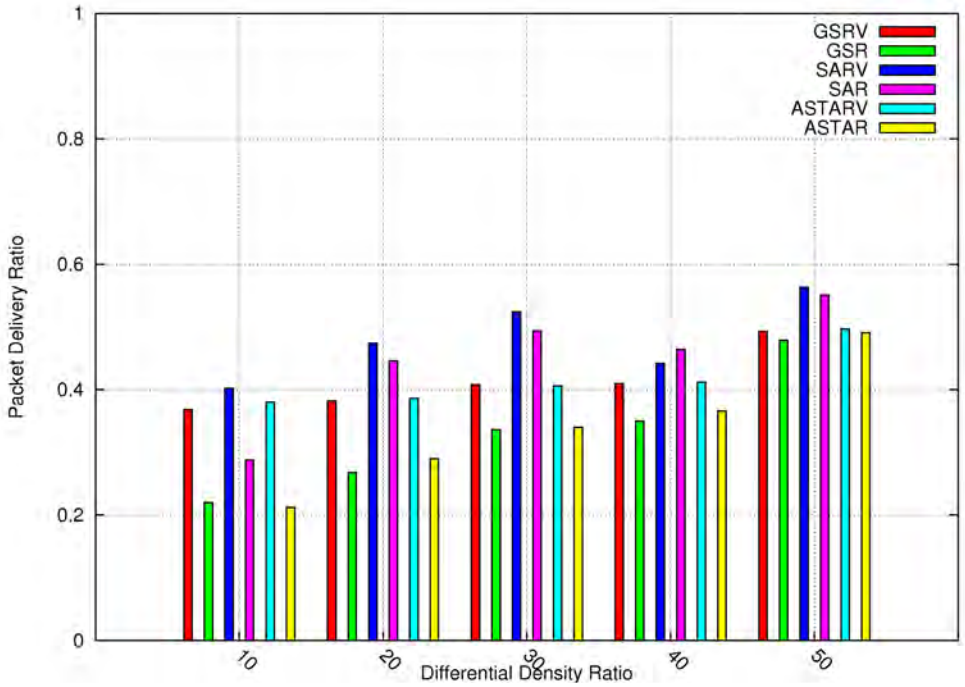


Figure 4.3 Packet delivery ratio

	10	20	30	40	50
GSRV	0,37 (0,03)	0,38 (0,04)	0,41 (0,04)	0,41 (0,04)	0,49 (0,04)
GSR	0,22 (0,02)	0,27 (0,04)	0,34 (0,03)	0,35 (0,05)	0,48 (0,04)
SARV	0,40 (0,03)	0,47 (0,05)	0,52 (0,04)	0,44 (0,03)	0,56 (0,05)
SAR	0,29 (0,05)	0,45 (0,05)	0,49 (0,03)	0,46 (0,05)	0,55 (0,05)
A-STARV	0,38 (0,03)	0,38 (0,04)	0,41 (0,05)	0,41 (0,04)	0,50 (0,04)
A-STAR	0,21 (0,02)	0,29 (0,048)	0,34 (0,03)	0,37 (0,05)	0,49 (0,03)

Table 4.2 Packet delivery ratio (CI for 95% confidence)

Our analysis of the PDR is supported by the causes of drops depicted in Figure 4.7 (data in Table 4.4), which shows that the total number of drops is lower using VISIONS. We classify packet losses as: “Drops MAC” if they are caused by the MAC layer (e.g. collisions); “Drops TTL” if the Time To Live (TTL) of the packet expired; “Drops PEX” if packets expired (PEX) in the buffer before finding a forwarding opportunity (it only make sense in SAR) and “Drops NBV” when a packet reaches a vehicle with no neighbours (NBV) to forward the message to.

The main reason for packet drops without VISIONS is NBV as is shown in Figure 4.4 (data in Table 4.3). However, using VISIONS, losses due to bad path selection are very few, because the information provided in the PHOTO ensures that the selected streets have always connectivity. The increase in the number of drops at the MAC layer is because the VANET protocols tend to choose neighbours that are very close to the OBU's radio range, and these have a higher probability of being lost. That is, the performance of VISIONS is limited by the performance of the protocols themselves. For the reason explained before, as the DDR increases, VANET protocols exhibit very similar performance with and without VISIONS. The Confidence Interval (CI) for a 95% confidence level is approximately 10%.

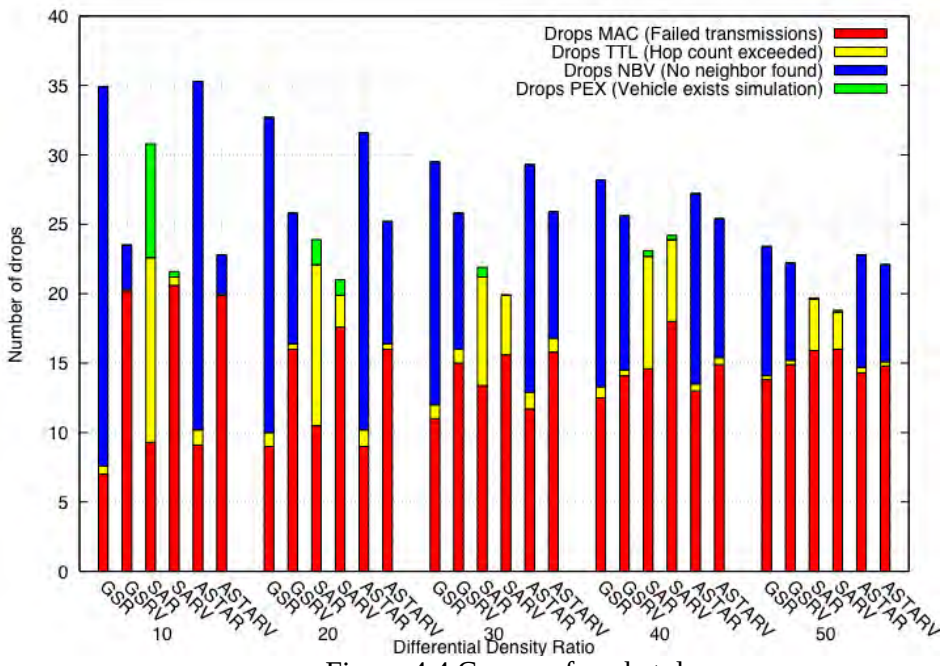


Figure 4.4 Causes of packet drops

	10	20	30	40	50
GSRV	23,70 (1,95)	27,20 (1,92)	25,90 (1,84)	25,60 (1,86)	22,4 (2,15)
GSR	34,90 (1,47)	32,70 (1,82)	29,50 (2,04)	28,20 (2,17)	23,40 (2,22)
SARV	22,80 (2,01)	22,40 (2,24)	20,00 (2,04)	24,2 (1,92)	19,00 (2,23)
SAR	30,80 (2,23)	23,90 (2,53)	21,90 (2,28)	23,10 (2,22)	19,70 (1,78)
A-STARV	22,50 (2,08)	27,00 (1,85)	26,00 (2,45)	25,50 (1,72)	22,3 (2,11)
A-STAR	35,30 (1,61)	31,6 (2,07)	29,30 (1,97)	27,2 (2,23)	22,8 (1,87)

Table 4.3 Total drops (CI for 95% confidence)

Figure 4.5 (data in Table 4.4) shows the end-to-end delay between the sending vehicle and the destination one with traffic being routed through the VANET. We can see that the impact of using VISIONS is only significant in the SAR routing protocol, because it is the only protocol in our analysis that buffers temporarily data messages when a route is not available. SARV considerably reduces the end-to-end delay because connected routes are chosen more often thanks to VISIONS, and therefore there is no need to buffer the packets. Note that the CI for 95% confidence in SAR/SARV are poor (30-60%) which implies uncertainty with DDR>20. The GSR and A-STAR protocols show a very similar delay with and without VISIONS, although it is slightly lower without VISIONS. The reason is that VISIONS tends to choose longer paths to ensure connectivity, which typically implies higher delay to reach the destination.

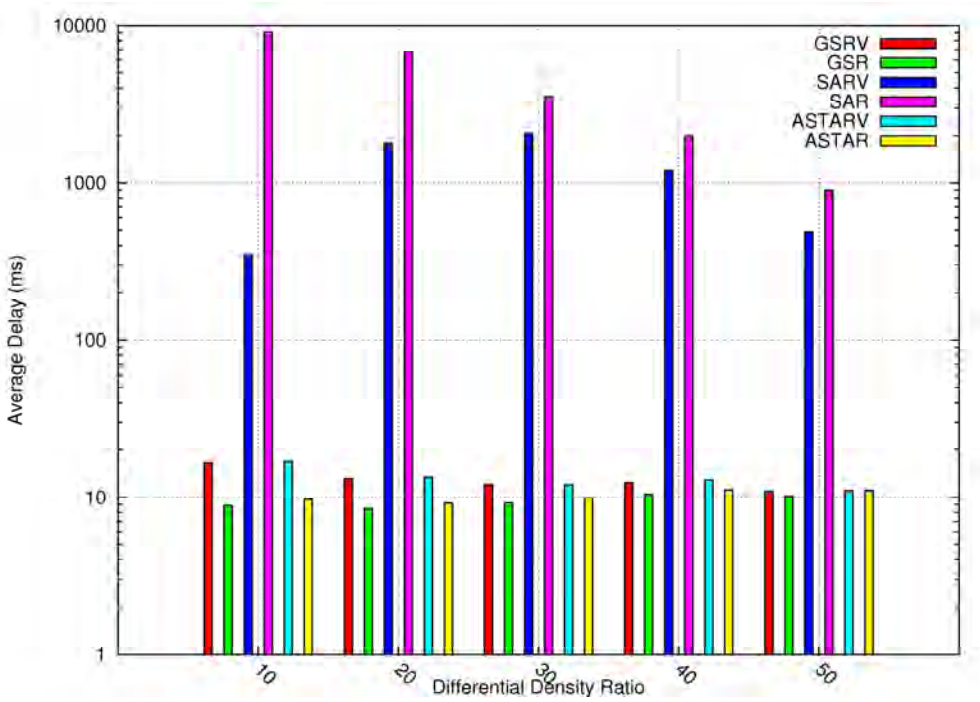


Figure 4.5 Average delay

	10	20	30	40	50
GSRV	16,49 (1,20)	13,15 (1,77)	12,06 (1,06)	12,41 (0,74)	10,92 (0,96)
GSR	8,88 (1,11)	8,51 (0,78)	9,17 (0,77)	10,42 (0,71)	10,15 (0,48)
SARV	348,20 (288,56)	1794,04 (704,04)	2063,92 (922,72)	1198,98 (712,95)	488,06 (277,74)

SAR	9066,44 (3948,53)	6749,17 (2025,0,9)	3544,03 (1002,73)	1995,17 (708,75)	897,78 (426,88)
A-STARV	16,94 (1,22)	13,39 (2,00)	12,01 (1,01)	12,91 (0,92)	11,02 (0,89)
A-STAR	9,67 (1,28)	9,18 (0,76)	9,87 (1,06)	11,17 (1,20)	11,05 (1,04)

Table 4.4 Average Delay (CI for 95% confidence)

Figure 4.6 shows the network load associated to the operation of VISIONS. We measure the utilization of a channel as the total amount of bits successfully delivered over the channel divided by the maximum capacity of the channel.

The figure shows that in our scenarios, VISIONS uses between 42 and 60% of the capacity of one of the 16 MBMS channels available. For the unicast traffic we can see that both for uplink and downlink the utilization of the HSDPA and HSUPA common channels are below a 3% and 15% respectively. In scenarios with more density the traffic is higher. In the case of unicast channels it is because there are more vehicles reporting data. In the case of MBMS it is due to the larger size of the PHOTOs. This utilization can be strongly lowered using higher rates as well as if the macrocell is divided in sectors. According to our formal model:

$$DL < 3\%, \qquad UL < 15\%, \qquad CU_MBMS < 60\%$$

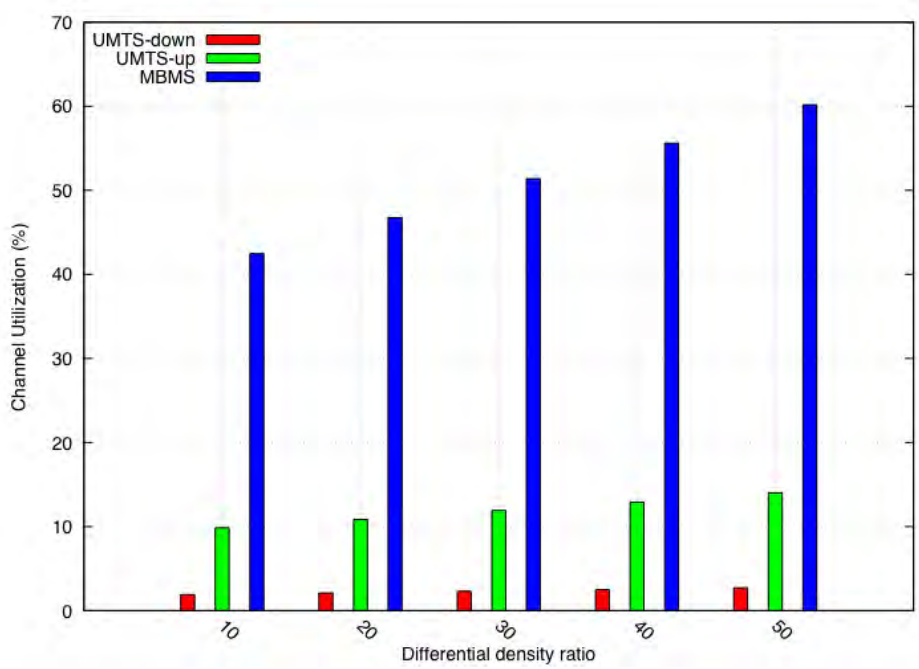


Figure 4.6 Network load due to VISIONS traffic

	10	20	30	40	50
DL	2,1	2,3	2,6	2,8	3,0
UL	9,9	11,0	12,1	13,1	14,2
CU_MBMS	42,4	46,7	51,4	55,6	60,2

Table 4.5 Resource consumption (in %)

In summary, our simulation results show that the proposed solution is able to improve VANET routing performance in urban scenarios, where there are important density differences among streets. In such a scenario, VISIONS increases the packet delivery ratio in all the protocols under analysis and reduces the end-to-end delay in SAR. As our results show, VISIONS is viable in terms of bandwidth consumption even in the low performance configurations of existing HSUPA, HSDPA and MBMS technologies.

The objective in this section has been demonstrating that VISIONS can significantly improve the performance of routing protocols using comprehensive configuration values in VISIONS formal model. Future works will leverage on this results to fine-tune this parameters in order to optimize the efficiency of VISIONS in different traffic scenarios.

4.5 Conclusions

The use of existing mobile telecommunication networks in the context of vehicular communications is quite common nowadays. However, their use is mainly oriented towards the provision of data communication services for vehicles. These networks have evolved very rapidly within the last few years, and nowadays operators have many efficient communication mechanisms that may become very valuable in the vehicular environment, much beyond providing connectivity.

In this context, based on the most relevant operator enablers applicable to the vehicular context, we have designed and evaluated a solution that can help VANET routing protocols take more accurate routing decisions. VISIONS provides a mechanism that automatically collects data from the vehicles (including position, speed and heading) in a certain region, analyzes the information and prepares an aggregated file that is distributed in the geographic areas in which the information is valuable.

The qualitative analysis predicts that routing protocols can optimize the performance by accessing the info of the rest of the nodes provided by VISIONS, so the decision of the path to traverse can be taken more accurately without sending any message through the VANET.

Our simulations in realistic urban scenarios demonstrate that using our solution the routing protocols achieve a higher delivery ratio. In some cases, protocols using VISIONS almost double the number of packets successfully delivered to the destination. It is also probed that VISIONS does not require a lot of bandwidth from the operator network; in our scenarios, the mechanism uses between 42% and 60% of the capacity of one of the 16 MBMS channels available, less than 15% and 3% of the Up Link and Down Link capacities respectively.

In our view, this idea of operator-enabled vehicular communications has a much wider applicability and opens up an interesting research direction. We believe that VISIONS can help other applications in the vehicular environment, which include efficient location service, access control for VANETs and management of certificate revocation lists. In fact, in the next chapter, we study and propose solutions to enhance security in VANETs using VISIONS.

5 VANET SECURITY IMPROVEMENT

5.1 Introduction

In the previous chapter we showed how VISIONS improve the performance of existing routing protocols. As we mentioned our solution can also be used to improve critical security issues in vehicular communications. Although we think that the potential of VISIONS is high in most of the topics related with security (authentication, authorization, privacy, non repudiation, etc) we are going to focus in the problem of efficiently revoking certificates of malicious and compromised nodes.

The objective is making sure that vehicles receive the new list of revoked certificates (CRL) before they can be compromised. Unlike previous solutions, our work revolves around the idea of making use of VISIONS capabilities to improve the efficiency of the distribution of the CRL.

Our proposed solution gathers positions and headings of vehicles using a location service enabler. Based on that information, the infrastructure can decide when and in which areas to broadcast the incremental CRL using any of the existing technologies such as the Multimedia Broadcast Multicast Service (MBMS). Our simulation results show that the proposed solution outperforms existing solutions based on the use of Road-side Units (RSUs) for disseminating CRLs. The improvement in terms of

distribution delay and number of compromised nodes is noticeably better as the density of vehicles decreases.

The remainder of the chapter is organized as follows: In section 2 we describe the related work in the literature. In section 3 we review those modules in the architecture that participate in the process and describe the benefits of using VISIONS to distribute CRLs in those areas where an attack occurs. In section 4 we make a quantitative evaluation of the performance of our solution using the KPIs defined in the requirements section; Window of Vulnerability, effectiveness and resource consumption. In section 5 we summary the conclusions of the Chapter.

5.2 Related Work

For the effective deployment of vehicular communications, a comprehensive set of security mechanisms must be implemented, especially because of the life-critical nature of the vehicular network operation. This is clearly reflected on the IEEE 1609.2 standard [113] for securing Wireless Access in Vehicular Environments (WAVE), whose features are detailed in Table 5.1. The standard addresses the issues of securing V2V communications basically against eavesdropping and spoofing. They propose authenticating vehicles using a Public Key Infrastructure (PKI). In this scheme the most critical entity is the Certification Authority (CA). It is responsible for the generation and management of digital certificates, which are data structures that probe that a node is legitimate.

Identity Management Features of 1609.2/1609.3	Purpose
Classes of Digital Certificates	There are **classes of certificates** identified by the ***'subject_type'*** field which describes what kind of entity owns the certificate. It is used to determine the scope of the certificate and the means of identifying the signer
Securing Transactions	Transactions are exchanges in which multiple messages are sent by one or both peers. Transactions are initiated when a user receives a Provider Service Table (PST) as advertised by a WSA
Securing Messages Structures	A secure message type is a published 1609.2 data structure with these options - ***unsecured message***, ***signed message*** or ***signed WSM***.
Bootstrapping Trust	All WAVE equipment are provisioned with a public key that can be used to validate root certificate updates. ***OBEs*** can generate their own key pairs.
Signed Messages	**ECDSA** signed messages provides authentication for WAVE multicast messages.
Encrypting Message Structures	The ***EncryptedMessage*** type is a subtype of the ***SecuredMessage*** type. A WAVE certificate contains one or two keys for either encryption, verification or both functions.
Provider Service ID (PSID)	**A** signed number that identifies a service provided by an application and announced in the WAVE Service Announcement (WSA) PSID
Certificate Requests	Device provisioning with certificates use either 'push' or 'pull' model. 'Pull Model' is supported with a CSR message type in 1609.2.
Certificate Revocation Lists	1609.2 defines a CRL type. The 'Security Manager' annex describes the CRL functionality on a 1609 device (i.e. OBE)

Table 5.1 Features of 1609.2

The IEEE 1609.2 standard proposes sending certificates containing the public key of the node within the V2V messages. When the destination receives the message checks the validity of the certificate (using the public key of the CA) to decide if it accepts or rejects the message. The certificate of the sender could be invalid for two reasons: the certificate has expired, or has been revoked by the CA.

When a CA revokes a certificate, it is crucial to inform the other participating vehicles of this revocation. The time since a compromised node has been identified until the rest of the vehicles are informed results in a window of vulnerability for these other vehicles.

There are several schemes like Certificate Revocation Tree (CRT) and Online Certificate Status Protocol (OCSP) [114] which are used to propagate revocation. However, the most effective and commonly adopted is based on Certificate Revocation Lists (CRLs). A CRL is a file, created and signed by a CA, which contains serial numbers of certificates that have been revoked. There are two types of CRLs [95]: Base CRLs keep a complete list of revoked certificates while delta CRLs include only those certificates that have been revoked since the last publication of a base CRL.

In the literature we find many proposals for the distribution of revocation information in vehicular networks.

In [62] the objective is achieving a simple and scalable mechanism for the distribution of large CRLs across wide regions by utilizing a very low bandwidth at each RSU. They propose the encoding of CRLs into numerous self-verifiable pieces, so vehicles only get from the RSUs those pieces of the CRL that are not on-board.

In [63] authors proposed a similar solution based on partitions of the CRLs in to pieces and the use of an epidemic V2V communication for distributing CRLs. This solution spreads revocation information very quickly even with a minimal deployment of RSUs.

In [64] a solution based on RSU-aided certificate revocation is presented. Each RSU has the complete and updated base-CRL and it is continuously checking the status of the certificates contained in all the messages broadcasted by passing vehicles. If a certificate has been revoked, the RSU broadcasts a warning message such that approaching vehicles can update their CRLs and avoid communicating with the compromised vehicle.

Another RSU-based solution is proposed by [116]. Nodes who want to transmit through RSUs validate the sender's certificate based on the concept of "freshness". That is, the receiver decides the acceptance or rejection of a message by taking into account the time at which the CA explicitly validated the sender's certificate.

Another line of related works are those in which the structure, size and computational costs of processing the CRLs is reduced.

Authors in [94] proposed the segmentation of an administrative area into several geographic subregions, and the assignment of region-specific certificates to the vehicles. With that assignment, the size of the CRLs in each region is reduced. Of course, this is thought of for scenarios with high spatial locality.

In [117], authors presented solutions for the improvement of the organization, the storage and the distribution of revocation information within CRLs. To reduce the computational cost they use (as proposed by [118]) a Bloom Filter. That is, a probabilistic data structure which requires a small amount of computational overhead when used for checking if a certificate is in the CRL. Another contribution of this work is a lightweight mechanism for exchanging CRL updates, similar to delta-CRLs [119].

The work in [120] proposes a compression technique based on Bloom filter to reduce the overhead of distribution of the CRL called Revocation Using Compressed Certificate Revocation Lists (RC2RL). Authors also present Distributed Revocation Protocol (DRP). When a compromised vehicle is detected and located, its neighbors can work together to temporally revoke its certificate.

While the related work mostly uses infrastructure equipments (RSUs), these works neglect the advantages that operator capabilities (e.g. effective broadcast distribution) can provide to deal with certificate revocation. In this thesis we show how operator-assisted CRL management can be very effective while still benefiting from individual contributions in terms of the reduction of the size of CRLs.

5.3 Certificate revocation improving using VISIONS

We propose a novel CRL distribution scheme based on the use of NGN capabilities offered by network operators. By making use of those functionalities together with the assistance of the infrastructure the CRL distribution service can distribute CRLs reactively (only when and where needed). Thus, vehicles are informed of revocations before they get close to those malicious or compromised nodes with revoked certificates. It results in a lower number of victims in the VANET and a reasonable network overhead compared to existing RSU-based solutions [64], [116].

In this section we describe the solution to efficiently distribute CRLs, with the following assumptions:

- In addition to the usual WAVE (IEEE 802.11p) interface, some vehicles can be equipped with a broadband cellular technology (3G/LTE) interface.

- The coverage of the cellular network is wide, but it may not be complete.
- Vehicles use X.509 digital certificates standardized by IEEE 1609.2 [113].
- Privacy issues are out of the scope of this paper, but our solution is compatible with extended pseudonymous based proposals in the literature [74], [75].
- As in any PKI deployment [121], we assume that the CA is a secure and reliable node.

5.3.1 Benefits

One of the novelties of our solution is that it takes advantage of the cellular connectivity of vehicles not only for data communications but as a signalling interface which can enhance the security of the VANET.

As stated in Chapter 3, to provide an efficient mechanism for disseminating revocation information, we must take into account four different requirements.

First of all, the distribution of the information must be fast to reduce the Window of Vulnerability (WoV) of nodes, which represents the time that a vehicle is vulnerable to certain attack. Secondly, the distribution must be effective. That is, vehicles susceptible of getting in contact with the compromised node must be informed in advance. Thirdly, the mechanism must be efficient in terms of resources required to distribute the CRL. Finally, the processing time of checking if a certificate is in the CRL must be fast and computationally light. In particular, for the latter there are already good solutions (see [117], [120]) to store CRLs in OBUs. Hence, we just benefit from using those schemes and focus our contribution on the efficient distribution of CRLs.

To the best of our knowledge all the solutions in the state of the art, distribute CRLs via RSUs and/or V2V dissemination mechanisms, which leads to an ineffective solution in terms of delay and unguaranteed delivery. This is mainly due to the highly partitioned nature of VANETs. We propose an innovative revocation mechanism based on the use of the Next Generation Networks (NGN) enablers and cellular connectivity from the vehicles. The concept is to use real time information gathered from the vehicles to optimize the distribution of revocation information through Multimedia Broadcast Multicast Service (MBMS) [46].

5.3.2 Architecture

Following the Design of VISIONS detailed in Chapter 3, we define the architecture of our system as illustrated in Figure 5.1. It consists of modules which are part of the OBU and others which are enablers deployed in the infrastructure.

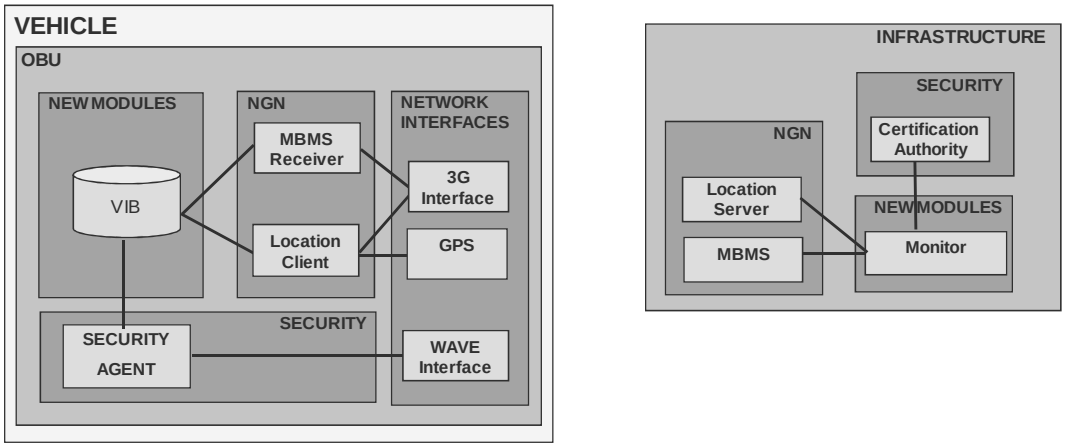


Figure 5.1 VISIONS modules for improving CRL distribution for VANET Security

The first module in the infrastructure is the Location Server. It stores position, speed and heading of the vehicles. This information is periodically gathered from the nodes through their cellular interface. The CA is also located in the infrastructure and it is responsible for the provision and maintenance of the certificates and the CRLs. Finally, the Monitor enabler is the element in the operator network that controls the revocation mechanism, monitors the location of the compromised nodes and manages the efficient dissemination of revocation information using the MBMS service, which is an operator enabler that provides multicast/broadcast communication capabilities.

On board the vehicles we can find a GPS receiver to get the position of the node. In addition, vehicles are equipped with a cellular network interface and a WAVE (IEEE 802.11p) interface for VANET communications. The Vehicle Information Base (VIB) consists of an information repository, which offers an API that the security module can use to get such data about the vehicle, VANET neighbours, etc. The Location client periodically obtains information from the GPS receiver (position, speed, heading and timestamp), stores it in the VIB and sends it to the location server using the cellular interface. The MBMS Receiver gets the CRLs broadcasted by the operator infrastructure and stores them in the VIB. The Security module is in charge of handling certificates and keys, ciphering/deciphering, attack detection and notification, etc.

5.4 Performance evaluation

To validate our proposal we provide two different studies. The first one focuses on the distribution of delta-CRLs and compares the performance of VISIONS against an RSU-based approach. The second one provides an analytical analysis of the Base-CRLs in a Pan-European vehicular network.

5.4.1 Delta-CRL distribution simulation

In order to validate the improvement of our solution, we have simulated a number of experiments using NS-2 [110] version 2.33 and SUMO [111] for the generation of traffic patterns. The simulations focus on the distribution of delta-CRLs assuming that every node has just received a base CRL. Once a malicious node is detected, those vehicles that identify the attack send a notification message to the CA announcing the identity of the malicious node. Afterwards, the CA generates the delta-CRL and finally broadcasts it to the nodes of the road.

We have compared the performance of our solution using VISIONS against an approach of the distribution of delta-CRLs based on RSUs. Both scenarios are detailed below in this section.

As we demonstrated in Chapter 3, the size of the delta CRL is 44 bytes for its header, 130 bits for the CA signature and 9 bytes per revoked certificate. Thus, the size of a delta-CRL reporting only one new revoked certificate is 183 bytes.

In both cases, we use a highway with a length of 4km and two lines of 4m in each direction with different speeds (50 and 80km/h). We have simulated different vehicle densities ranging between 1/75, and 1/5 vehicle/s per line. The number of malicious nodes has been varied between 5 and 15. We have simulated 20 independent experiments for each configuration (figures in simulation results section show the average of such experiments).

With respect to the performance, we align the requirements with the following metrics:

1. To quantify the Window of Vulnerability (WoV), we evaluate the time elapsed since a notification of a malicious node is sent from the CA until a node receives the delta-CRL.
2. For evaluating the effectiveness of the distribution must be effective we provide the number of victims of the malicious nodes.
3. To be efficient in terms of resources required we provide the resource consumption associated to UMTS channels: *UL, DL and CU_MBMS.*

5.4.1.1 VISIONS scenario

In this scenario, according to our architecture, we use an MBMS channel to distribute CRLs. For simulating this, we consider a single Node-B covering the whole highway (best case) and connected to the CA responsible for receiving the notifications and generating delta-CRLs. We consider the following configuration for VISIONS:

- Freq_Loc= 0,2 Hz
- Freq_Delta_CRL =0,016Hz
- *Size_Loc* = 105 bytes
- *Size_Loc_ACK* = 50 bytes
- The whole D-Zone is covered by 1 Node B: D-Zone = 4.000m x 20m
- Size_Delta_CRL= 183 bytes

For our proposal, the downlink capacity of UMTS is 1,8 Mbps using an HSDPA channel (most basic QPSK 1/4 coding rate, 15 codes, 1 sector, 5MHz bearer). The uplink capacity is 730Kbps using a common HSUPA channel (1 code SF=4 TTI = 10 ms, TBS = 7296 bits). And finally, for MBMS the capacity we use a single 64 kbps channel (out of the 16 available ones per cell and per sector) using SF=32, TBS=5120 bits, TTI = 40ms. VISIONS scenario is depicted in Figure 5.2.

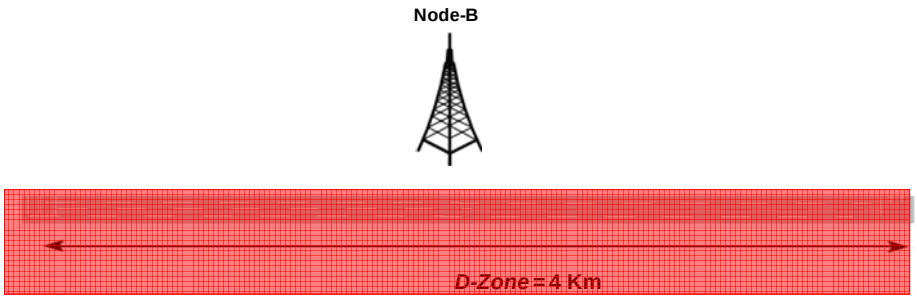


Figure 5.2 VISIONS scenario

5.4.1.2 RSU-based scenario

For the RSU-based scenario we have placed two RSUs, in the first and third kilometres, to cover the same 4KM road. Both RSUs are connected to a special fixed node that takes the role of the CA receiving the notifications and generating delta-CRLs. The RSU coverage range is fixed at 250 m. We use the two-ray-ground propagation model and 802.11p as the WAVE interface module by implementing the

enhanced ns-2 802.11 physical and MAC models. The frequency of the beacons is 2Hz. RSU-based scenario is depicted in Figure 5.3.

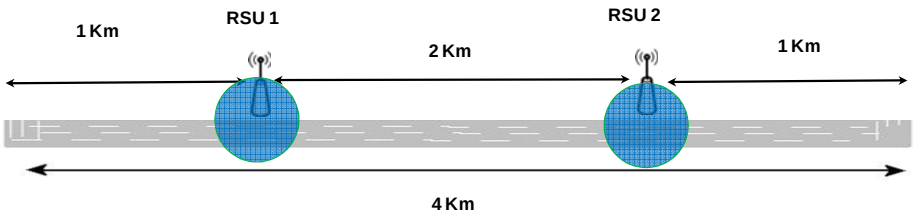


Figure 5.3 RSU-based scenario

5.4.1.3 Simulation results

Figure 5.4 (data in Table 5.2) shows the average delay in receiving the new delta-CRLs as a metric to evaluate WoV. For the same reason as above, for higher density the VANET is more connected. Thus, in the RSUs based scheme better connectivity means lower distribution delays. As density decreases many disconnected nodes must wait until they are by an RSU to get the updated CRL. This increases pretty much the delay. The delay in our proposal only depends on the time slot rate used by MBMS. Thus, the delay is constant in our simulations.

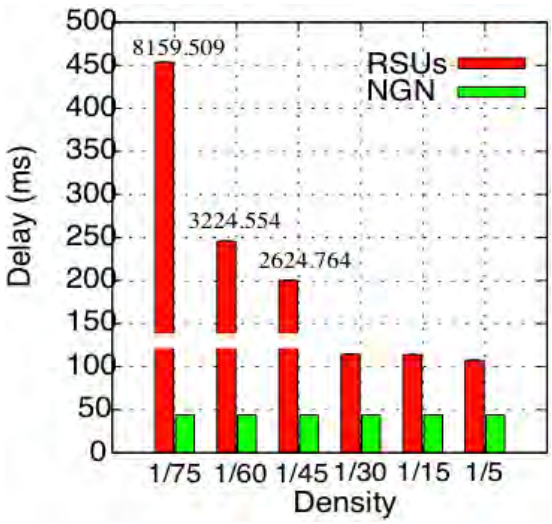


Figure 5.4 Average delay in receiving the new delta-CRLs

	1/75	1/60	1/45	1/30	1/15	1/5
RSUs	8159,51 (1372,12)	3224,55 (946,39)	2624,76 (744,43)	114,36 (44,28)	113,90 (49,72)	107,31 (45,16)
NGN	43,99	43,99	43,99	43,99	43,99	43,99

Table 5.2 Average delay (ms) in receiving the new delta-CRLs (95% confidence interval)

Figure 5.5 (data in Table 5.3) shows the number of victims with both schemes considering 10 malicious nodes and different densities. We consider a node to be a victim when it gets into the radio range of a malicious node without having received the delta-CRL including its certificate. We can see that our proposal obtains a better performance than the RSU-based scheme. The lower the density of vehicles the bigger is the advantage of our solution. This is due to the fact that with lower densities the networks tend to be more partitioned. Thus, the flooding initiated by the RSU cannot reach all vehicles.

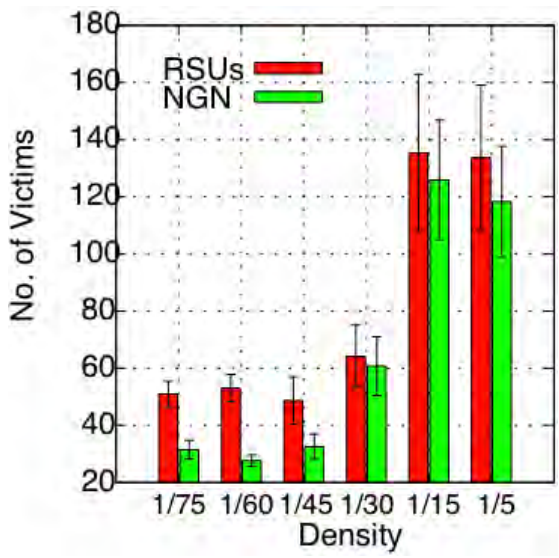


Figure 5.5 Number of victims

	1/75	1/60	1/45	1/30	1/15	1/5
RSUs	51,00 (4,49)	53,05 (4,72)	48,70 (8,32)	64,20 (10,92)	135,35 (27,53)	133,75 (25,39)
NGN	31,5 (3,28)	27,65 (2,02)	32,55 (4,37)	60,70 (10,27)	125,90 (20,94)	118,3 (19,40)

Table 5.3 Number of victims (95% confidence interval)

Figure 5.6 (data in Table 5.4) measures the utilization of the cellular networks. Here "Uplink" and "Downlink" refer to unicast UMTS channel usage (*CU_UL* and *CU_DL*) while MBMS refers to the use of the downlink MBMS multicast channel (CU_MBMS). In all cases we can see that our solution has a very small channel usage (less than 5%). *CU_MBMS* is similar for all densities (approximately 3%). For *CU_UL* and *CU_DL* the consumption is incremented for higher densities because there are more vehicles updating information.

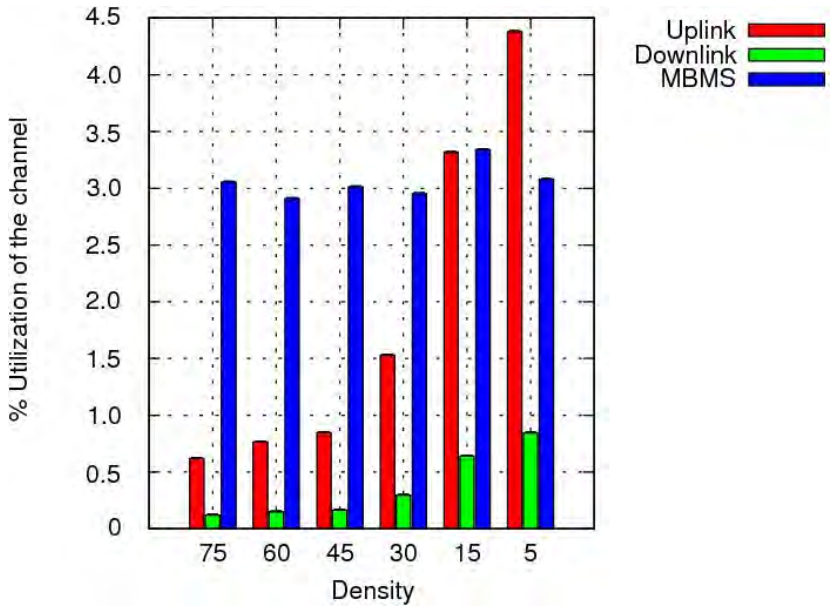


Figure 5.6 Utilization of the channel percentage

	1/75	1/60	1/45	1/30	1/15	1/5
Uplink	0,6185	0,7648	0,8482	1,5277	3,3159	4,3781
Downlink	0,1194	0,1477	0,1638	0,2950	0,6404	0,8455
MBMS	3,0535	2,9081	3,0125	2,952	3,3398	3,078

Table 5.4 Utilization of the channel percentage (95% confidence interval)

5.4.2 Base-CRL analytical study

For determining the size of a base CRL in regions we make the following assumptions, similar to the ones proposed by [94] and [96]. We consider a European VANET with N=200 million vehicles, T=1 year as the lifetime of a legitimate certificate and p=5%=0,05 the revocation rate per year. As certificates expire, we also consider that a revoked certificate will remain in a CRL for approximately half its lifetime. Thus the number *n* of certificates to be included in the base-CRL in any moment and can be calculated by:

*n = (N*p*T)/2* = (200 10exp6 * 5. 10exp-2) /2 = 5 10 exp6 certificates

If we partition the whole VANET into Z=10.000 AAs, *Attackers_AA=n/10.000*

Then, following our formal model the average size of base-CRL in each region is:

Size_Base_CRL = 44 + (Attackers_AA) 9 + 130* = 44 + 5 10exp6*(9)/10exp4 + 130

Size_Base_CRL = 4,5 Kbytes.

As the WoV is effectively reduced by the Delta-CRL distribution in D-Zones, the distribution of a new Base CRL is only needed to optimize the resource consumption because it eliminates all the Delta CRLs distribution within the AA. *Freq_Base_CRL* basically depends on the number of malicious nodes identified in the AA since the last Base CRL distributed, this means that it can be significantly lower than *Freq_Delta_CRL.*

With this analysis we conclude that the size base CRLs with our solution are appropriate for disseminating them through the cellular channel. The size is also appropriate for the reduction of computational and temporal costs for processing the CRL in the OBUs. These costs are not significant when using Bloom filter [100] to store certificate identifiers. [117] and [120] provide exhaustive quantitative and qualitative analysis about the performance of this technique in OBUs.

5.5 Conclusions

In this chapter we propose a CRL dissemination mechanism that reduces the time for informing the vehicles of malicious nodes, decreasing significantly the window of vulnerability by using an operator enabler to broadcast fresh revocation information only to specific geographic areas using MBMS and a cellular channel. It also optimizes the processing time of the CRLs in each vehicle because the base-CRLs only includes the identifiers of the compromised nodes in an area, not all of the network.

Our solution considerably reduces the consumption of processing resources as the node always has an updated and reduced list of revoked certificates, including only those within the subarea it is passing through. When the OBU receives a message from a new node, it can easily go through this reduced list to know if the sender has a valid certificate, so it can trust the information. The delay between the detection of the attack and the moment when all the nodes are informed is also dramatically reduced from those solutions in which vehicles must communicate with a RSU to obtain an updated CRL.

Other important benefit of our proposal is that it dispenses two expensive elements in the vehicle and in the infrastructure in terms of cost and deployment; a tamper proof device and the RSUs. Tamper proof device is not needed as every vehicle can obtain its own certificate through a secure channel and if a malicious user modifies it and misbehaves, the certificate will be immediately revoked and won´t be able to communicate with other nodes of the VANET. RSUs are not necessary to transmit the CRL as every vehicle can communicate with the CA directly and do not need any unit to act as a gateway.

The usage of the cellular channel is very low, making the proposed solution very reasonable in existing deployments. The delay between the detection of the attack and the moment when vulnerable nodes are informed is also reduced compared to solutions in which vehicles must communicate with a RSU to obtain an updated CRL. The reduction is particularly important in scenarios with lower densities, resulting also in a reduction in terms of the number of compromised nodes.

The objective in this Chapter has been to probe that VISIONS can significantly improve the performance of CRL distribution using comprehensive configuration values in VISIONS formal model. Future works will leverage on this results to fine-tune this parameters in order to optimize the efficiency of VISIONS in urban and interurban scenarios.

After demonstrating how VISIONS improves crucial aspects of the VANETS like routing and security; in the next Chapter we will explore how can also act as a reliable platform for service delivery in vehicular environments.

6 A SERVICE BASED ON VISIONS

6.1 Introduction

As we have seen in previous chapters, using VISIONS to assist vehicular communications can provide performance benefits. However, the potential of VISIONs goes beyond improving the intrinsic challenges of vehicular communications. It is a valid framework for the effective deployment of real services. In this chapter, we propose an innovative Social Network service for in-vehicle use based on our solution. As OBUs and Car PCs are still not widely adopted, we choose only the Smartphone as the on-board equipment in which the front-end of the service runs.

Social Networks are one of the latest revolutions in networking, allowing users with common interests to stay connected and exchange information. They have enjoyed great success not only for traditional Internet users, but also for mobile users. Recent efforts are also being done to make social networks available within vehicles. However, to exploit social networks at their full potential in a vehicular context a number of technical challenges and design issues need to be faced. In this chapter, we analyze those challenges and present an innovative solution for providing Social services on

our solution (VISIONS). To demonstrate the viability of the proposed scheme we present a social network service called Drive and Share (DaS) which offers relevant information to vehicles using our proposed architecture.

Social networking is the integration of individuals into groups based on interests, profiles, etc. It has recently gained a lot of interest, especially for on-line members. Social Networks (SN) are mainly oriented to meet other people and to share first-hand information or experiences about common interests. As human interaction facilities, they also allow developing friendships and professional alliances. SN have provoked a real revolution in the way people provide and consume information, converting SN in one of the most important tendencies in today's society.

The basic human need to connect and share information is the basis for well-known success cases such as Facebook, MySpace, and Twitter (with 400, 130 and 75 million registered users respectively in January 2010 [123]). They are so popular nowadays, that users want to use them anytime and everywhere regardless of their communication device. In fact, SN have been recently ported to mobile devices with a great acceptance from users and high penetration rates. Their use within vehicles has been identified [124] as one of the more interesting and with higher potential for the application of these technologies. However, this scenario brings up a number of technical challenges that need to be tackled.

In this chapter, we shed some light onto the problem of providing efficient support for the provision of social network services in the vehicular environment. In particular, we study and compare the current approaches to deliver social networks to vehicles environments, analyzing the type of information and the mechanisms they use to share it. We also highlight the main technical challenges and design issues, giving a number of design guidelines to face them and proposing a novel approach based on VISIONS. As a proof of concept, we present an innovative social network application for the vehicular context called DaS, which follows our design principles.

The remainder of this Chapter is organized as follows: Section 2 analyzes existing efforts to bring social networks to vehicular environments. Section 3 describes the main technical challenges and provides design guidelines to build social network solutions. Section 4 presents a comparative study of existing and emerging application platforms. We present the Drive and Share application based on VISIONS in section 5. Finally, in section 6 we summarize our lessons learned and provide some conclusions of our work.

6.2 Social networks in vehicular scenarios

Social networks have a lot of potential for vehicular scenarios. Vehicles are the third place, after home and office, where a regular citizen spends more time daily. Sharing information while driving can enable a number of extremely useful applications not only for drivers of private vehicles but also for public transportation, road authorities and transportation companies.

There is a lot of valuable information on board vehicles that can be posted and shared with other users. It includes personal information (including location, destination, recommendations, voice notes or even pictures), traffic information (accidents, roadwork, congestion, road systems failures) and vehicle information gathered through on board sensors (icy/slippery roads, heavy rain/snow, fog, vehicle failures or even legal documentation). In the case of electric vehicles, it can also include the battery level, allowing electric companies to efficiently manage the distribution of energy and suggesting drivers with the optimal charging point.

With this information, service providers can offer an extended portfolio of applications. For private drivers we point out two relevant ones: calculating accurately the best route for reaching the destination and searching the most rated/recommended places (gas stations, parkings, restaurants, hotels, stores, villages, monuments and routes). Authorities can also benefit managing efficiently the traffic, sparing it accordingly to the load of the roads (current and future, as they can gather the destination of the vehicles and estimate which roads would suffer congestion). These all represent a perfect opportunity for SN to break into the transportation sector.

The main trend to make social networks available for mobile users is Mobile Social Software (MoSoSo), also known as Location-based Mobile Social networks (LoMoSo). It is a class of mobile applications whose scope is to support social interaction among interconnected mobile users. The concept is to consider the location and time dimension to share information and generic architectures for this purpose have been proposed [125]. They enable users to find one another, in particular vicinity, for data sharing. However, vehicular environments have a larger application domain that goes beyond this basic interaction scheme.

Some companies have recently realised of the potential of social networks in the vehicle and are currently adding basic social features to navigation applications and services. From the applications found during our study and listed in Table 6.1, we point out two relevant ones; Aha mobile [126] and Waze [127]. Aha is an application that adapts web content, social messages and traffic reports to the vehicle environment. Basically it is an interactive personalized radio that converts the web contents and messages from existing social networks into audio streaming. Waze is an application

for the exchange of traffic information within a community of drivers. Using this application, drivers can report traffic incidents, so other drivers can benefit from the information on their routes in real time. The information posted is geographically referenced and it is only valid for a period of time, depending on the severity of the incident. A post can include pictures or pre-recorded voice notes.

	Personal Info	Traffic Info	Photos	Voice Notes	Integration with other SN
Navigon	Location, destination				Facebook, Twitter
ALK Technologies	Location				Facebook
Telmap & GyPSii	Location, recommendations, events, news, advertisement		Yes	Yes	Facebook, Twitter
Aha Mobile	Location, destination, recommendations, events, songs	Congestion		Yes	Facebook, Twitter
Waze	Comments	Hazards, accidents, speed cams, map inconsistencies congestion, parkings.	Yes	Yes	Facebook, Twitter

Table 6.1 Classification of social services for vehicular environments

As we can see in Table 6.1, despite their potential, there are still few relevant initiatives that bring social services to vehicles. Most of them consist in adding existing social networks (mainly Facebook and Twitter) APIs in order to post location and route information to their contacts. The more advanced ones are focused on sharing personal content or traffic information (but not both) and none of them automatically posts information generated by on board sensors without human intervention. We also miss the possibility of rating and enriching previously posted content, which should be an interesting feature in order to validate traffic incidents reported by other users. Exploiting SN at their full extent in vehicular scenarios poses a number of technical challenges that we analyze in the following section.

6.3 Challenges and Design issues

There are diverse reasons for the lack of successful social services in vehicular environments. In this subsection we identify those aspects related with technical challenges and design issues and propose specific solutions to address all of them.

1. Effective content generation. Unlike traditional SN users, drivers are concentrated on the road. There is no clear incentive for them to report or share information while driving. Thus, posting mechanisms must be extremely easy and efficient, requiring minimal or even no intervention. In many countries drivers cannot manipulate external devices due to regulatory aspects. This leads to challenging requirements for Human-Machine Interface (HMI) [128]. These are not properly addressed by existing solutions, where drivers use the same mechanism to post information as if they were using laptops in classical environments like home or office. Our recommendation is to optimize the application interface, avoiding as much as possible touching the screen, presenting the information according to the size of the screen, using speech technologies for command recognition and to communicate to drivers about incoming information.
2. Large time to market. Most car manufacturers are moving towards the vision of networked vehicles and vehicles equipped with on-board computers. However, even if they are able to produce those cars nowadays, there is still some time required until all vehicles in our roads are equipped with that technology. So, social services and application developers must find alternatives to make these applications available before then, by including their software in Personal Navigator Devices (PND), Smartphones, etc. In particular, we promote the use of Smartphones because they have already a high penetration rate, and many of their features (location, touch screen, advanced connectivity, remotely upgradeable, etc) are relevant for the vehicular scenario.
3. Heterogeneity of application devices. The wide variety of existing devices with low penetration rate is other relevant aspect that is slowing down the massive exploitation of services on board. We can classify these devices basically on: External Car PCs, embedded On Board Units (OBU), Personal Navigator Devices (PND), Tablet PCs and Smartphones. Each of these categories includes several operating systems, programming languages and hardware features. Thus the cost of migrating a service to more than one device is significant for service providers and final users are often forced to choose a device according to the kind of services offered (i.e. navigation, multimedia content streaming, fleet management, etc). Taking into account that services should be brought to several platforms, to reduce the migration costs our approach is to extend the use of standardized facilities like Location Server, AAA or Identity Provider.
4. Privacy and anonymity. For some services drivers do not want to be tracked by external bodies even when they share information with social networks. Users should be able to easily configure the desired level of anonymity and privacy for the

gathered and shared data. This includes allowing tracking through anonymization services that hide the real identity of drivers while still being useful for statistical data processing. These issues are still an important requirement [129] not properly covered by existing services. Delegating the management of user's identity to IMS enablers like Identity Provider (IdP) [35] can overcome this problem.

5. Usage of network resources. As the service will be accessed by many mobile users sending and receiving information, the design must take into account the usage of network resources. Moreover, many basic functionalities like getting vehicle location, authentication and information delivery are often developed exclusively for each service. This duplicates effort, information and makes expensive the development and deployment of services. To solve this issue we propose implementing these functionalities through VISIONS, reducing the consumption of network resources and facilitating component reuse. It also minimizes the impact of migrating services to several application platforms or devices, as the logic is basically in the network.

Summing it up, our recommendations to provide social networking in vehicular scenarios are as follows. First of all, make use of standardized APIs and existing operator enablers to develop applications. This simplifies and reduces the development cost of your services because many of the required functionalities are already implemented and made available by operators. Second, put as much intelligence in the network. This allows services and applications to be easily ported and maintained across multiple devices. Third of all, study carefully the HMI aspects of your application to make it usable and accepted. Finally, select proper application devices for your development so that you can reduce time to market. This last aspect is very important and requires a deep knowledge and hands-on experience on the different platforms. We analyze and share our experiences on that in the next section.

6.4 Application Platforms

We left behind those times when the only solution for bringing ICT services to vehicles was using OBUs and Car PCs. Due to the emergence of a variety of mobile and nomadic devices; nowadays there are multiple possibilities to deploy services in vehicular scenarios.

Diversification among different platforms usually involves a big effort to migrate applications. To face these difficulties we propose to base applications on remote services as much as possible, thanks to IMS and NGN capabilities. This reduces the migration costs. Furthermore, we think that there are ways to avoid many of the

specific details of implementations for the different platforms, thanks to the presence of advanced browsers with HTML5 [131] support in almost all those platforms. Given the fact that applications for vehicular environments access intensively to devices, one could argue whether browser based development is an option for this environment. However, with the appearance of JavaScript libraries like PhoneGap [55] that abstract the access to devices, wrapping the specific libraries for each platform, this is certainly a viable option. In addition, relevant standardization organizations such as W3C [82] are also moving in that direction. With the growing demand for this kind of functionality in mobile browsers we will likely see complete integrated stacks to access all devices in a standard way. This line of work is not suitable for time-critical (safety related) services but it is certainly interesting for bringing SN to vehicles.

Due to the enormous fragmentation of this market, several aspects should be taken into account in order to select the most appropriate application platform. The most relevant aspects that we analyze in Table 6.2 are maturity of the technology and penetration rate, developing and deployment costs for several platforms, built-in technical capabilities, and openness.

APPLICATION PLATFORMS	Integration	Open platform	Time to market	Technical capabilities						
	Indicates whether the platform have been deployed (or it's planned to be deployed) on **smartphones** or on **in-vehicle** equipment.	A platform increases its value when provides some degree of openness in terms of: • be based on **standards** • allowing the development of **third party applications** • **open source code**, and allowing its modification	• **Mature:** available some time ago • **Recent:** available, but growing • **Short term**: available in the next year • **Medium-Long term**: not available in the next two years	**Accelerometer**	**GPS**	**Multitouch screen**	**Speech technology**	**Customizable**	**Cellular communications**	**Multiple hardware platforms**
iPhone OS	Smartphone	Third party applications	Mature	Yes	Yes	Yes	Yes		Yes	
Android	Smartphone and in-vehicle	Open source	Recent	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Microsoft Auto	In-vehicle	Close source	Recent	Yes	Yes		Yes		Yes	
MeeGo	Smartphone and in-vehicle	Open source	Short term	Yes	Yes	Yes		Yes	Yes	Yes
European ITS Reference Platform	In-vehicle	Standard	Long term	Yes	Yes				Yes	Yes

Table 6.2 Comparison of application platforms

The comparison of application platforms in Table 6.2 shows that there are many options that satisfy SN requirements for being used within vehicles. Maturity, time to market and capabilities indicators have been identified as the most relevant. So, in that regard the use of Smartphones is a viable alternative. The presence of a significant number of these devices today and decreasing cellular bandwidth costs allows deploying interesting and affordable applications for final users. On the contrary, the technical and normative related difficulties to integrate specific on-board equipment with car manufactures, makes the option of in-vehicle devices not recommendable for efficiently bring SN to the vehicular market in a short term.

6.5 Drive and Share

To support our recommendations and get hands-on experience we have developed a social network application called Drive and Share (DaS) for one of the most relevant application platforms nowadays: iOS [130]. DaS allows drivers and passengers to efficiently exchange traffic and personal information on board vehicles. In this section we present the overall architectural design of the service including the functionality of every component, and explain the operation of the overall service. We conclude with a usability analysis of the service.

6.5.1 Service features

DaS provides a framework for generating and distributing traffic and personal information for vehicles in real time. With traffic information users are informed of traffic events in their routes and their navigation systems can choose the best path accurately. It can estimate the travel time of different alternatives calculated with real-time data gathered from the vehicles that are currently moving along those roads. If an incident in the route happens, the service will automatically inform the driver and the route will be recalculated. DaS also allows drivers and passengers to post personal information (including pictures and voice notes), recommend places and associate this information to a route or a travel.

One of the most innovative features in DaS is that it allows not only drivers and passengers but also vehicles to participate in the social network autonomously. Vehicles themselves can generate and automatically post information. This is possible because vehicle behaviour is continuously monitored by tracking functionalities, so they can gather real time situation of roads without human intervention. Once the content is generated, enablers on the infrastructure distribute efficiently the information to other vehicles and servers. After detecting an incident, DaS can continuously track the

evolution by gathering information from vehicles in the area. This allows us to accurately determinate the duration of the incident notification and/or to reduce the area of influence of the incident.

When dealing with automatically posted traffic information, the system calculates the target receivers of the information, including not only users on the same road but those whose routes trespass the affected zone. For personal information DaS offers the possibility of posting it on specific groups of the two main social networks: Facebook and Twitter.

DaS also allows users who receive information to easily rate it, voting the relevance of the notification in case of traffic incidents or the interest aroused if it is personal information. Moreover, users can enrich the information by adding new multimedia content (voice notes, pictures, songs or movie clips).

6.5.2 Architectural design

In this subsection, we describe the architecture of the back-end of the service, and the functionalities carried out by each element. The proposed architecture follows the design principles of VISIONS described in chapter 3 in the operator infrastructure and while bringing a new approach by using the smartphone instead an OBU in the vehicle. As we recommended before, most of the intelligence of the service in the operator infrastructure, which offers standardized APIs and enablers to the application.

6.5.2.1 Infrastructure modules

These modules depicted in Figure 6.1, mainly based on operator enablers, represent the logic of the service. They receive data from application devices (automatically or through posting mechanisms), process the information and send it back efficiently to those users (vehicles, drivers or passengers) susceptible to be interested in the information. That interest is determined by considering the relation with the sender or the geographical location (i.e. close to an incident detected). These modules are grouped into two categories: the VISIONS enablers and the specific backend of the DaS service.

VISIONS enablers offer through standardized protocols and open APIs many facilities and functions needed for social services in mobile environments. The location server is based on the Mobile Location Server [122] standardized by OMA and stores real time information from the vehicles (location, speed, heading) gathered automatically by the system. The Identity Provider (IdP) [35] module is based on the enabler standardized

by Liberty Alliance and it is in charge of authentication and authorization functions, covering privacy issues. Once the user has been authenticated, IdP sends some credentials to the DaS server, avoiding third parties to obtain the real identity and track the user movements. MBMS [46] enabler, standardized by 3GPP, provides broadcasting and multicasting capabilities that are used to efficiently distribute new content posted to those vehicles interested in the information.

The Location Server generates the mobility patterns for the roads in different hours automatically. This mobility patterns can be continuously updated (improved) and queried at any time. Due to the design of the enabler, VISIONS also tracks the location and trajectory of the vehicles in real time fuelling the database of the Location Server.

DaS Backend provides all the specific logic of the service not covered by VISIONS. Incident Detection is in charge of matching the current vehicle density and speeds with the estimated mobility patterns of the roads. If any anomalous behaviour is detected, it triggers an incident posting mechanism. The DaS Backend also collects the information generated by drivers and passengers. The Geographic Posting module is in charge of referencing the information geographically, and selecting the target group of vehicles that will be informed through MBMS. The reference mechanism is done using inverse geocoding functionalities standardized by Open Geospatial Consortium (OGC) [115]. When dealing with traffic information, the Incident Revocation module monitors the situation accessing the Location Server, and stops disseminating the information to the geographic area when the incident disappears.

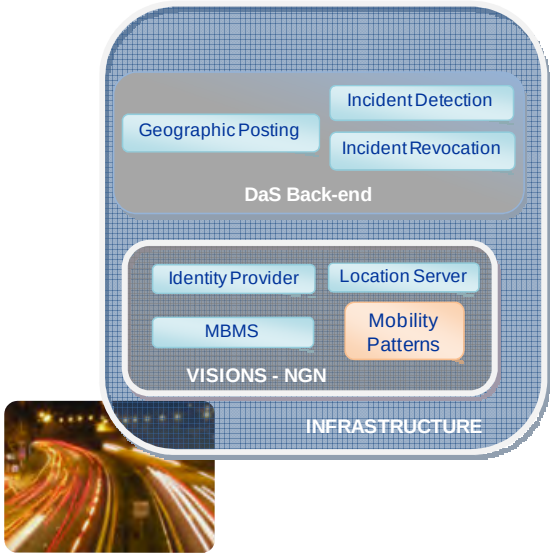


Figure 6.1 Infrastructure modules

6.5.2.2 Application platform modules

As the service is provided as a web application, modules in the application platform basically consist of application platform adapters, middleware and specific DaS functionalities instantiated on the browser. These modules are responsible for collecting information from drivers, passengers or vehicle behaviour, sending it to the infrastructure and efficiently informing the driver or navigation systems with the received traffic or personal information. The architectural design of these modules is depicted in Figure 6.2.

The hardware modules needed are a GPS receiver, a camera, an advanced HMI (including touch screen, microphone and speaker) and a 3G cellular communication interface. An accelerometer is also recommended to accurately detect incidents.

The application platform adapters consist in JavaScript wrappers on the web browser that give the application access to the native resources of the Smartphone. There are adapters for the four resources used in our service available in the web browser.

In the Middleware category, we include those functionalities that may be used by many applications, promoting component reuse. These functionalities are: An IMS client acting as the front-end for VISIONS enablers. The Speech Technology module (ST) module is in charge of both converting incoming messages into human recognizable speeches and recognizing voice commands for using the service without touching the screen. There is also a front-end API for accessing Facebook and Twitter.

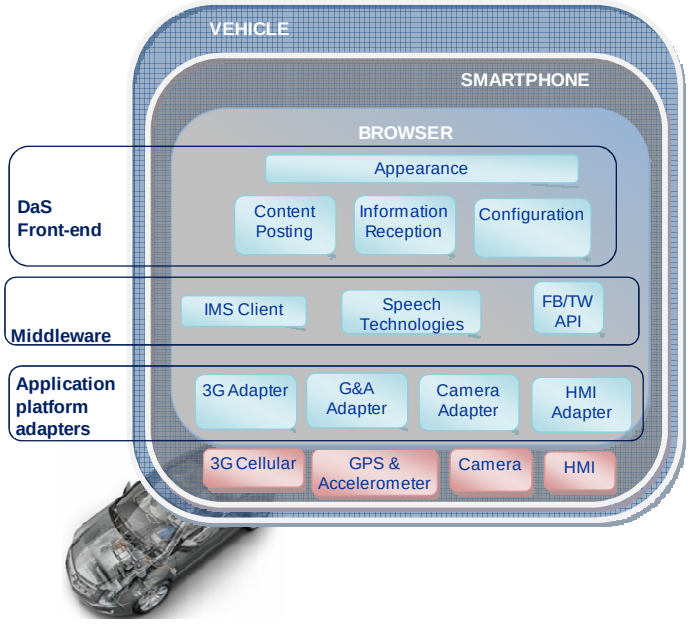


Figure 6.2 On-board modules (Smartphone)

Regarding the iOS platform we have used, as the platform is neither open nor customizable, we found some difficulties integrating external modules needed for the service as Speech Technologies and IMS client.

The specific modules developed for DaS consist of the functionalities for receiving information, post new content or enrich previously received notifications and a configuration module. Using the configuration module, users can establish their preference, including their groups of contacts and the maximum distance to an incident for receiving information. One of the most important aspects of these modules is the appearance and usability of them. This issue is deeply analyzed in the next subsection.

6.5.2.3 Connectivity

Dealing with vehicular communications, there are currently several initiatives that aim to provide the best connectivity attending the service´s needs and the traffic situation. These initiatives present a set of mechanisms that control the connectivity through several network interfaces. These interfaces are typically cellular (3G, EDGE, GPRS...), nomadic (WiFi, WiMAX) and even Vehicular Ad-hoc Networks (VANET) in which vehicles communicate among them and with roadside elements through multihop relaying without any infrastructure mediation.
VANETs offer a lot of potential as the networking support for social networks because of their distributed architecture and decentralized operation. Messages and information could be distributed on demand as vehicles and users meet along the streets or roads. Nowadays, providing some of the abovementioned requirements (e.g. security, privacy, reduced time to marked, etc.) using VANETs is very difficult. They require a minimum penetration rate and many platforms (including iOS) do not support yet implementations of those multihop data communication protocols.

Due to these factors and the reasonable coverage of 3G networks among interurban roads and cities, we propose to use 3G connectivity between vehicles and infrastructure modules for an initial phase. Fortunately, the proposed system can work with different networking solutions with minimal changes.

6.5.2.4 Usability analysis

Although it is out of the scope of this thesis, due to the importance of usability aspects for in-vehicle services we provide the interface design approaches followed in DaS. For the HMI design we take into account that drivers should be able to access the service while the vehicles are on the move. HMIs for vehicular services must efficiently cover

challenging requirements in order to avoid driver distraction (misunderstanding displayed information, lack of clarity, ambiguity, non-intuitive interface or difficult to handle, etc).

We group all the service functionalities into two different categories: related to posting and related to receiving information. In order to graphically explain this case, we include several screen captures of the DaS service in Figures 6.3, 6.4 and 6.5.

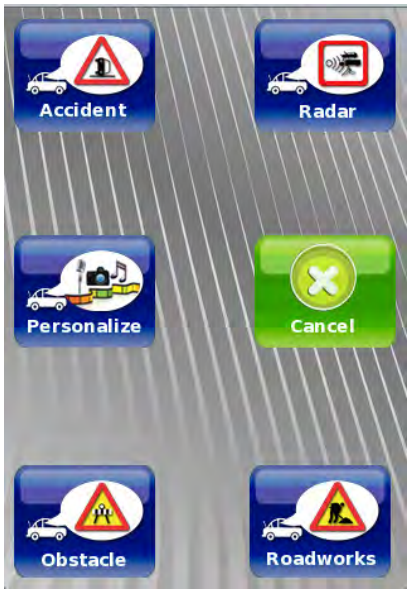


Figure 6.3 DaS main view

Figure 6.4 Traffic info display

Figure 6.5 Rating and enriching info

6.5.2.5 Information reception related functionalities

One of the key design decisions is not mixing traffic and personal information in the same view to avoid driver misunderstandings. In the main view of the service (Figure 6.3) all the space not occupied by buttons is to present traffic or personal messages (but not both). There is a button for switching between categories: from traffic to personal or vice versa.

As the application receives events, it automatically reproduces the content through Speech technologies to inform the driver, sorts them chronologically and displays (Figure 6.4) the type of information (congestion, radar, roadwork, accident or obstacle in traffic related), location (road-number in urban environments or road-kilometre in interurban), distance to incident, rate and an icon to easily identify the type.

To obtain more information or rate a received event, users must select the event and a new window (Figure 6.5) will appear displaying all the information and several buttons to rate or enrich it. Users can also access to a help menu for operating instructions through an information button (the round green one at the bottom).

6.5.2.6 Posting related functionalities

When dealing with posting related functionalities, we follow two different approaches. The first one is to automatically generate and post traffic incidents detected autonomously by the system (i.e. an unexpected hard braking detected by the accelerometer or an abnormally slow speed detected by tracking module and mobility patterns). In this approach, drivers do not interact with the system at all.

The other approach is allowing users to generate traffic or personal information. For this purpose we provide four different buttons in the main view of DaS (Figure 6.3). Two specific shortcuts to request DaS to post a congestion incident or a personal voice note and two more generic for selecting unspecified personal/traffic information. These last two buttons link to other views in which users can specify the type of information to post or add multimedia content. Buttons are grouped by categories (traffic at top, personal at bottom), have a significant size and are placed in the corners of the screen to avoid clicking failures, making the interface intuitive and easy to handle. Notice that there is a compromise between the size of the buttons and the space available to present received events to the driver.

It is important to highlight that DaS buttons can be selected not only by touching them but also by voice commands. Every time that some information is successfully posted

(even when it is traffic info automatically generated) the service reproduces a sound to inform the driver.

6.6 Conclusions

In order to demonstrate the potential of VISIONS as a Service Delivery Platform, we design a Smartphone Application that leverages on the enablers of our solution. It is a Social Network service called Drive and Share (DaS).

Social Networks have a lot of potential in vehicular scenarios but the few initiatives nowadays have not succeeded, because many restrictive technical challenges and normative issues have not been covered. It is crucial considering the usability of the service, privacy and anonymity features as well as technical aspects related to dealing with multiple types of devices in an effective and cost-efficient manner.

Regarding the usability of services for drivers, applications must work to reduce device manipulation to accomplish the restrictive regulation in many countries. This basically involves adapting the contents (info, messages, icons) to the size of the screen and using speech technologies to communicate/gather information to/from drivers. We also envision automatic traffic content posting, obtained by on-board sensors, to avoid driver reluctance to manually generate information. In order to assess the validity of traffic information generated by other vehicles, rating mechanisms for these events should be provided.

Privacy is a very relevant aspect for users of vehicular services, and must be covered allowing users to configure the anonymity level of the information posted. It is necessary to provide mechanisms to avoid third parties to track user identity and trajectory. On the other hand, users often desire to post personal content that deeply impact their contact groups. Thus social services in vehicles should also link with extended and successful SN like Facebook and Twitter through their APIs rather than creating a new SN only for vehicular services.

In order to provide a massively accepted service, providers should be able to deploy it over a large number of devices and application platforms. There are many devices (Car PCs, embedded OBUs, PNDs, tablet PCs, Smartphones, etc) susceptible to bring SN to vehicular environment. The most relevant and extended application platforms associated to Smartphones are iOS and Android. Thus, most of the service logic should not depend on specific capabilities of these devices or platforms. We propose the use of VISIONS in which the functionalities reside in accessible servers in the infrastructure. Our solution is based on extended enablers, standard protocols and

open APIs, becoming ideal candidates to host service functionalities. By following this approach, we reduce service development, maintenance and migration costs.

Social Networks will definitely emerge in vehicular scenarios: users are interested on them and providers can deploy interesting services based on social networking data sharing. DaS is an illustrative example. Moreover, sharing traffic information through SN on vehicles can lead to new business models (i.e. based on the number and quality of the reports provided by users) and novel applications and services.

7 CONCLUSIONS

Vehicular services provide the essential communication to improve road transport in several ways like reducing accidents, improving efficiency pathways, optimizing the transport of goods and people. These services present very challenging requirements in terms of communications that haven´t been successfully fulfilled nowadays. Many technological and business challenges need to be faced before launching to the mass market communication systems and services on the road.

Vehicular communications typically rely on VANET (Vehicular Ad hoc Networks). In these networks vehicles communicate with each other without the need of infrastructure. VANET are mainly oriented to disseminate information to the vehicles in certain geographic area for time critical services like safety warnings.

The use of existing mobile telecommunication networks in the context of vehicular communications is quite common. However, their use is mainly oriented towards the provision of data communication services for vehicles. These networks have evolved very rapidly within the last few years, and nowadays operators have many efficient communication mechanisms that may become very valuable in the vehicular environment, much beyond providing connectivity.

The main motivation of this thesis is to improve the operation of vehicular communications and services with an efficient and inexpensive solution based on IMS subsystem and NGN enablers. The objectives are:

1. Improving efficiency in VANET networks using a cellular channel to access valuable information from IMS and its capabilities. This efficiency improvement will occur in two relevant areas currently unresolved: Routing and Security.
2. Providing an Intelligent Transport System (ITS) service delivery platform based on IMS and NGN capabilities. The aim is to facilitate the development and deployment of new services on the vehicular environment by using standardized enablers.

7.1 Novelties

We analyze the most relevant initiatives worldwide proposing vehicular platforms. This includes reference research projects in the EU and also national initiatives from US and Japan to deploy ITS solutions in real environments. Regarding standardization activities, we remark the architecture of CALM, proposed by ISO and followed by the majority of the proposals. We analyze four of the most important topics of vehicular platforms; architectures, the use of multiple radio technologies, routing performance and security issues.

Most of the requirements for our solution are technical; basically related with Safety, security, mobility support and reliability, but there is also a socio-economic parameter which is feasibility: The solution must be deployable in terms of costs and profitable supporting the use of 3rd party service providers with several business models.

After analyzing current challenges, establishing the objectives and defining the requirements we design a solution called VISIONS “Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services”. It is a modular system which includes elements on board vehicles in the infrastructure based in standard elements of network operators. We justify the three most important design decisions of VISIONS which are:

a) Leverage on IMS subsystem and NGN standard enablers. For showing the potential of IMS and NGN enablers in the vehicular scenario; we provide explicit examples of the applicability of the enablers to both vehicular communications and services.
b) Follow the CALM reference architecture in the On board elements. CALM supports mobility through different communication interfaces seamlessly attending service requirements and status of the networks. Different radio technologies can be easily

integrated reducing the risk of obsolescence. It also helps to reduce obsolescence issues which are typical as automotive and digital sectors have completely different production cycles.

c) Avoid the use of Road Side Units (RSUs) as basic infrastructure elements. This significantly reduces deployment and maintenance costs without compromising communication performance.

There are two different units on board vehicles; the OBU, responsible for all the communications of the vehicle and the AU in which is the service environment container and the Human Machine Interface (HMI). Services communicate with external elements through the OBU using the CALM API. As many 3rd parties develop their services following the CALM reference architecture, it facilitates the execution of those services from external providers in our platform.

Modules in the infrastructure leverage on standard elements of the network operator like IMS subsystem, NGN enablers and basic security servers like Certification Authority (CA) and Authentication, Authorization and Accounting (AAA). Two modules have been designed from scratch, while the others are used as standardized or only require light adaptations.

We propose and formally define the mechanism used for both purposes; improving VANET routing performance and enhancing the distribution of Certificate Revocation Lists. We define the Key Performance Indicators (KPI) for the different phases of the process and justify the modus operandi. In summary, the process collects data from the vehicles, analyzes the information in the infrastructure, prepares an aggregated file, selects dynamically geographic regions in which the file is valuable and distributes the file in those regions.

7.2 Fulfillment of the objectives

After studying the current trends and challenges in routing protocols for VANET we justify our decision of bringing a new mechanism that enriches the information that protocols use for taking local decisions. This mechanism periodically distributes topology and networking information from the VANET. We demonstrate through simulations how our platform improves efficiency in some relevant routing protocols. In some cases we obtain an increase of 18% in the delivery rate, which corresponds to almost doubling the number of packets being delivered. It is also demonstrated it does not require a lot of bandwidth from the operator network; in our scenarios, VISIONS uses between 42 and 60% of the capacity of one of the 16 MBMS channels available.

We also study security problems and particularly those related with certificate revocation mechanisms when one of the members of the VANET becomes an attacker. We propose a CRL dissemination mechanism based on VISIONS that reduces the time for informing the vehicles of malicious nodes, decreasing significantly the window of vulnerability by broadcasting fresh revocation information only to specific geographic areas. We simulate an interurban scenario and demonstrate how our platform is more efficient neutralizing these attackers than the widely proposed solutions based on RSUs. In the best case, VISIONS reduces the number of victims by 48% compared with the RSU-based process. The usage of the cellular channel is very low, always below 5%, making the proposed solution feasible for real deployments.

Finally, we describe the limitations for bringing vehicular services and applications to the vehicle in a mass-market approach and propose an innovative Social Network service based in our platform. We justify how VISIONS facilitate the deployment and usage of complex services. In particular, it is of paramount importance considering the usability of the service, privacy and anonymity features as well as technical aspects related to dealing with multiple types of devices in an effective and cost-efficient manner. Thus, most of the service logic should not depend on specific capabilities of these devices or platforms. VISIONS functionalities reside in accessible enablers in the infrastructure and use standard protocols and APIs, becoming ideal candidates to host service functionalities. By following this approach, we reduce service development, maintenance and migration costs.

7.3 Future Work lines

The main objective of this thesis has been demonstrating that our solution VISIONS can significantly improve the performance of VANET routing protocols and security using comprehensive configuration values in VISIONS formal model. Future works will leverage on these results to fine-tune this parameters in order to optimize the efficiency of VISIONS in different traffic scenarios. This will demonstrate that there is margin for improvement in terms of performance while reducing even more the use of operator´s resources.

Moreover, this idea of operator-enabled vehicular communications has a much wider applicability and opens up an interesting research direction. We believe that VISIONS can help other applications in the vehicular environment, which include efficient

configuration mechanism for VANET, access control, QoS provisioning and security improvements in other fields like privacy violation or denial of service.

VISIONS has been designed to reduce obsolescence issues which are typical in this area because automotive and digital sectors have completely different production cycles. In fact, during the elaboration of this thesis, 4G technologies like LTE have been commercially launched in many countries. The higher data rates and lower latencies of LTE will significantly improve the performance of VISIONS.

7.4 List of Publications

All the publications authored and co-authored by the author, including journals, conference proceedings and book chapters are listed in this section.

7.4.1 Journal Papers

- **I. Lequerica**, M. García Longarón, P. M. Ruiz. “Drive and Share: Efficient provisioning of Social Networks in Vehicular Scenarios”. IEEE Communications Magazine, v. 48, n.11, p.90-97, November 2010.
- **I. Lequerica**, P. M. Ruiz, Victor Cabrera, “Improvement of vehicular communications by using 3G capabilities to disseminate control information”. IEEE Network, v.24 n.1, p.32-38, January-February 2010

7.4.2 Conference Proceedings

- **I. Lequerica,** J.A. Martinez, P. M. Ruiz. "Efficient Certificate Revocation in Vehicular Networks Using NGN Capabilities," Vehicular Technology Conference Fall (VTC 2010-Fall), 2010 IEEE 72nd , vol., no., pp.1,5, 6-9 Sept. 2010
- **I. Lequerica,** A.J.R. Ruiz, A.S.G. Ruiz, A.F.G. Skarmeta. "An IMS Based Vehicular Service Platform," Vehicular Technology Conference Fall (VTC 2010-Fall), 2010 IEEE 72nd , vol., no., pp.1,5, 6-9 Sept. 2010
- **I. Lequerica**, L. García, A. Alonso, J. Rodríguez, M.A. Guijarro, D. Quesada. “Prototipo M:Vía – Hacia el despliegue de comunicaciones y servicios en el vehículo”. X Congreso Español sobre Sistemas Inteligentes de Transporte. Madrid. Mayo 2010
- L. García, C. Pinart, **I. Lequerica**, A. Alonso, J. J. Rodríguez, J. M. González, D. Quesada, "m:Vía, an IMS-based approach for turning vehicles and roads into a smart envinronment", in Proc. 16th World Congress and Exhibition on

Intelligent Transport Systems and Services. Stockholm (Sweden), September 21-25, 2009

- J.A. Olivera, I. Cortazar, C. Pinart, A. Los Santos, **I. Lequerica.** "VANBA: A Simple Handover Mechanism for Transparent, Always-On V2V Communications," Vehicular Technology Conference, 2009. VTC Spring 2009. IEEE 69th , vol., no., pp.1,5, 26-29 April 2009
- C. Pinart, P. Sanz, **I. Lequerica**, D. García, I. Barona, D. Sánchez-Aparisi. DRIVE: a reconfigurable testbed for advanced vehicular services and communications, Proceedings of the 4th International Conference on Testbeds and research infrastructures for the development of networks & communities, March 18-20, 2008, Innsbruck, Austria
- D. Quesada, **I. Lequerica**, A. Alonso, JM. González, J. Rodríguez. "mVía, creación de un entorno inteligente en las carreteras". VIII Congreso Español sobre Sistemas Inteligentes de Transporte" 2008
- C. Pinart, P. Sanz, I. Barona, D. García, **I. Lequerica**. "El automóvil conectado: Servicios avanzados para desplazamientos más seguros, eficientes y cómodos". Revista Ingenieros de Telecomunicación de Cataluña. 2008
- **I. Lequerica**. "Comunicaciones en el entorno del automóvil". XI Congreso Nacional de Internet, Telecomunicaciones y Sociedad de la Información. Málaga 2007
- **I. Lequerica**, I. Cortázar. "Rendimiento de VANETs en escenarios de uso realistas". VII Congreso Español sobre Sistemas Inteligentes de Transporte. Valencia 2007
- **I. Lequerica**, C. Miguel Nieto. "Plataforma de comunicaciones B3G para medios de transporte". Telecom I+D Madrid 2006

7.4.3 Book Chapters

- **I. Lequerica** (Telefónica I+D Ed.), chapter "Comunicaciones y servicios en el automóvil habilitados por la operadora" of the book "Tecnologías para las comunicaciones del futuro", published by Bubok. July 2010
- M.ª E. López Lambas (Ed.), A. Monzón, I. Barona, D. García, F. Jiménez, L. Lavara, **I. Lequerica**, C. Pinart (Telefónica I+D Ed.), chapter "Hacia una red inteligente de transporte público urbano" of the book "La contribución de las TIC a la sostenibilidad del transporte en España", reviewed by L. Lada and published by the Real Academia de Ingeniería of Spain. July 2009

7.5 Patents

As a result of this thesis, the author has patented the solution. The patent application has been submitted to the following countries: Spain, European Patent Office (EPO), Argentina, Brasil, Chile, El Salvador, Guatemala, México, Peru, Venezuela and Morocco.

The patent by July 2013 has already been approved in Spain (ES 2346277 B1) and Morocco and it is still under evaluation in the rest of the geographies.

- **I. Lequerica**. System and method for improving communications between vehicles. EP Patent 2,146,486, 2010

REFERENCES

[1] International Traffic Safety Data and Analysis Group. International Transport Forum, “Road Safety Annual Report 2011”. [Online]. Available: http://internationaltransportforum.org/irtadpublic/pdf/11IrtadReport.pdf

[2] IP Multimedia Subsystem, 3G Partnership Program (3GPP) [Online]. Available: http://www.3gpp.org/.

[3] E. Schoch, F. Kargl, M. Weber and T. Leinmuller, "Communication patterns in VANETs," Communications Magazine, IEEE, vol.46, no.11, pp.119-125, November 2008.

[4] L. Franck and F. Gil-Castineira, "Using Delay Tolerant Networks for Car2Car communications," Industrial Electronics, 2007. ISIE 2007. IEEE International Symposium on, pp.2573-2578, 4-7 June 2007

[5] T. Kosch, C.J. Adler, S. Eichler, Schroth and M. Strassberger, "The scalability problem of vehicular ad hoc networks and how to solve it," Wireless Communications, IEEE, vol.13, no.5, pp.22-28, October 2006

[6] S. Ni, Y. Tseng, Y. Chen, and J. Ping, “The Broadcast Storm Problem in a Mobile Ad Hoc Network,“ in Proc. of the 5th annual ACM/IEEE international conference on Mobile computing and networking, pp.151-162. 1999.

[7] S. Chakrabarti, and A. Mishra, "QoS issues in ad hoc wireless networks," Communications Magazine, IEEE, vol.39, no.2, pp.142,148, February 2001

[8] M. Boban, G. Misek, and O. K. Tonguz, “What is the best achievable QoS for unicast routing in VANET?,” in Proc. of the IEEE Globecom Workshops, pp. 1–10, New Oreleans, La, USA, December 2008.

[9] B. Parno and A. Perrig, “Challenges in securing vehicular networks,” in Proc. of the Workshop on Hot Topics in Networks (HotNets-IV), 2005.

[10] M. Raya and J.-P. Hubaux, “Securing vehicular ad hoc networks,” Journal of Computer Security, vol. 15, no. 1, pp. 39–68, 2007.

[11] IEEE 802.11 Standard. [Online]. Available: http://www.ieee802.org/11/

[12] IP European Project CVIS, “Cooperative Vehicle-Infrastructure Systems”. [Online]. Available: http://www.cvisproject.org/

[13] IP European Project SAFESPOT, “Cooperative vehicles and road infrastructure for road safety”. [Online]. Available: http:// http://www.safespot-eu.org/

[14] IP European Project COOPERS, “Cooperative Systems for Intelligent Road Safety”. [Online]. Available: http://www.coopers-ip.eu

[15] IP European Project PREVENT, “Preventive and Active Safety Applications”. [Online]. Available: http://www.prevent-ip.org/

[16] STREP European Project Com2react. [Online]. Available: http://www.com2react-project.org/

[17] Car to Car Communication Consortium. [Online]. Available: http://www.car-to-car.org/

[18] Imobility Forum. [Online]. Available: http://www.icarsupport.eu/esafety-forum/

[19] European Commission Recommendation on eCall. [Online]. Available: http://ec.europa.eu/information_society/activities/esafety/ecall/index_en.htm

[20] US Department of Transportation (DOT), Research and Innovative Technology Administration (RITA) Safety Pilot. [Online]. Available: http://www.its.dot.gov/safety_pilot/index.htm

[21] Japanese Ministry of Land, Infrastructure and Transportation (MLIT), “ITS initiatives in Japan”. [Online]. Available: http://www.mlit.go.jp/road/ITS/pdf/ITSinitiativesJapan.pdf

[22] ISO TC204 WG16. [Online]. Available: http://calm.its-standards.info/

[23] Y. Zhang, J. Zhao, G. Cao, “On Scheduling Vehicle-Roadside Data Access,” in Proc. of 4th ACM International Workshop on Vehicular Ad Hoc Networks (VANET-07), Montréal, Québec, Canada, pp. 9-18, September 2007.

[24] N. Banerjee, M.D. Corner, D. Towsley and B.N. Levine, “Relays, base stations, and meshes: Enhancing mobile networks with Infrastructure,” in Proc. of the 14th Annual ACM International Conference on Mobile Computing and Networking. MOBICOM 2008, San Francisco, California, USA, September 2008.

[25] Project Cooperative Cars: CoCar. [Online]. Available: http://www.aktiv-online.org/english/aktiv-cocar.html

[26] E. Gustafsson y A. Jonsson, “Always best connected,” Wireless Communications, IEEE, vol. 10, pp. 49-55, February 2003.

[27] IEEE Standard for Local and Metropolitan Area Networks- Part 21: Media Independent Handover," IEEE Std 802.21-2008, pp.c1,301, January 2009

[28] ISO/DIS 21217, “Intelligent transport systems-Communications access for land mobiles (CALM)-Architecture”, 2010.

[29] ISO/DIS 21210, “Intelligent transport systems-Communications access for land mobiles (CALM)-Networking Protocols”, 2008.

[30] E. Weiss, M. Muehleisen, C.H. Rokitansky, B. Walke and L. Georgi, "Architecture of an Always Best Connected Vehicular Communication Gateway," Vehicular Technology Conference, 2006. VTC-2006 Fall. 2006 IEEE 64th, pp.1-5, 25-28 September 2006.

[31] J.-H. Seol, J.-M. Chung, "IEEE 802.21 MIH based Handover for Next Generation Mobile Communication Systems," in Proc. Innovations in Information Technology, 2007. IIT '07. 4th International Conference on, pp.431-435, 18-20 November 2007.

[32] J. A. Olivera, I. Cortázar, C. Pinart, A. Los Santos, I. Lequerica, "VANBA: a simple handover mechanism for transparent, always-on V2V communications", in Proc. IEEE 69th Vehicular Technology Conference (VTC2009-Spring). pp.1-5, Barcelona (Spain), April 26–29, 2009.

[33] N. Wisitpongphan, F. Bai, P. Mudalige, and O.K. Tonguz, “On the Routing Problem in Disconnected Vehicular Ad-hoc Networks”, in Proc. of 26th IEEE International Conference on Computer Communications (INFOCOM '07), pp. 2291-2295, Montreal, Quebec, Canada, May 2007.

[34] V. Cabrera, F.J. Ros, P.M. Ruiz, "Simulation-based Study of Common Issues in VANET Routing Protocols," in Proc. IEEE 69th Vehicular Technology Conference (VTC2009-Spring), pp. 1-5. Barcelona, Spain, April 26-29, 2009.

[35] Liberty Alliance, "Liberty ID-WSF Authentication, Single Sign-On, and Identity Mapping Services Specification, Version v2.0-16”. [Online]. Available: http://www.projectliberty.org/liberty/content/download/350/2563/file/draft-liberty-idwsf-authn-svc-v2.0-16.pdf

[36] Y.-C. Hu, A. Perrig and D. Johnson, “Ariadne: a secure on-demand routing protocol for ad hoc networks,” in Proc. of Mobicom’02, pp. 12–23. September 2002.

[37] T. Paila, R. Walsh, M. Luby, V. Roca, R. Lehtonen, “FLUTE - File Delivery over Unidirectional Transport, RFC 6726,” November 2012. [Online]. Available: http://tools.ietf.org/rfc/rfc6726.txt

[38] P. Agrawal, Y. Jui-Hung, Ch. Jyh-Cheng and Z. Tao, "IP multimedia subsystems in 3GPP and 3GPP2: overview and scalability issues," Communications Magazine, IEEE, vol.46, no.1, pp.138,145, January 2008.

[39] J. Rosenberg, H. Schulzrinne, G. Camarillo, A. Johnston,J. Peterson, R. Sparks, M. Handley, E. Schooler, “SIP: Session Initiation Protocol, RFC 3261,” June 2002. [Online]. Available: http://www.ietf.org/rfc/rfc3261.txt

[40] M. Handley, V. Jacobson, C. Perkins, “SDP: Session Description Protocol, RFC 4566,” July 2006. [Online]. Available: http://www.ietf.org/rfc/rfc4566

[41] H. Schulzrinne, S. Casner, R. Frederick and V. Jacobson, “RTP: A Transport Protocol for Real-Time Applications, RFC 3550,” July 2003. [Online]. Available: http://tools.ietf.org/rfc/rfc3550.txt

[42] P. Calhoun, J. Loughney, E. Guttman, G. Zorn and J. Arkko, “Diameter Base Protocol, RFC 3588,” September 2003. [Online]. Available: http://tools.ietf.org/rfc/rfc3588.txt

[43] 3GPP TS 29.198-7 V4.5.0, “Terminal Capabilities Technical Specification,” March 2003.

[44] B. Campbell, R. Mahy and C. Jennings, “MSRP: The Message Session Relay Protocol, RFC 4975,” September 2007. [Online]. Available: http://tools.ietf.org/rfc/rfc4975.txt

[45] M. Bakhuizen and U. Horn, “Mobile Broadcast/Multicast in Mobile Networks,” Ericsson Review, vol. 82 no. 1, pp. 6–13, 2005.

[46] 3GPP, “Multimedia Broadcast/Multicast Service (MBMS); Architecture and functional description,” 3rd Generation Partnership Project (3GPP), TS 23.246 v6.10.0, 2006.

[47] D.N. Cottingham, I.J. Wassell and R.K. Harle, "Performance of IEEE 802.11a in Vehicular Contexts," In Proc. Vehicular Technology Conference, VTC2007-Spring. IEEE 65th, pp.854-858, 22-25 April 2007.

[48] V. Bychkovsky, B. Hull, A. Miu, H. Balakrishnan and S. Madden, “A measurement study of vehicular internet access using in situ Wi-Fi networks,” in Proc. of the 12th annual international conference on Mobile computing and networking (MobiCom '06). ACM, New York, NY, USA, 50-61. 2006.

[49] C. Wewetzer, M. Caliskan, K. Meier and A. Luebke, "Experimental Evaluation of UMTS and Wireless LAN for Inter-Vehicle Communication," in Proc. of 7th International Conference on ITS, pp.1-6, 6-8 June 2007.

[50] J.B. Kenney, “Dedicated Short-Range Communications (DSRC) Standards in the United States,” in Proc. of the IEEE. Volume: 99, no. 7,pp. 1162–1182. July 2011.

[51] J. Zhao and G. Cao, “VADD: Vehicle-Assisted Data Delivery in Vehicular Ad Hoc Networks”, In Proc. of 25th IEEE International Conference on Computer Communications (INFOCOM ’06), Barcelona, Spain, April 2006.

[52] I. Leontiadis, and C. Mascolo, “GeOpps: Opportunistic Geographical Routing for Vehicular Networks”, in Proc. of the IEEE Workshop on Autonomic and Opportunistic Communications, Helsinki, Finland, 2007.

[53] J. Lebrun, C.N. Chuah, and D. Ghosal, “Knowledge-Based Opportunistic Forwarding in Vehicular Wireless Ad Hoc Networks,” in Proc. of 61st Vehicular Technology Conference, Spring 2005, pp. 2289–2293, Vol. 4, May 2005.

[54] Official Journal of the European Communities, “Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data,” 1995.

[55] PhoneGap. [Online]. Available: http://phonegap.com

[56] John O'Connell, "Service Delivery within an IMS Environment", IEEE Vehicular Technology Magazine, March 2007.

[57] Y. Zang, B. Walke, S. Sories, G. Gehlen, "Towards a European Solution for Networked Cars-Integration of Car-to-Car Technology into Cellular Systems for Vehicular Communication in Europe," in Proc. Of IV Fully Networked Car Workshop, Geneva, Switzerland, pp. 14, March 2009.

[58] Joyn Initiative. [Online]. Available: http://www.joynus.com

[59] K. Rikitake, Y. Araki, Y. Kawahara, M. Minami and M. Hiroyuki, "NGN/IMS-Based Ubiquitous Health Monitoring System," in Proc. of the 6th IEEE Conference on Consumer Communications and Networking Conference, pp. 56-57, Las Vegas, NV, USA, January 2009.

[60] P. Yanlin, Z. Abichar and J.M. Chang, "Roadside-Aided Routing (RAR) in Vehicular Networks," IEEE International Conference on Communications, ICC '06, vol: 8, pp. 3602-3607, 2006.

[61] K. Mershad, H. Artail, and M. Gerla, "ROAMER: Roadside Units as message routers in VANETs," Ad Hoc Networks , vol. 10, no. 3, pp.479–496, 2012.

[62] P. Papadimitratos, G. Mezzour, J.P. Hubaux, "Certificate revocation list distribution in vehicular communication systems," in Proc. Of the fifth ACM international workshop on VehiculAr Inter-NETworking, San Francisco, California, USA. September 2008.

[63] P. Laberteaux, J. Haas, and Y.C. Hu, "Security certificate revocation list distribution for vanet," in Proc. of the fifth ACM international workshop on Vehicular Internetworking, VANET '08, pp. 88-89, 2008.

[64] X. Lin, R. Lu, C. Zhang, H. Zhu, P.H. Ho and X. Shen, "Security in Vehicular Ad Hoc Networks," IEEE Communications Magazine, vol. 46, no. 4, pp. 88-95, 2008.

[65] J. Barrachina, P. Garrido, M. Fogue, F. J. Martinez, J.-C. Cano, C. T. Calafate and P. Manzoni, "D-RSU: A Density-Based Approach for Road Side Unit Deployment in Urban Scenarios," in Proc. of the1st International Workshop on IPv6-based Vehicular Networks (Vehi6), Alcalá de Henares, Madrid, Spain. June 3, 2012.

[66] V. Namboodiri, M. Agarwal and L. Gao, "A study on the feasibility of mobile gateways for vehicular ad-hoc networks", in Proc. of the 1st ACM international workshop on Vehicular ad hoc networks, pp. 66-75, 2004.

[67] D.R. Tufano. “Automotive HUDs: The Overlooked Safety Issues,” Human Factors: The Journal of the Human Factors and Ergonomics Society, vol. 39, no. 2, pp. 303-311, June 1997.

[68] ISO 11898-1:2003, “Road vehicles -- Controller area network (CAN) -- Part 1: Data link layer and physical signaling”. [Online]. Available: http://www.iso.org/iso/catalogue_detail.htm?csnumber=33422

[69] LIN Consortium. [Online]. Available: http://www.lin-subbus.org

[70] MOST Cooperation Initiative. [Online]. Available: http://www.mostcooperation.com

[71] T. ElBatt, C. Saraydar, M. Ames and T. Talty, "Potential for Intra-Vehicle Wireless Automotive Sensor Networks," in Proc. of Sarnoff Symposium, IEEE, pp.1,4, 27-28 March 2006.

[72] S. Thomson, T. Narten, T. Jinmei, “IPv6 Stateless Address Autoconfiguration, RFC 4862,” September 2007. [Online]. Available: http://tools.ietf.org/rfc/rfc4862.txt

[73] R. Droms, J. Bound, T. Lemon, B. Volz, C. Perkins and M. Carney, “Dynamic Host Configuration Protocol for IPv6 (DHCPv6), RFC3315,” July 2013. [Online]. Available: http://tools.ietf.org/html/rfc3315

[74] L. Buttyan, T. Holczer and I. Vajda, "On the Effectiveness of Changing Pseudonyms to Provide Location Privacy in Vanets," in Proc. of the 3rd Euro. Wksp. Sec. and Privacy in Ad Hoc and Sensor Networks, vol. 4572, pp.129-141, 2007.

[75] E. Fonseca, A. Festag, R. Baldessari, and R. Aguiar, "Support of anonymity in VANETs-Putting pseudonymity into practice," in Proc. of the IEEE WCNC, March 2007.

[76] J. Walker and A. Kulkarni, “The COPS (Common Open Policy Service) Protocol, RFC 4261,“ December 2005. [Online]. Available: http://tools.ietf.org/rfc/rfc4261.txt

[77] O. Wolfson and H. Yin, “Accuracy and Resource Consumption in Tracking and Location Prediction,” Lecture Notes in Computer Science Volume 2750, pp. 325-343, 2003.

[78] A. Civilis, C.S. Jensen and S. Pakalnis, “Techniques for Efficient Road-Network-Based Tracking of Moving Objects,” IEEE Transactions on Knowledge and Data Engineering, vol.17, no.5, pp.698-712, May 2005.

[79] Genius service of Apple Inc. [Online]. Available: http://www.apple.com/legal/internet-services/itunes/es/genius.html

[80] M. Garcia-Martin, M. Belinchon, M. Pallares-Lopez, C. Canales-Valenzuela, K. Tammi, “Diameter Session Initiation Protocol (SIP) Application, RFC 4740,” November 2006. [Online]. Available: http://www.ietf.org/rfc/rfc4740.txt

[81] S. Farrell, J. Vollbrecht, P. Calhoun, L. Gommans, G. Gross, B. de Bruijn, C. de Laat, M. Holdrege and D. Spence, "AAA Authorization Requirements", RFC 2906,” August 2000. [Online]. Available: http://tools.ietf.org/rfc/rfc2906.txt

[82] W3C, “Ubiquitous Web Applications Activity Statement”. [Online]. Available: http://www.w3.org/2007/uwa/Activity.html

[83] Open Mobile Alliance, “OMA Mobile Location Service Architecture,” Candidate Version 1.0, 07 June 2005. [Online]. Available: http://technical.openmobilealliance.org/Technical/release_program/docs/MLS/V1_0-20050607-C/OMA-AD-MLS-V1_0-20050607-C.pdf

[84] A.H. Sayed, A. Tarighat, N. Khajehnouri. “Network-based wireless location: challenges faced in developing techniques for accurate wireless location information,” Signal Processing Magazine, IEEE. vol. 22, no. 4, pp. 24-40, July 2005.

[85] Open Mobile Alliance, “OMA Secure User Plane Location Architecture,” Candidate Version 1.0, 27 Jan 2006. [Online]. Available: http://docbox.etsi.org/STF/Archive/STF321_TISPAN3_EC_Emergency_Call_Location/Public/Library/OMA%20Documents/oma-ad-supl-v1_0-20060127-c.pdf

[86] 3GPP, “Multimedia Broadcast/Multicast Service (MBMS); Architecture and functional description,” 3GPP Specification detail TS 23.246. [Online]. Available: http://www.3gpp.org/ftp/Specs/html-info/23246.htm

[87] M. Watson, A. Begen and V. Roca “Forward Error Correction (FEC) Framework, RFC 6363,” October 2011. [Online]. Available: http://tools.ietf.org/rfc/rfc6363.txt

[88] D. Gomez-Barquero, A. Fernandez-Aguilella and N. Cardona, “Multicast Delivery of File Download Services in Evolved 3G Mobile Networks With HSDPA and MBMS,” IEEE Transactions on Broadcasting, vol. 55, no 4, pp. 742-751, December 2009.

[89] Open Mobile Alliance, “OMA User Plane Location Protocol,” Candidate Version 3.0, 20 Sep 2011. [Online]. Available: http://technical.openmobilealliance.org/Technical/release_program/docs/SUPL/V3_0-20110920-C/OMA-TS-ULP-V3_0-20110920-C.pdf

[90] A. Soares, N. Souto, J.C. Silva, P. Eusebio and A. Correia, “Effective Resource Management for MBMS in UMTS Networks,” Wireless Personal Communications, no.42, pp.185-211, 2007.

[91] M.S. Bouassida, G. Guette, M. Shawky and B. Ducourthial, “Sybil Nodes Detection Based on Received Signal Strength Variations within VANET,” International Journal of Network Security, vol.9, no.1, pp. 22-33, July 2009.

[92] T. Zhou; R.R. Choudhury, P. Ning and K. Chakrabarty, "P2DAP — Sybil Attacks Detection in Vehicular Ad Hoc Networks," Selected Areas in Communications, IEEE Journal on, vol.29, no.3, pp.582-594, March 2011.

[93] T. Zhou; R.R. Choudhury, P. Ning and K. Chakrabarty, "Privacy-Preserving Detection of Sybil Attacks in Vehicular Ad Hoc Networks," Mobile and Ubiquitous Systems: Networking & Services, Fourth Annual International Conference on, pp.1-8, 6-10 Aug. 2007.

[94] B. Bellur, “Certificate Assignment Strategies for a PKI-Based Security Architecture in a Vehicular Network”. Global Telecommunications Conference, 2008. IEEE GLOBECOM 2008, pp. 1-6, December 2008.

[95] R. Housley, W. Polk, W. Ford and D. Solo, “Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile. RFC 3280,” April 2012. [Online]. Available: http://www.ietf.org/rfc/rfc3280.txt

[96] L. Batten, and R. Safavi-Naini, “Information Security and Privacy,” Lecture Notes in Computer Science, LNCS 4058, Springer-Verlag, pp. 1-446, Germany, 2006.

[97] Li and Y. Wang, "Routing in Vehicular Ad Hoc Networks: A Survey," IEEE Vehicular Technology Magazine, vol. 2, no. 2, pp. 12-22, June 2007.

[98] J. Bernsen, D. Manivannan, “Greedy Routing Protocols for Vehicular Ad Hoc Networks,” in Proc. of the Wireless Communications and Mobile Computing Conference, pp. 632-637, Crete, Greece, August 2008.

[99] ISO 21217:2010, “Intelligent transport systems-Communications access for land mobiles (CALM)-Architecture”. [Online]. Available: http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=50507

[100] Burton H. Bloom, “Space/time trade-offs in hash coding with allowable errors,” Communications ACM, 13(7), pp. 422-426, July 1970.

[101] F. Li, Y. Wang, "Routing in vehicular ad hoc networks: A survey," IEEE Vehicular Technology, Vol. 2, no. 2, pp.12–22, June 2007.

[102] B. Paul, Md. Ibrahim and Md. Abu Naser Bikas. "VANET Routing Protocols: Pros and Cons," International Journal of Computer Applications, Vol. 20, no.3, April 2011.

[103] C. Lochert, M. Mauve, H. Fusler and H. Hartenstein, "Geographic routing in city scenarios," ACM SIGMOBILE Mobile Computing and Communications Review, Vol 9, no. 1, pp. 69–72, January 2005.

[104] V. Naumov and T.R. Gross, "Connectivity-Aware Routing (CAR) in Vehicular Ad-hoc Networks," in Proc. of the 26th IEEE International Conference on Computer Communications (INFOCOM '07), pp. 1919–1927, Anchorage, Alaska, USA, May 2007.

[105] C. Lochert, H. Hartenstein, J. Tian, H. F¨ußler, D. Hermann and M. Mauve, "A Routing Strategy for Vehicular Ad Hoc Networks in City Environments", In Proc. of the IEEE Intelligent Vehicles Symposium 2003, Columbus, OH, USA, pp. 156–161, June 2003.

[106] B. C. Seet, G. Liu, B.S. Lee, C. H. Foh, K. J. Wong and K. K. Lee, "A-STAR: A Mobile Ad Hoc Routing Strategy for Metropolis Vehicular Communications, In Proc. of the 3rd International Networking Conference IFIP-TC6 (IFIP '04), Athens, Greece, pp. 989–999, Vol. 3042, Lecture Notes in Computer Science, December 2004.

[107] K. Wong, B. Lee, B. Seet, G. Liu and Lijuan Zhu. "BUSNet: Model and Usage of Regular Traffic Patterns in Mobile Ad Hoc Networks for Inter-Vehicular Communications," in Proc. of International Conference on Information and Communication Technologies, 2003.

[108] J. Tian, L. Han, K. Rothermel, and C. Cseh, "Spatially Aware Packet Routing for Mobile Ad Hoc Inter-Vehicle Radio Networks", In Proc. of IEEE Intelligent Transportation System Conference (ITSC '03), Shanghai, China, pp. 1546-1551, October 2003.

[109] J. Holis and P. Pechac, "Simulation of UMTS capacity and quality of coverage in urban macro- and microcellular environment," Radioengineering, vol. 14, no. 4, p. 21-26, 2005.

[110] Network Simulator 2. [Online]. Available: http://www.isi.edu/nsnam/ns/

[111] Simulation of Urban Mobility (SUMO). [Online]. Available: http://sumo.sourceforge.net/

[112] Traffic and Network Simulation Environment (TRANS). [Online]. Available: http://trans.epfl.ch/

[113] IEEE, “Trial-Use Standard for Wireless Access in Vehicular Environments - Security Services for Applications and Management Messages,” IEEE Std 1609.2-2006, pp.1-105, 2006.

[114] P. Wohlmacher, “Digital Certificates: A Survey of Revocation Methods,” in Proc. of the ACM Workshop Multimedia, Los Angeles, CA, pp. 111–14, October 2000.

[115] Open Geospatial Consortium (OGC). [Online]. Available: www.opengeospatial.org/

[116] A. Rao, A. Sangwan, A. Kherani, A. Varghese, B. Bellur, and R. Shorey, “Secure V2V Communication With Certificate Revocations,” IEEE Infocom 2007, Mobile Networking for Vehicular Environments workshop, pp. 127–132, 2007.

[117] J. Haas, Y. Hu and K.P. Laberteaux, “Design and analysis of a lightweight certificate revocation mechanism for VANET,” In Proc. of the sixth ACM international workshop on VehiculAr InterNETworking (VANET '09), pp. 89-98, Beijing, China, September 2009.

[118] M. Raya, P. Papadimitratos, I. Aad, D. Jungels and J.-P. Hubaux, “Eviction of misbehaving and faulty nodes in vehicular networks,” Selected Areas in Communications, IEEE Journal on, vol. 25, pp. 1557–1568, October 2007.

[119] D. Cooper, “A More Efficient Use of Delta-CRLs,” in IEEE Symposium on Security and Privacy 2000, pp. 190–202, May 2000.

[120] M. Raya, D. Jungels, P. Papadimitratos, I. Aad and J.-P. Hubaux, “Certificate Revocation in Vehicular Networks,” Technical Report LCA-Report-2006-006, 2006.

[121] S. Choudhury, K Bhatnagar and W Haque, “Public key infrastructure implementation and design,” Published by M&T Books, edition 2002. [Online]. Available: http://www.e-reading-lib.org/bookreader.php/142115/Public_Key_Infrastructure_implementation_and_design.pdf

[122] OMA, "Enabler Release Definition for Mobile Location Service (MLS)," v. 1.2, February 2010.

[123] J. Owyang, “A Collection of Social Network Stats for 2010”. [Online]. Available: http://www.web-strategist.com/blog/2010/01/19/a-collection-of-social-network-stats-for-2010

[124] A. Tolve, “Social Networking and the Connected Car”. [Online]. Available: http://social.telematicsupdate.com/industry-insight/social-networking-and-connected-car

[125] M. Li and Z. Du, “Dynamic Social Networking Supporting Location-Based Services,“ Intelligent Human-Machine Systems and Cybernetics, IHMSC '09. International Conference on, pp. 149–152, 2009.

[126] Aha Radio. [Online]. Available: http://aharadio.com/

[127] WAZE. [Online]. Available: http://world.waze.com/

[128] G. de Melo, F. Honold, M. Weber, M. Poguntke and A. Berton, “Towards a flexible UI model for automotive human-machine interaction,“ 1st International Conference on Automotive User Interfaces and Interactive Vehicular Applications, 2009, pp. 47-50, September 2009.

[129] Ch. Guanling and F. Rahman, “Analyzing Privacy Designs of Mobile Social Networking Applications,“ Embedded and Ubiquitous Computing, 2008. EUC '08. IEEE/IFIP International Conference on, vol. 2, pp. 83–88, 2008.

[130] Apple Inc. iOS. [Online]. Available: http://www.apple.com/ios/

[131] S.J. Vaughan-Nichols, "Will HTML 5 Restandardize the Web?," Computer, vol.43, no.4, pp.13-15, April 2010.

Glossary of Terms & Acronyms

AA *Administrative Areas*

AAA *Authentication, Authorization and Accounting*

ABC *Always Best Connected*

ADAS *Advanced Driver Assistance System*

AKA *Authentication and Key Agreement*

API *Application Programming Interface*

AS *Application Server*

A-STAR *Anchor-based Street and Traffic Aware*

AU *Application Unit*

BGCF *Breakout Gateway Control Function*

CA *Certification Authority*

CALM *Continuous Air interface Long and Medium range*

CAN *Controller Area Network*

CAR *Connectivity-Aware Routing*

CDMA *Code Division Multiple Access*

CI *Confidence Interval*

CME *CALM Management Entity*

CRL *Certificate Revocation List*

CRT *Certificate Revocation Tree*

DaS *Drive and Share*

DDR *Differential Density Ratio*

DoS *Denial of Service*

DOT *Department of Transportation (US)*

DRIP *Dynamic Route Information Panel*

DRP *Distributed Revocation Protocol*

DSRC *Dedicated Short Range Communications*

DTN *Delay/Disruption Tolerant Networking*

ETSI *European Telecommunications Standards Institute*

FEP *Forward Error Correction*

FLUTE *File Delivery over Unidirectional Transport*

GDP *Gross Domestic Product*

GEOOPS *Geographical Opportunistic Routing for Vehicular Networks*

GL *Global Location*

GPCR *Greedy Perimeter Coordinator Routing*

GPS *Global Positioning System*

GSR *Geographic Source Routing*

GYTAR *Greedy Traffic Aware Routing*

GUI *Graphical user interface*

GUP *Generic User Profile*

HMI *Human Machine Interface*

HSDPA *High-Speed Downlink Packet Access*

HSUPA *High-Speed Uplink Packet Access*

HTTP *Hyper Text Transfer Protocol*

HUD *Head-Up Display*

ICT *Information Communications Technology*

I-CSCF *Interrogating Call Session Control Function*

IdP *Identity Provider*

ID-WSF *Identity Web Service Framework*

IEEE *Institute of Electrical and Electronics Engineers*

IETF *Internet Engineering Task Force*

IGMP *Internet Group Management Protocol*

IM *Instant Messaging*

IME *Interface Management Entity*

IMS *IP Multimedia Subsystem*

IP *Internet Protocol*

ISO *International Organization for Standardization*

ITS *Intelligent Transport Systems*

ITU *International Telecommunication Union*

LIN *Local Interconnect Network*

LoMoSo *Location-based Mobile Social networks*

LTE *Long Term Evolution*

M2M *Machine to Machine*

MAC *Media Access Control*

MBMS *Multimedia Broadcast Multicast Service*

MEGACO *Media Gateway Control Protocol*

MGC *Media Gateway Control Function*

MGW *Media Gateway*

MIH *Media Independent Handover*

MLIT *Ministry of Land, Infrastructure and Transportation (Japan)*

MLP *Mobile Location Protocol*

MLS *Mobile Location Services*

MM *Multimedia Messaging*

MNO *Mobile Network Operator*

MoSoSo *Mobile Social Software*

MOST *Media Oriented Systems Transport*

MoVe *Motion Vector protocol*

MSISDN *Mobile Station International Subscriber Directory Number*

MRF *Multimedia Resource Function*

MRFC *Multimedia Resource Function Controller*

MRFP *Multimedia Resource Function Processor*

MSRP *Message Session Relay Protocol*

NAS *Network Access Server*

NFC *Near Field Communications*

NGN *Next Generation Networks*

NHTSA *National Highway Traffic Safety Administration*

NME *Network Management Entity*

NS2 *Network Simulator 2*

NVB *Neighbour Void*

OBD-II *On Board Diagnostics*

OBU *On Board Unit*

OCSP *Online Certificate Status Protocol*

OEM *Original Equipment Manufacturer*

OGC *Open Geospatial Consortium*

OMA *Open Mobile Alliance*

OSI *Open Systems Interconnection*

PC *Personal Computer*

P-CSCF *Proxy Call Session Control Function*

PDR *Packet Delivery Ratio*

PKI *Public Key Infrastructure*

PND *Personal Navigator Devices*

POI *Point of Interest*

PSTN *Public Switched Telephone Network*

QoS *Quality of Service*

QPSK *Quaternary Phase Shift Keying*

RFC *Request for Comments*

RITA *Research and Innovative Technology Administration*

RLP *Roaming Location Protocol*

RLS *Reactive Location Service*

RSU *Road Side Unit*

RTP *Real-Time Transport Protocol*

SAP *Service Access Point*

SCF Store-Carry-Forward

S-CSCF Serving Call Session Control Function

SDK Software Development Kit

SDP Session Description Protocol

SDP Service Delivery Platform

SF Spreading Factor

SIMPLE SIP for Instant Messaging and Presence Leveraging Extensions

SIP Session Initiation Protocol

SLA Service Level Agreement

SN Social Networks

SOAP Simple Object Access Protocol

ST Speech Technology

SUMO Simulation of Urban Mobility

SUPL Secure User Plane Location

T&C Terms and Conditions

TBS Transport Block Size

TCP Transmission Control Protocol

TDM Time Division Multiplexing

TRANS Traffic and Network Simulation Environment

TTI Transport Transmission Interval

TTL Time To Live

TTS Text To Speech

UDP User Datagram Protocol

UE User Equipment

UMTS Universal Mobile Telecommunications System

UP User Profile

URI Universal Resource Identifier

V2I Vehicle to Infrastructure

V2R Vehicle to Roadside Units

V2U Vehicle to User

V2V Vehicle to Vehicle

VADD Vehicle-Assisted Data Delivery in Vehicular Ad Hoc Networks

VANET Vehicular Adhoc Network

VHO Vertical Handover

VIB Vehicle Information Base

VICS Vehicle Information and Communication System

VISIONS Vehicular communication Improvement: Solution based on IMS Operational Nodes and Services

WAVE Wireless Access Vehicular Environment

WoV Window of Vulnerability

XML EXtensible Markup Language